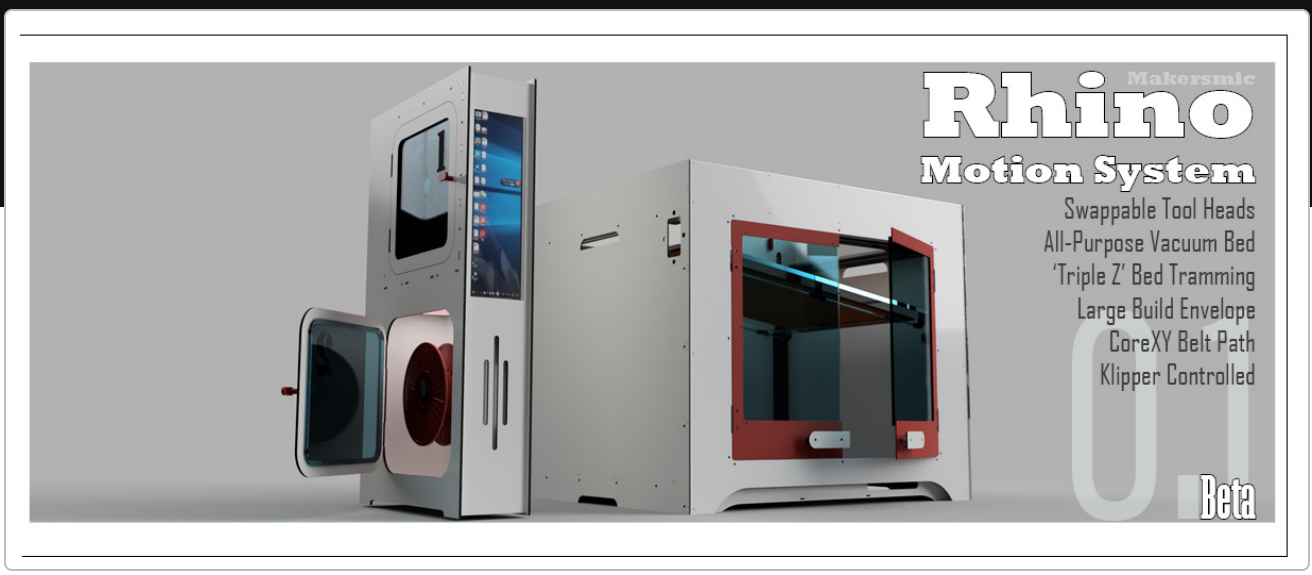
MAKERSMIC · USER MANUAL

Rhino Multi-Tool Motion System

Release 1.9.18 · Klipper, Rhino Studio and the Rhino Portal



01
BlockOne
Print head

02
SwitchFly
Dual-filament print head

03
LightSaber
Diode laser

04
HotJoe
CNC spindle

05
DragKnife
Drag knife

One machine, five swappable toolheads: print, laser-cut, mill and drag-cut on the same motion system. This manual covers the machine, its tools, the software that runs them, and how to keep it all in shape. Source and updates: the Makersmic GitHub page.

2026
A Makersmic Project

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PART

01

The Rhino

The machine, what it needs, and how it is wired.

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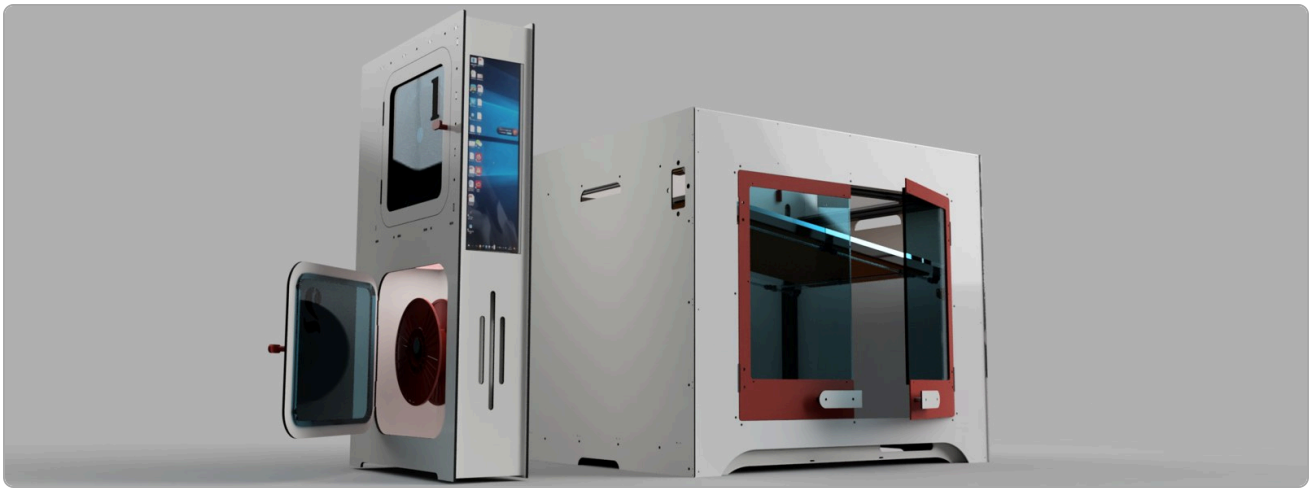
About the Rhino

One motion system, five built-in tools, and room for more: the Rhino changes discipline by changing the tool on its X-carriage, and Klipper changes behaviour to match.



Full assembly render: the enclosed machine and its side tower with display and filament bay

Feature	Rhino
Motion	CoreXY, 15 mm GT2 belts on X and Y (3 m per belt, 6 m total)
Axis travel (from `printer.cfg`)	X 576 mm, Y 406 mm, Z 295 mm
X axis	MGW7 linear rail with MGW7H carriage on an aluminium-tube gantry
Z axis	Three leadscrews (rear T12×2, front T8×2) on kinematic couplings; automatic tramming with `Z_TILT_ADJUST`
Build plate	Aluminium heat spreader with silicone heater mats, air-gap insulated; designed as an all-purpose vacuum bed
Enclosure	3 mm ACP panels with twin front doors
Tool change	Tool slides onto the carriage's V-groove, held by one M3×50 bolt and knurled nut; one 21-pin umbilical plug
Tools	Print heads (BlockOne, SwitchFly), LightSaber laser, HotJoe CNC spindle, DragKnife, plus tools you add
Air	“Fresh Air” lines: print part cooling, laser blow-air, spindle chip evacuation
Controller	BTT Octopus (STM32F446)
Software	Klipper, Moonraker, Mainsail, the Rhino Tool Manager; OrcaSlicer for printing, LaserWeb 4 for laser and drag knife (Kiri:Moto is being looked at)

The design files, drawings and build notes are in the Makersmic `Rhino-3d-Printer` repository on GitHub (see *Build reference*). Still in development there: the “Cheeky Monkey” filament box (filament monitoring, air handling, chip exhaust) and a needle-cutter tool.

System necessities

Klipper runs on a wide range of controller boards, so the Rhino is not tied to one. This manual and the configuration are written for the Big Tree Tech Octopus v1.1, but any controller will do if its microcontroller and mainboard provide what the two tables below list. The third table is the additional hardware the build needs: power, switching, protection and wiring for the electronics cabinet, plus the mechanical parts.

	Microcontroller
1	Capable of running Klipper firmware
2	1 × 24 V hotend PWM channel
3	1 channel capable of controlling a 5 V servo
4	7 × 24 V PWM channels
5	3 × 24 V on/off channels
6	3–5 thermistor input channels
7	6–8 individual stepper outputs
8	1 RGB LED channel
9	3 endstop inputs

	Mainboard
	Capable of running Klipper firmware
	2 USB channels

Qty	Recommended additional hardware
3	DIN-mount SSR rated 3–30 V / 220 V
3	DIN-mount 24 V / 24 V SSR
1	21W4 female socket connector
1	40 mm double-stack Delta fan
1	40 mm fan
2	DIN-mount 20 A circuit breaker, 12–120 V
1	DIN-mount CP40-1P 275 V surge protection
1	DIN-mount DZ47-63 3P breaker
24	211B terminal block
50	Fusable-style terminal block
4	DIN-mount UK10-DREHSI connector
6	300 mm DIN rail
3	Solid-state relay heatsinks
1	WJ175 JQX-38F high-current relay
3	12 V white LED strips
1	12 V RGB LED strip

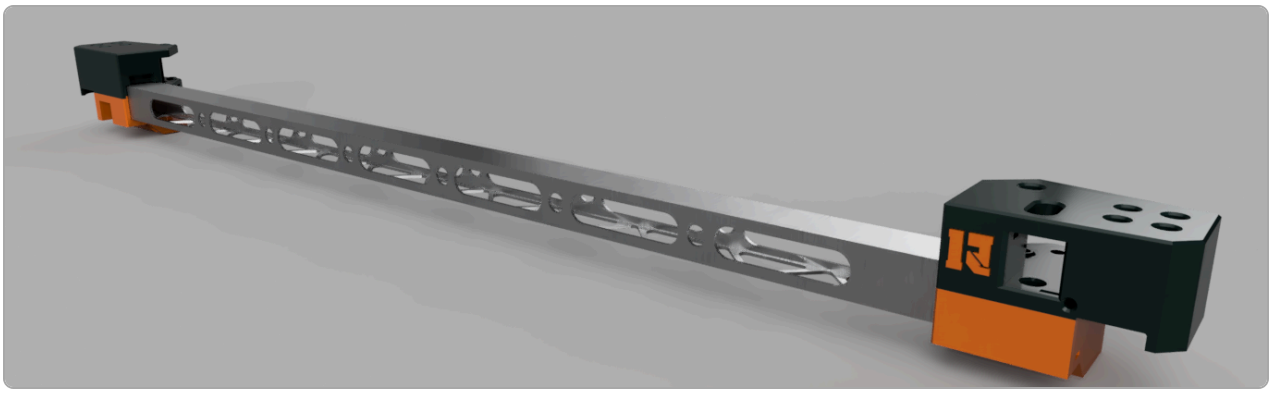
Qty	Recommended additional hardware
1	12 V power supply
1	24 V power supply
1	T12 leadscrew
2	T8 leadscrew with the same lead as the T12
4	20T 15 mm belt 2GT pulleys (5 mm bore)
12	608zz ABEC-7 bearing
4	12 mm bearings
16	15 mm idler pulleys
6	5 mm × 120 mm stainless steel toy shafts
2	8 mm × 200 mm stainless steel toy shafts
4	M/F 8 mm × 20 mm camera rosette
1	110 V fused power plug
1	220 V power plug
6–8	NEMA 17 stepper motors
3	Optical endstops
1	8 ft CPAP hose
1	PC817 12 V optocoupler

The machine

Every tool rides the same gantry and plugs into the same 21-pin umbilical; only the wiring of each individual tool travels with it.

Gantry

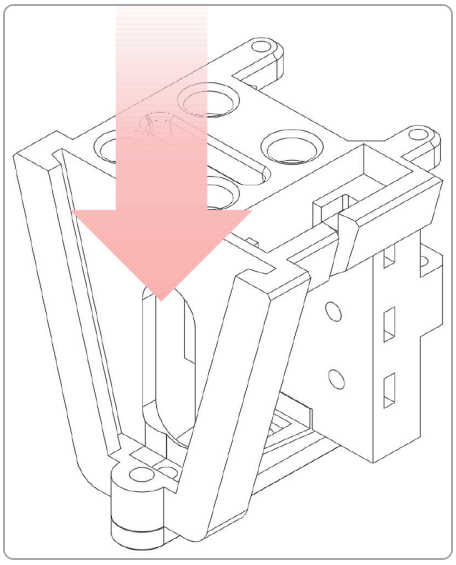
The X beam is 1″ aluminium square tube (1/16″ wall), lightened with cut-outs made from 3D-printed router templates. The “Bulkup” XY joiners (April 2023) carry extra material at the pulleys and stopped the beam sagging where it meets the joiners. The A/B stepper motors must be no more than 48 mm long with a shaft under 30 mm; they stand on M3×30 mm aluminium standoffs.



X gantry beam with the Bulkup XY joiners

X-carriage and tool slot

The carriage runs on an MGW7 rail and carries the tool lock, the belt clamps and the optical X endstop. A tool slides down into the V-groove on the carriage face, which aligns it, and one M3×50 bolt with a knurled nut (or wing nut) clamps it.



X-carriage detail: the tool drops into the wedge-type slot

Z axis

The bed rides three leadscrews — one T12×2 at the rear, two T8×2 at the front — joined to the plate by kinematic couplings. Each screw is belt-driven through a 2:1 reduction on closed-loop 15 mm GT2 belts of 180–190 mm, and printer.cfg matches it with `gear_ratio: 2:1` on all three Z steppers. Because each corner has its own motor, Klipper trams the bed with `Z_TILT_ADJUST`.

Electrical cabinet

The BTT Octopus controller switches every heavy load through a DIN-rail relay, which keeps high currents off the board and makes a failed relay a quick swap.

Circuit	Switched by
12 V low-current: LED and NeoPixel lighting, fans	700-TBR24 6 A relay
12 V high-current: laser module, vacuum pump	JQX-13FL 10 A relay
24 V spindle (ESC power)	JQX-38F 40 A relay
Heated bed mats	Solid-state relays

Power comes from separate 24 V, 12 V and 5 V supplies. DIN-rail fuse holders protect the stepper motor lines against shorts. Bridgeable distribution blocks spread power; Wago-style blocks (the “Hub”) terminate the umbilical inside the cabinet.

The umbilical

An essential assembly of the Rhino, the umbilical provides all the connections each toolhead needs. It uses a 21W4 socket connector for every tool connection, with automatic reconfiguration between tool changes, and the umbilical LED strip signals user-specified alarms.

Each tool ends in a male 21W4 D-sub (four high-current contacts plus seventeen signal pins); the female half is fixed at the top rear of the frame with the Hub below it. The cable is braided Cat5 plus two 14 AWG conductors. Pins 1, 2, 11 and 12 are the high-current contacts.

The pin-by-pin assignment of the umbilical, with the board pin behind each contact, is in *Wiring and pin mapping*. Each tool's own page in *The tools* shows which contacts that tool uses.

Why every tool has a thermistor. Klipper's extruder expects a temperature reading at all times. Rather than editing `printer.cfg` for each tool, every tool carries the same thermistor type on pins 3–4 (the laser's is on its heatsink). The swap wizard reads it to confirm the umbilical is seated.

Constant connections do not pass through the umbilical; they run to the carriage through the *Back Bone*: the X endstop (3 wires), the umbilical LED strip, the chamber temperature sensor and two 20 mm air tubes.

Wiring and pin mapping

These pages record how the Rhino is wired to its Big Tree Tech Octopus v1.1 controller: the 21 contacts of the umbilical, the board pins that stay on the machine, every stepper driver, and the wiring diagrams for the special cases. Blank rows and cards are for your own additions.

> Where these sheets and `printer.cfg` disagree, Klipper follows `printer.cfg`. The differences found while preparing this manual are listed in *Still to check on the machine*.

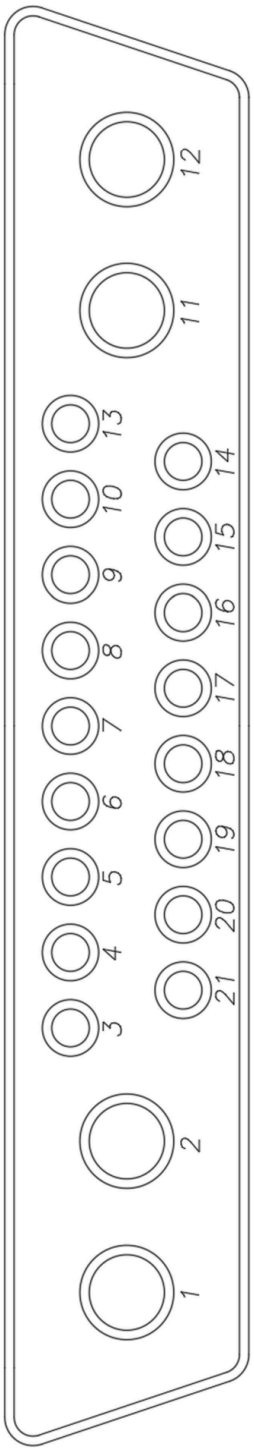
Rhino Wiring: Pin Mapping

Board: Big Tree Tech Octopus v1.1

Umbilical pin	Board pin	Function	Board label
1	**PA2 +**	Hotend heater	HE0
2	**–**	Hotend heater	
3	**PF4 –**	Hotend thermistor	T0
4	**+**	Hotend thermistor	
5	**PE5**	Hotend fan	Fan0
6	**–**	Hotend fan	
7	**A1**	Stepper A1	Motor2
8	**A2**	Stepper A2	
9	**B1**	Stepper B1	
10	**B2**	Stepper B2	
11	**PA1 +**	24 V power activate	Bed Out
12	**–**	24 V power activate	
13	**–**	Probe	Probe
14	**+**	Probe	
15	**PG10**	Probe signal	
16	**PB0**	Laser PWM (5 V required)	RGB
17			
18	**PB11**	PWM signal 24 V –, ESC signal to GND pin	
19			
20	**PD12 +**	12 V power activate +	Fan2
21	**–**	12 V power activate –	

Umbilical constant connections	Board pin
X-limit switch	**PG6**
Chamber temperature	**PF5**
Umbilical LED	**PD14**

Umbilical colour coding



Pin	Wire colour
1	
2	
3	
4	
5	
6	
7	
8	
9	
10	
11	
12	
13	
14	
15	
16	
17	
18	
19	
20	
21	

Record the wire colour on each contact of your umbilical as you build it.

Rhino Wiring: Pin Mapping

Other board pins · Big Tree Tech Octopus v1.1

Pin	Description	Board label
PA8	Part cooling fan PWM *Power provided through an always-on port of the Octopus board*	Fan5
	Driver cooling fan PWM *Power provided through an always-on port of the Octopus board*	
PA3	Bed 0 power	HE1
PD14	Umbilical LED	Fan4
PF3	Bed temperature	TB
PF5	Chamber temperature	T1
PG6	X limit	
PG9	Y limit	
	Z max	

Rhino Wiring: Auxiliary Board

Aux board: _____

Model no: _____

Pin	Description			

Rhino Wiring: Stepper Assignments

Big Tree Tech Octopus v1.1

X

Driver type and manufacturer	BTT TMC2240
SPI / UART	
Run current	*1.1 A*
Board socket	0
Step	**PF13**
Dir	**PF12**
Enable	**PF14**
CS / UART	**PC4**

Y

Driver type and manufacturer	BTT TMC2240
SPI / UART	
Run current	*1.1 A*
Board socket	1
Step	**PG0**
Dir	**PG1**
Enable	**PF15**
CS / UART	**PD11**

Rhino Wiring: Stepper Assignments

Big Tree Tech Octopus v1.1

E0

Driver type and manufacturer	BTT TMC2240
SPI / UART	
Run current	*0.8 A*
Board socket	4
Step	**PF9**
Dir	**PF10**
Enable	**PG2**
CS / UART	**PF2**

Pins and current from printer.cfg [extruder]

Z0

Driver type and manufacturer	BTT TMC2208 v3.0
SPI / UART	
Run current	*0.65 A*
Board socket	5
Step	**PE6**
Dir	**PA14**
Enable	**PE0**
CS / UART	**PD3**

Rhino Wiring: Stepper Assignments

Big Tree Tech Octopus v1.1

Z1

Driver type and manufacturer	BTT TMC2208 v3.0
SPI / UART	
Run current	*0.8 A*
Board socket	6
Step	**PC13**
Dir	**PF0**
Enable	**PF1**
CS / UART	**PE4**

Z2

Driver type and manufacturer	BTT TMC2208 v3.0
SPI / UART	
Run current	*0.8 A*
Board socket	7
Step	**PE2**
Dir	**PE3**
Enable	**PD4**
CS / UART	**PE1**

Rhino Wiring: Stepper Assignments

Big Tree Tech Octopus v1.1

X1		
	Driver type and manufacturer	BTT TMC2240
	SPI / UART	
	Run current	*0.8 A*
	Board socket	
	Step	
	Dir	
	Enable	
	CS / UART	

Y1		
	Driver type and manufacturer	BTT TMC2240
	SPI / UART	
	Run current	*0.8 A*
	Board socket	
	Step	
	Dir	
	Enable	
	CS / UART	

Rhino Wiring: Stepper Assignments

Big Tree Tech Octopus v1.1

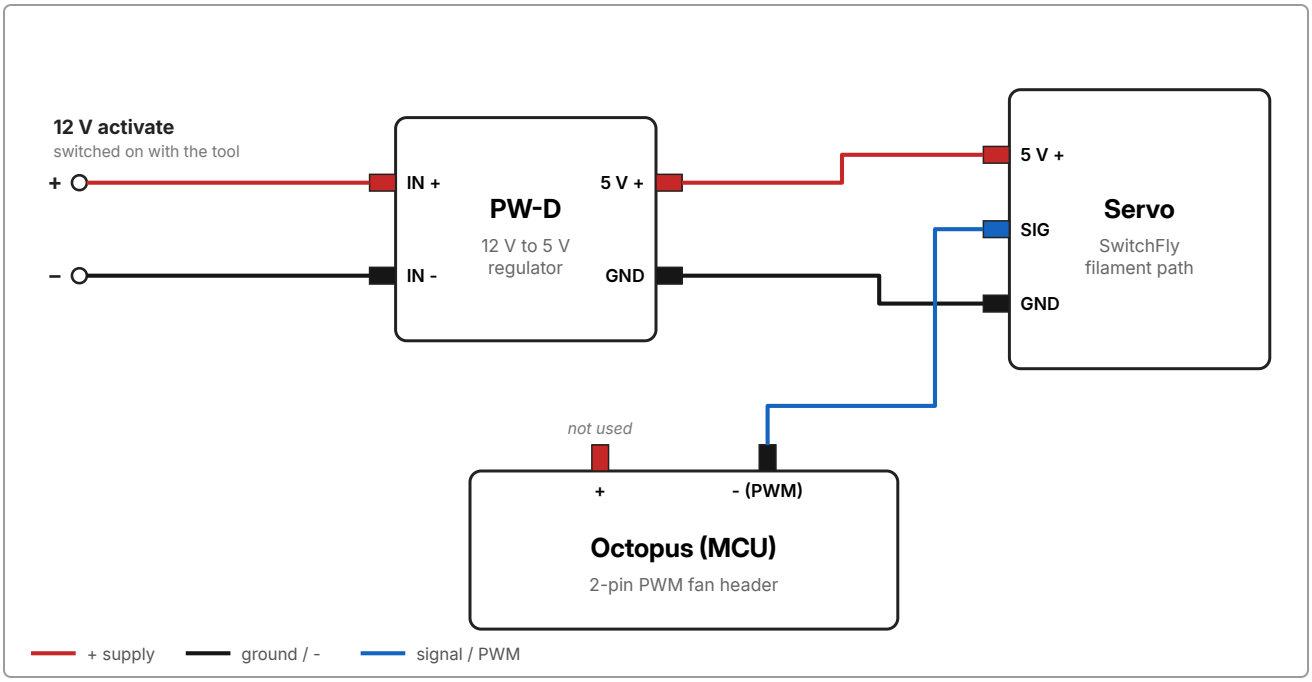
Driver type and manufacturer	
SPI / UART	
Run current	
Board socket	
Step	
Dir	
Enable	
CS / UART	

Driver type and manufacturer	
SPI / UART	
Run current	
Board socket	
Step	
Dir	
Enable	
CS / UART	

Rhino Wiring: Diagrams

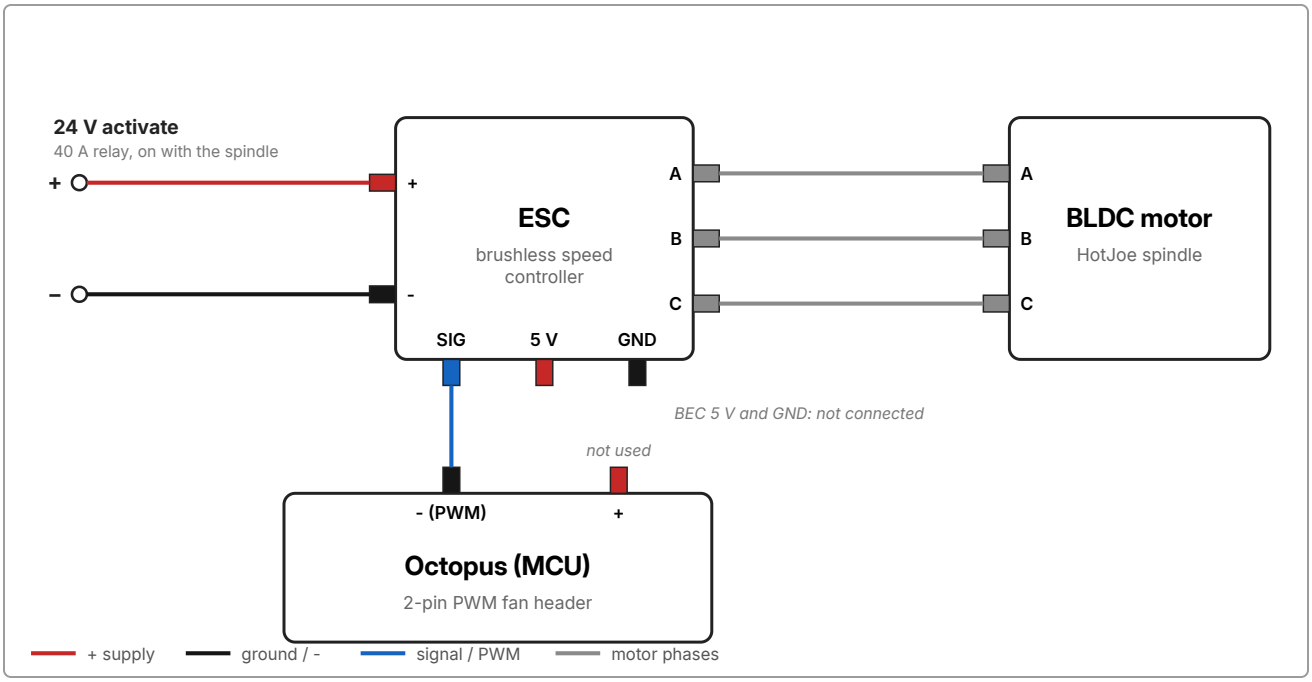
Red: + supply · black: ground / – · blue: signal or PWM · grey: motor phases

SwitchFly servo



A PW-D regulator turns the switched 12 V into 5 V for the servo; the board's PWM output drives its signal line.

HotJoe BLDC spindle

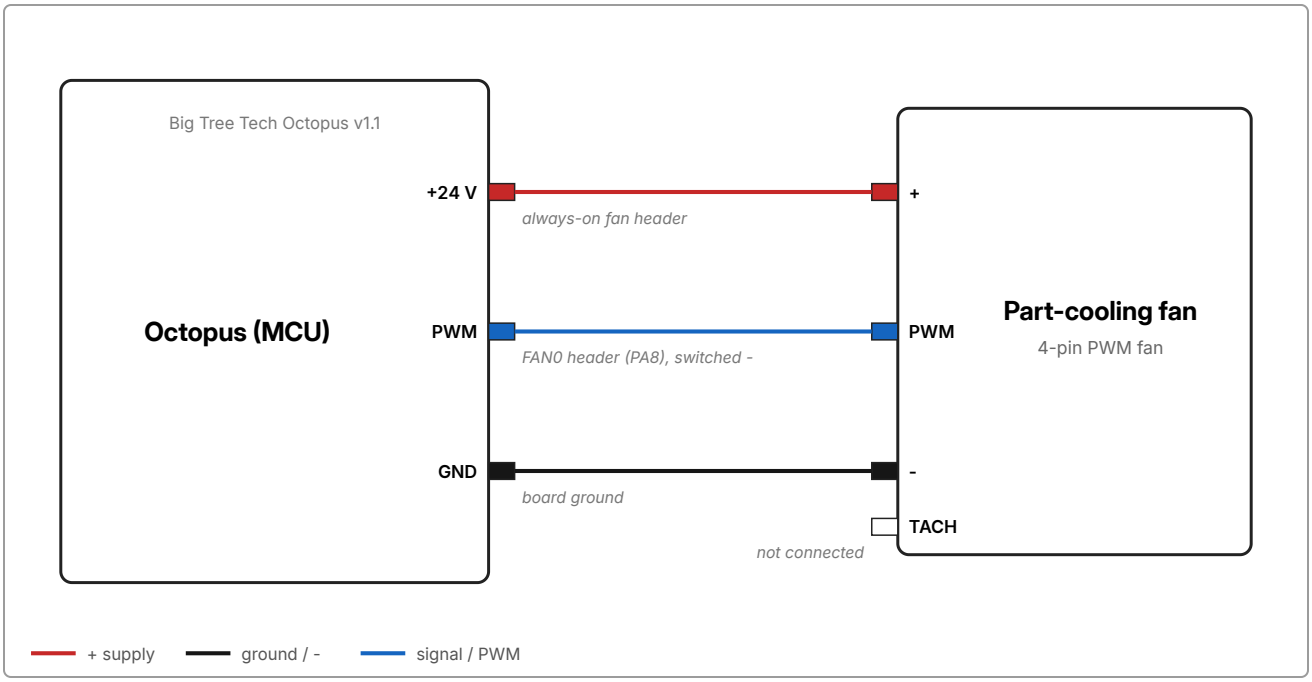


The relay-switched 24 V powers the ESC; the board's PWM output (SPINDLE_SPEED) drives its signal lead.

Rhino Wiring: Diagrams

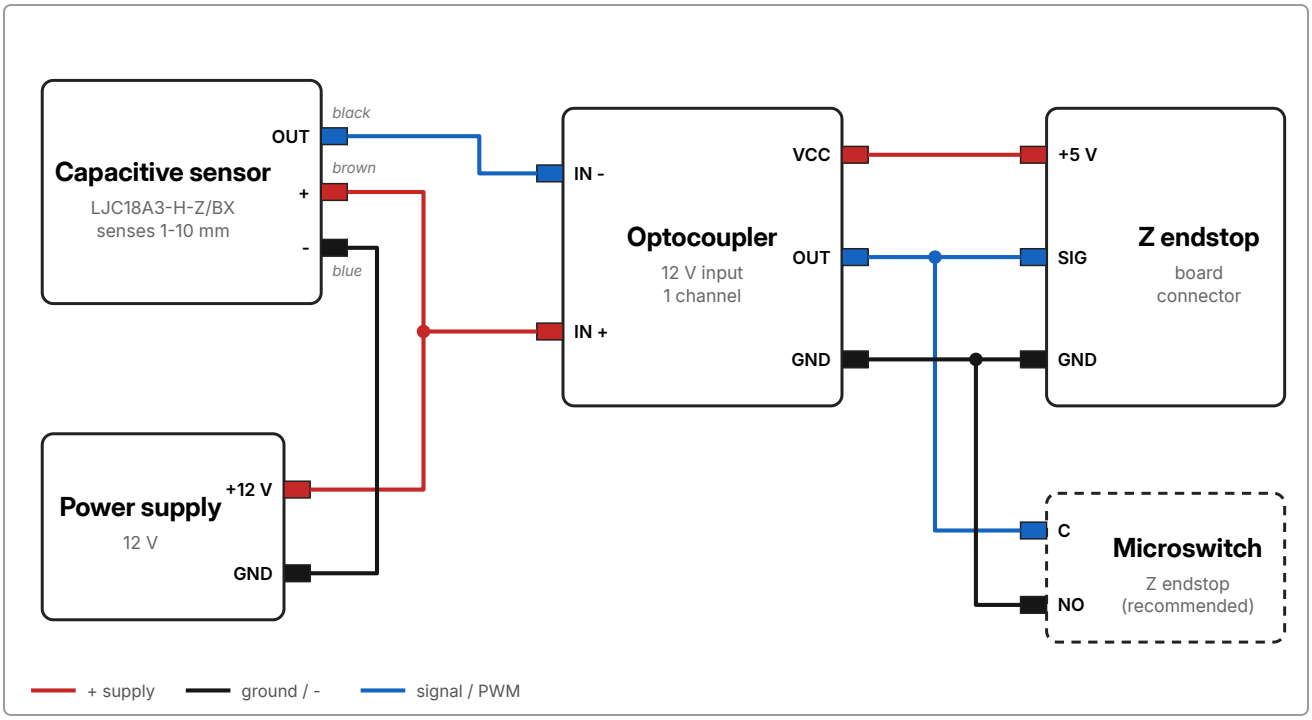
Red: + supply · black: ground / – · blue: signal or PWM

4-pin part-cooling fan



Power from an always-on fan header; the PWM line from the FAN0 header (PA8). The tachometer wire is not used.

Capacitive probe



The 12 V sensor switches an optocoupler, which gives the Z endstop input a clean 5 V signal. A microswitch in parallel is recommended as a backup.

PART

02

The Tools

The built-in tools, their umbilical contacts and parts, and sheets for the tools you build.

The tools	**22**
BlockOne	**24**
SwitchFly	**28**
LightSaber	**32**
HotJoe	**36**
DragKnife	**40**
Tools you add	**44**
Rhino tools log	**52**

The tools

Every tool has a slot number. Slots 1–5 are the built-in tools defined in `variables.cfg`; tools you add in the Tool Manager get slot 6 and up.

Slot	Tool	Hardware	Set up a job with	Power outputs
1	BlockOne	Single-extruder print head	Slicer start G-code (`START_JOB`)	Hotend heater, fans
2	SwitchFly	Two filaments, one stepper, one nozzle	Slicer start G-code (`START_JOB`)	Hotend heater, fans, path servo
3	LightSaber	12 V 80 W 450 nm blue diode laser	`LASER_JOB_SETUP FILE=<name>.gcode`	`LASER_INITIALIZE` rail, `SERVO_LASER` intensity
4	HotJoe	24 V brushless spindle, ER8 collet	`CNC_JOB_SETUP FILE=<name>.gcode`	`Spindle_power` relay, `SPINDLE_SPEED` ESC
5	DragKnife	Drag knife for Roland No. 9 and No. 10 blade holders	`DRAGKNIFE_JOB_SETUP FILE=<name>.gcode`	None
6+	Your tools	Powered, passive or print-head clone	`TOOL_JOB_SETUP FILE=<name>.gcode` (clones use the slicer)	Own pin or a shared output

Built-in tools are changed by editing `variables.cfg`. Tools you add are changed in the Tool Manager or with console commands, never by hand-editing `custom_tools.cfg`.

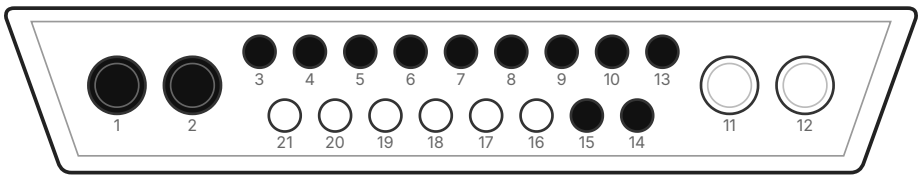
Notes

Rhino Tools: BlockOne

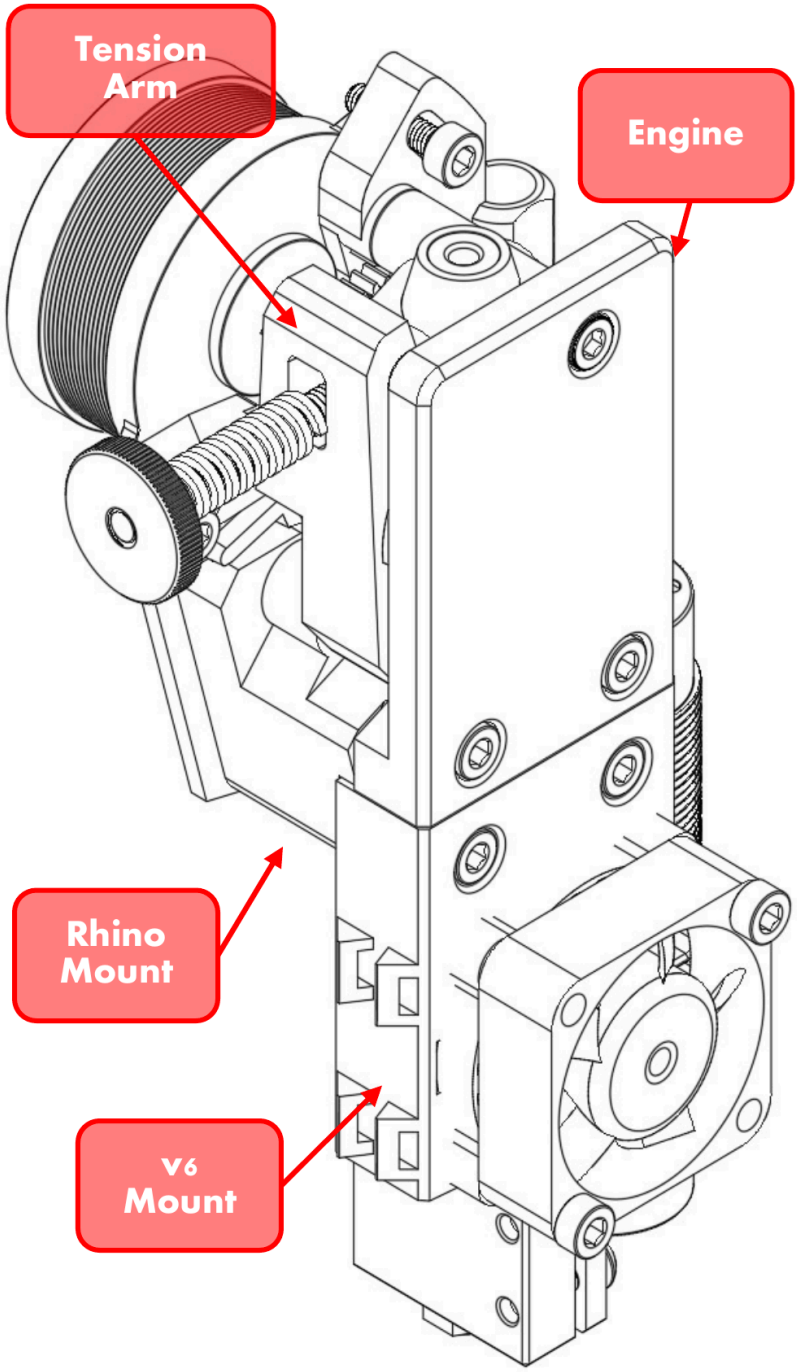
Direct-drive 3D printing, with profiles for 0.4 mm and 0.8 mm nozzles in OrcaSlicer. E3D v6 Volcano hotend at 24 V, BMG drive gear and 1.75 mm filament.

Function	3D printing (0.4 and 0.8 mm nozzles)
Slot	1
Set up a job with	Slicer start G-code (`START_JOB`)
Outputs in Klipper	Hotend heater, hotend fan, part-cooling fan

Umbilical connector



Pin no	Description	Board pin
1	24 V hotend	**PA2**
2	24 V hotend	
3	Thermistor	**PF4**
4	Thermistor	
5	24 V hotend fan	**PE5**
6	24 V hotend fan	
7	Stepper	**A1**
8	Stepper	**A2**
9	Stepper	**B1**
10	Stepper	**B2**
11		
12		
13	Probe	
14	Probe	
15	Probe	**PG10**
16		
17		
18		
19		
20		
21		



*Part cooling provided by the universal blow-air system.

BlockOne supersedes the Jack Rabbit print head in the repository; the Jack Rabbit files there are outdated. Its materials cover 0.4 and 0.8 mm nozzles.

Rhino Tools: BlockOne

Parts list · 3D printing (0.4 and 0.8 mm nozzles)

Qty	Description
	3D-printed pieces:
1	BlockOne Rhino Mount
1	BlockOne Engine
1	BlockOne Tension Arm
1	V6 Mount
	Off-the-shelf hardware:
1	BMG big gear and shaft
1	625zz sealed bearing
2	MR185 sealed bearing
3	3 × 5 × 7 brass sleeve bearing
1	24 V E3D v6 Volcano hotend assembly
1	J4-HV inductive proximity sensor
1	25 mm NEMA 14 stepper motor
1	30 × 3 mm toy shaft
1	4 × 8 × 25 mm medium-duty (bed) spring
1	21W4 male socket connector
	Fasteners:
6	M3 × 20 mm socket head cap bolt
1	M3 × 30 mm socket head cap bolt
6	2 × 6 × 6 mm M3 square nut
1	M5 × 12 mm socket head cap bolt
1	M4 × 50 mm socket head cap bolt
1	M4 × 2 mm thumb nut
1	M4 washer
7	M3 washers
5	Small zip ties
1	Medium zip tie

Rhino Tools: BlockOne

Notes · settings that worked, changes, repairs

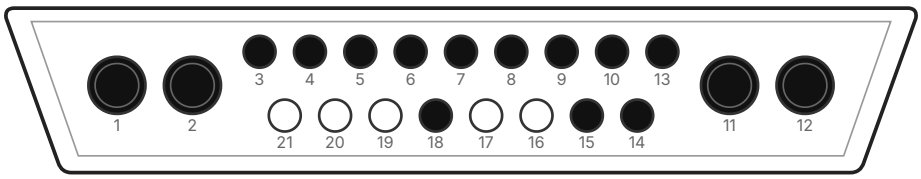
Date	Note

Rhino Tools: SwitchFly

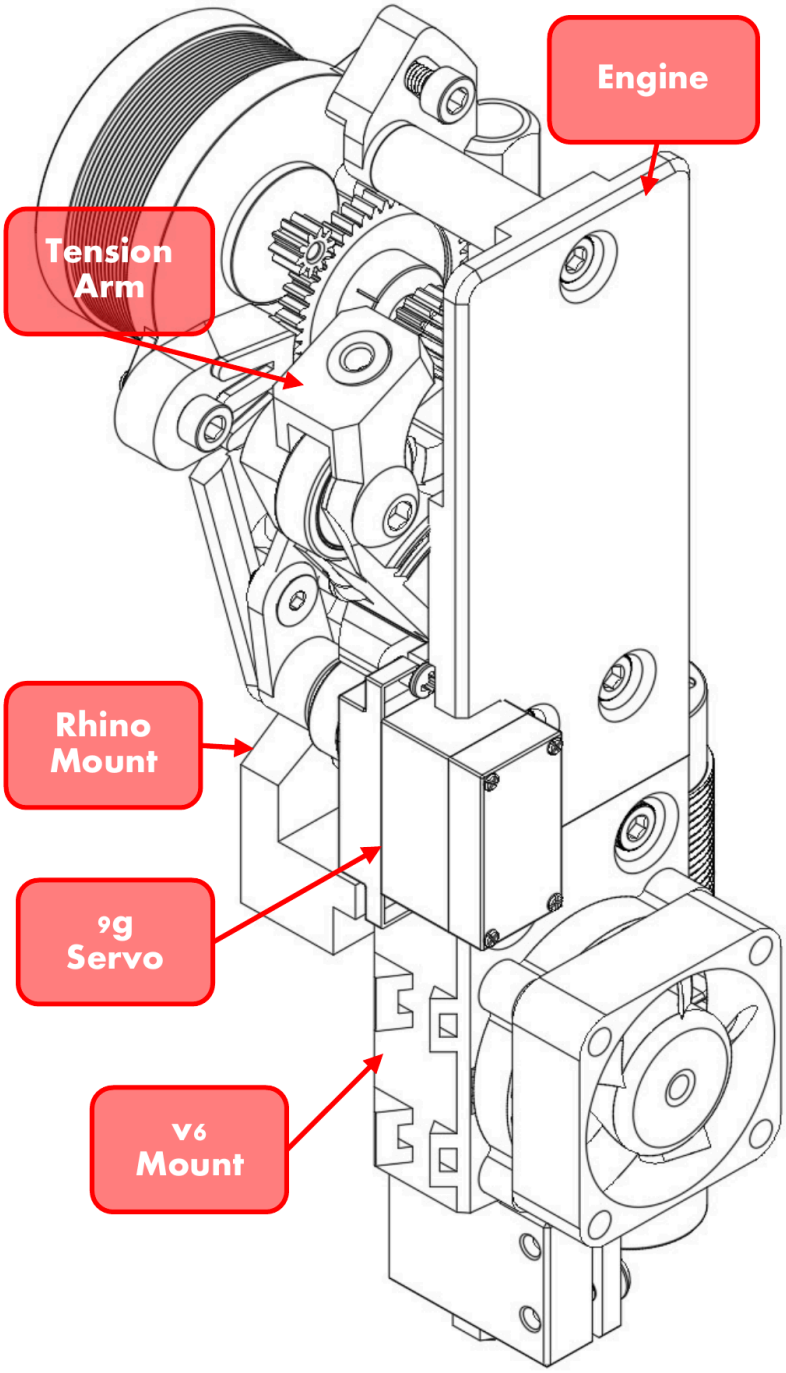
Dual extrusion at its finest: the first true direct-drive, dual-extrusion, single-nozzle extruder for 3D printing. Optimized for the E3D v6 Volcano hotend and BMG drive gear, and rated at 24 V for 1.75 mm filament.

Function	Dual-extrusion 3D printing (0.4 and 0.8 mm nozzles)
Slot	2
Set up a job with	Slicer start G-code (`START_JOB`)
Outputs in Klipper	Hotend heater, fans, filament-path servo (`SERVO_LASER`)

Umbilical connector



Pin no	Description	Board pin
1	24 V + hotend	**PA2**
2	24 V hotend	
3	Thermistor	**PF4**
4	Thermistor	
5	24 V hotend fan	**PE5**
6	24 V hotend fan	
7	Stepper	**A1**
8	Stepper	**A2**
9	Stepper	**B1**
10	Stepper	**B2**
11	24 V + power activate	**PA1**
12	24 V – power activate	
13	Probe	
14	Probe	
15	Probe	**PG10**
16		
17		
18	PWM – signal	**PB11**
19		
20		
21		



*Part cooling provided by the universal blow-air system.

SwitchFly feeds two filaments through one stepper and one hotend, so there is no second nozzle to align. `SWITCH_T0` and `SWITCH_T1` select the filament with a servo on the shared `SERVO_LASER` output.

Rhino Tools: SwitchFly

Parts list · Dual-extrusion 3D printing (0.4 and 0.8 mm nozzles)

Qty	Description
	3D-printed pieces:
1	SwitchFly Rhino Mount
1	Cam
1	Linkage Arm
1	Left Tension Arm
1	Right Tension Arm
1	SwitchFly Engine
1	V6 Mount
1	30 mm Fan Spacer
	Off-the-shelf hardware:
1	BMG big gear and shaft
2	625zz sealed bearing
2	MR185 sealed bearing
8	3 × 5 × 7 brass sleeve bearing
1	24 V E3D v6 Volcano hotend assembly
1	J4-HV inductive proximity sensor
5	Small zip ties
1	Medium zip tie
1	25 mm NEMA 14 stepper motor
1	Ecopower 827 12 g micro servo
1	12 mm control horn
1	35 × 3 mm toy shaft
2	30 × 3 mm toy shaft
1	3 mm × 25 mm expansion spring
1	21W4 male socket connector
1	3-pin servo extension cable
1	24 V 30 mm fan
	Fasteners:
3	M3 × 30 mm socket head cap bolt
4	M3 × 25 mm socket head cap bolt
6	2 × 6 × 6 mm M3 square nut
2	M3 × 6 mm flat head bolt
2	M5 × 12 mm button head bolt
2	M5 square nut
4	M3 × 12 mm socket head cap bolt
2	M2 × 6 mm Phillips head screw
7	M3 washers

*Shaft rotation is negated by the custom G0 and G1 Klipper commands.

*Ensure the servo default angle is set to 90 degrees in Klipper.

Rhino Tools: SwitchFly

Notes · settings that worked, changes, repairs

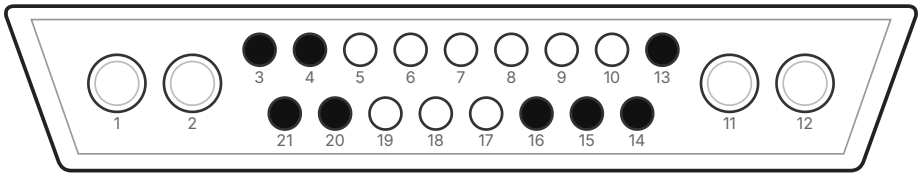
Date	Note

Rhino Tools: LightSaber

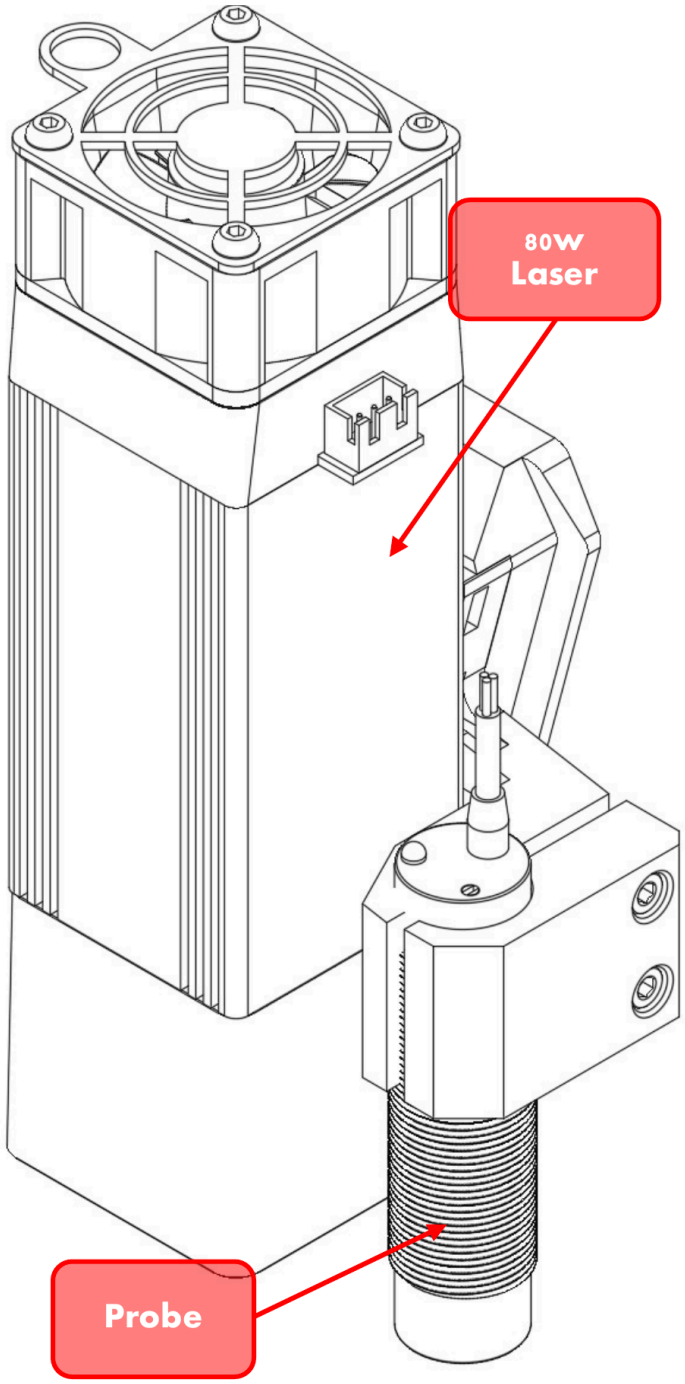
12 V, 80 W, 450 nm blue diode laser cutter and engraver.

Function	Laser cutting and engraving
Slot	3
Set up a job with	`LASER_JOB_SETUP FILE=<name>.gcode`
Outputs in Klipper	`LASER_INITIALIZE` 12 V rail, `SERVO_LASER` intensity

Umbilical connector



Pin no	Description	Board pin
1		
2		
3	Thermistor	**PF4**
4	Thermistor	
5		
6		
7		
8		
9		
10		
11		
12		
13	Probe	
14	Probe	
15	Probe	**PG10**
16	Laser PWM (5 V)	**PB0**
17		
18		
19		
20	+ 12 V power activate	**PD12**
21	– 12 V power activate	



*Blow-air provided by the universal blow-air system.

A relay switches the laser's 12 V supply and Klipper drives its intensity through the PWM contact. The printed height gauge in the build files sets the focus distance.

Safety: wear glasses rated for the laser's wavelength, run extraction, never cut PVC or vinyl, and stay with the machine for the whole cut.

Rhino Tools: LightSaber

Parts list · Laser cutting and engraving

Qty	Description
	3D-printed pieces:
1	Light Saber Rhino Mount
1	Laser Module Bracket
1	Sensor Clamp Arm
	Off-the-shelf hardware:
1	80 W blue diode laser
1	J4-HV inductive proximity sensor
	Fasteners:
4	M3 × 10 mm socket head cap bolt
2	M3 × 22 mm socket head cap bolt

Rhino Tools: LightSaber

Notes · settings that worked, changes, repairs

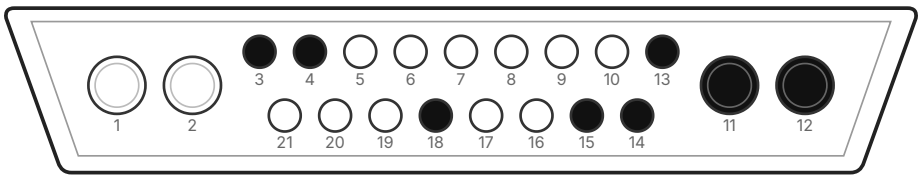
Date	Note

Rhino Tools: HotJoe

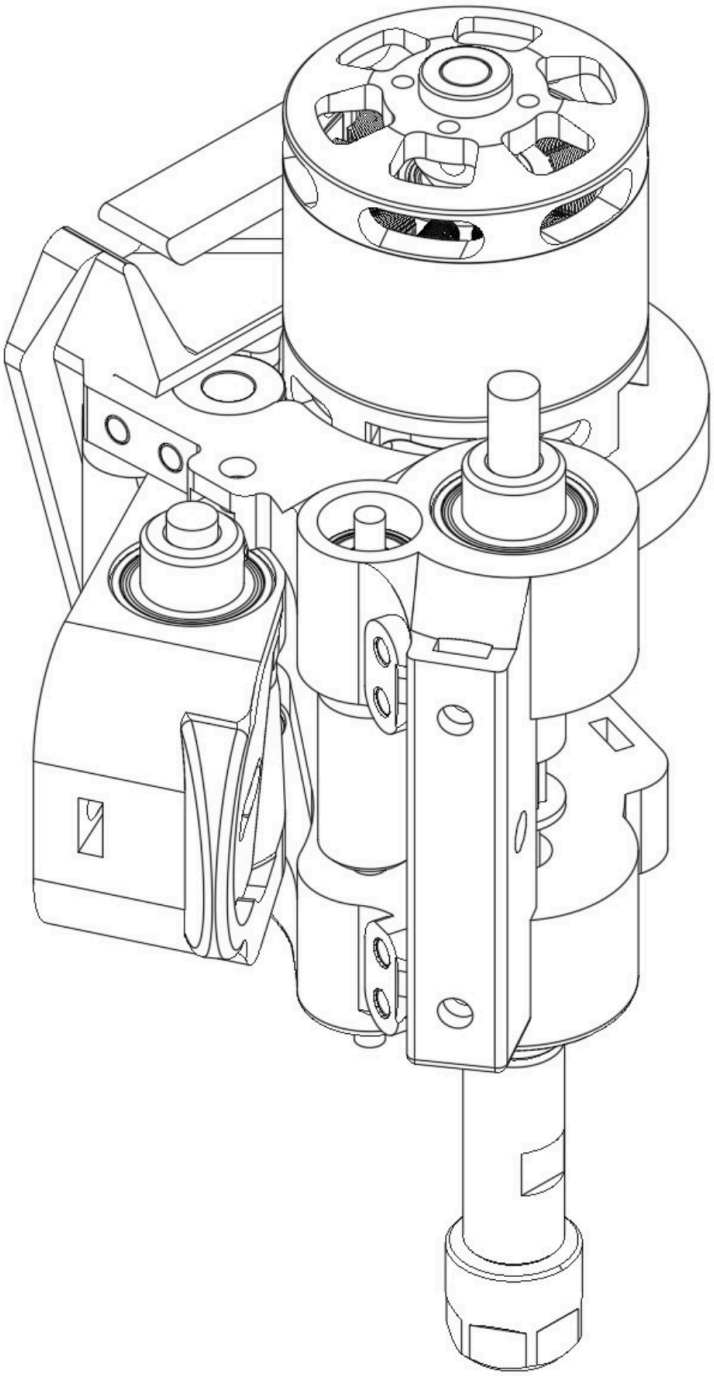
Spindle attachment with an ER8 collet and a 2814 1000 kV brushless motor, driven by an 80 A HobbyWing ESC on 24 V power.

Function	CNC spindle for light engraving and milling
Slot	4
Set up a job with	`CNC_JOB_SETUP FILE=<name>.gcode`
Outputs in Klipper	`Spindle_power` relay, `SPINDLE_SPEED` ESC signal

Umbilical connector



Pin no	Description	Board pin
1		
2		
3	Thermistor	**PF4**
4	Thermistor	
5		
6		
7		
8		
9		
10		
11	24 V relay power	**PA1**
12	24 V relay power	
13	Probe	
14	Probe	
15	Probe	**PG10**
16		
17		
18	PWM	**PB11**
19		
20		
21		



*Shown without safety covers.
*Blow-air provided by the universal blow-air system.

Meant for light-duty milling. A 40 A relay powers the ESC only while the spindle is in use; the speed signal runs at 400 Hz, where 0.3 is stopped (neutral) and about 0.99 is full throttle. Check real speeds at the collet with a tachometer before relying on them. Calibrate the ESC once with `CALIBRATE_ESC` (see *Maintenance*). The build files also include a 70 mm brush holder for dust collection, a blow-air clip and motor and gearbox shields.

Rhino Tools: HotJoe

Parts list · CNC spindle for light engraving and milling

Qty	Description
	Off-the-shelf hardware:
1	ER8 collet

Rhino Tools: HotJoe

Notes · settings that worked, changes, repairs

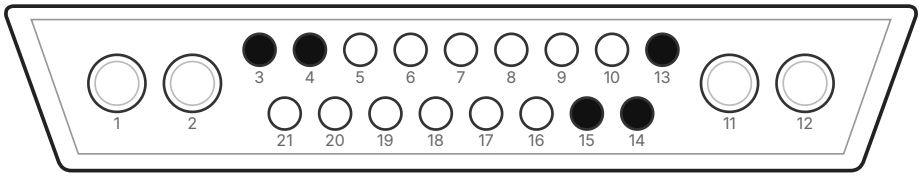
Date	Note

Rhino Tools: DragKnife

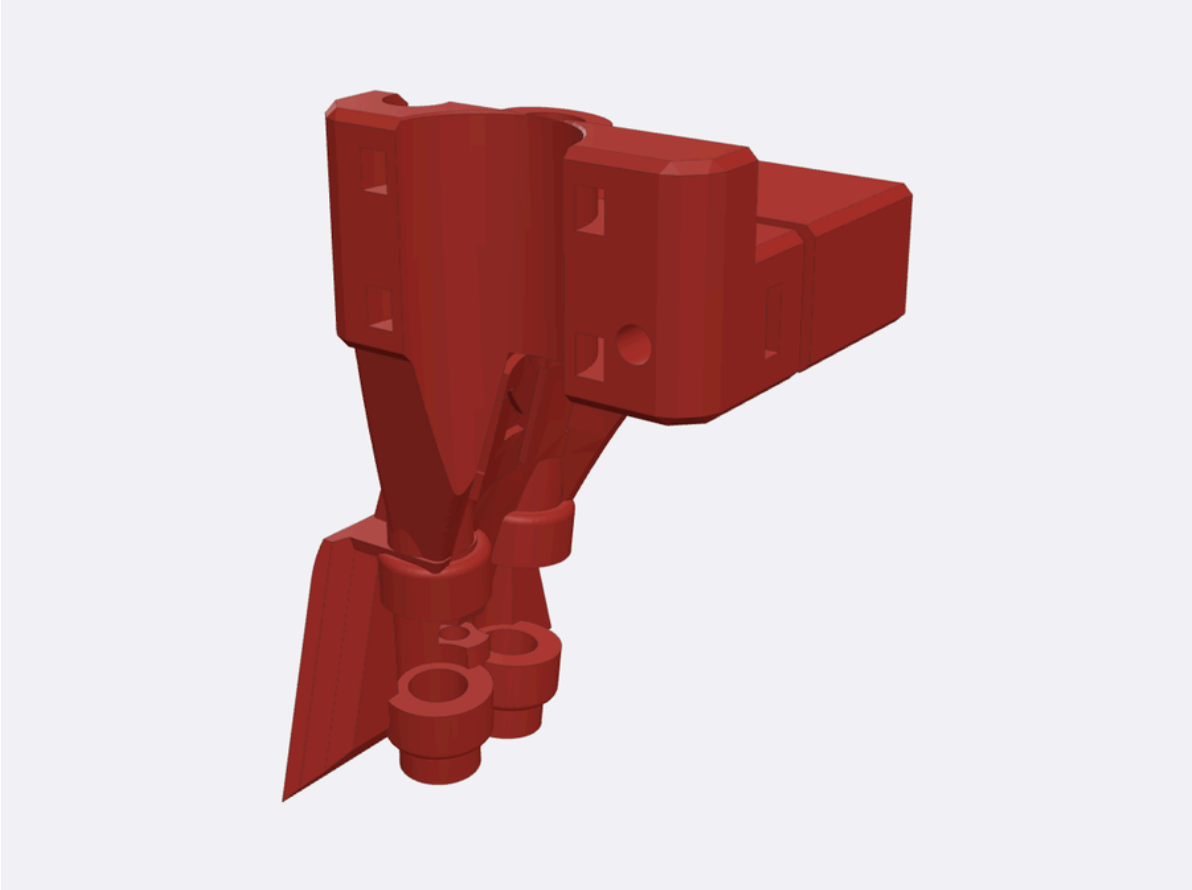
A drag-knife tool designed for Roland No. 9 and No. 10 blade holders.

Function	Drag-knife cutting (vinyl, cardstock, thin foam)
Slot	5
Set up a job with	`DRAGKNIFE_JOB_SETUP FILE=<name>.gcode`
Outputs in Klipper	None (passive tool)

Umbilical connector



Pin no	Description	Board pin
1		
2		
3	Thermistor	**PF4**
4	Thermistor	
5		
6		
7		
8		
9		
10		
11		
12		
13	Probe	
14	Probe	
15	Probe	**PG10**
16		
17		
18		
19		
20		
21		



*Rendered from the build files.

The drag knife has no power output; cut depth comes from the blade holder's dial and the Z zero set in the job wizard. Its thermistor and probe contacts let the swap wizard confirm the umbilical is seated.

Rhino Tools: DragKnife

Parts list · Drag-knife cutting (vinyl, cardstock, thin foam)

Qty	Description
	Off-the-shelf hardware:

Rhino Tools: DragKnife

Notes · settings that worked, changes, repairs

Date	Note

Tools you add

Hot-wire cutters, needle cutters, dispensers, pens and other holders, and extra print-head nozzle sizes are added from the Tool Manager (see *The Tool Manager portal*). A powered tool takes its main power from an umbilical contact it can share, such as the spindle ESC signal on contact 18, or from a new output on a spare pin, and can switch more outputs on with itself.

The next pages are blank tool sheets, two per tool, laid out like the built-in tools' pages. Fill one in for each tool you design. They face each other when the manual lies open: the left page records what the tool is and which umbilical contacts it uses, the right page the settings you enter in the portal (controls, macros, materials) and its parts. Keep the sheet with the tool.

> Before wiring a new tool, check **Pins & outputs** in the portal: it shows every contact of the umbilical and what drives it. Only free pins can be given to a new output, and the portal refuses a pin that anything in the config already uses.

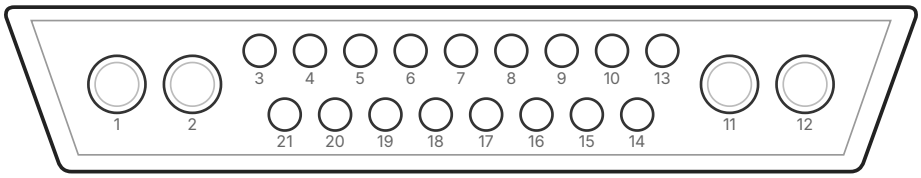
Notes

Rhino Tools: _____

New tool sheet 1 · identity and umbilical

Tool name	
Slot	
Category and kind	☐ Additive ☐ Subtractive ☐ Passive kind: _____
Function	
Main power (PWM)	contact ___ or new pin _____ output _____
Switch outputs	contacts / pins _____
Power cap / max on-time	_____ % / _____ s

Umbilical connector **(shade the contacts the tool uses)**



Pin no	Description	Board pin	Output / control
1			
2			
3			
4			
5			
6			
7			
8			
9			
10			
11			
12			
13			
14			
15			
16			
17			
18			
19			
20			
21			

Rhino Tools: _____

New tool sheet 1 · portal settings and parts

Controls

Control name	What it does	Pin	Off on safe-off	Macros
			☐	
			☐	
			☐	
			☐	
			☐	

Macros

Macro name	Purpose

Materials

Material	Power %	Feed mm/min	Z offset mm

Parts list

Qty	Part

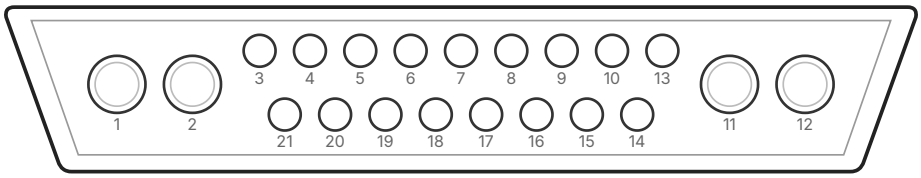
Notes

Rhino Tools: _____

New tool sheet 2 · identity and umbilical

Tool name	
Slot	
Category and kind	☐ Additive ☐ Subtractive ☐ Passive kind: _____
Function	
Main power (PWM)	contact ___ or new pin _____ output _____
Switch outputs	contacts / pins _____
Power cap / max on-time	_____ % / _____ s

Umbilical connector **(shade the contacts the tool uses)**



Pin no	Description	Board pin	Output / control
1			
2			
3			
4			
5			
6			
7			
8			
9			
10			
11			
12			
13			
14			
15			
16			
17			
18			
19			
20			
21			

Rhino Tools: _____

New tool sheet 2 · portal settings and parts

Controls

Control name	What it does	Pin	Off on safe-off	Macros
			☐	
			☐	
			☐	
			☐	
			☐	

Macros

Macro name	Purpose

Materials

Material	Power %	Feed mm/min	Z offset mm

Parts list

Qty	Part

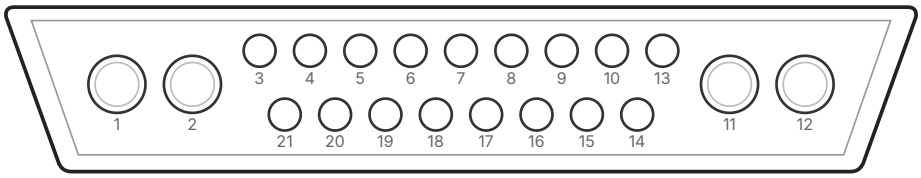
Notes

Rhino Tools: _____

New tool sheet 3 · identity and umbilical

Tool name	
Slot	
Category and kind	☐ Additive ☐ Subtractive ☐ Passive kind: _____
Function	
Main power (PWM)	contact ___ or new pin _____ output _____
Switch outputs	contacts / pins _____
Power cap / max on-time	_____ % / _____ s

Umbilical connector **(shade the contacts the tool uses)**



Pin no	Description	Board pin	Output / control
1			
2			
3			
4			
5			
6			
7			
8			
9			
10			
11			
12			
13			
14			
15			
16			
17			
18			
19			
20			
21			

Rhino Tools: _____

New tool sheet 3 · portal settings and parts

Controls

Control name	What it does	Pin	Off on safe-off	Macros
			☐	
			☐	
			☐	
			☐	
			☐	

Macros

Macro name	Purpose

Materials

Material	Power %	Feed mm/min	Z offset mm

Parts list

Qty	Part

Notes

Rhino Tools Log

Print heads and nozzles

Extruder 1	
Nozzle size	Nozzle type
0.2 / 0.25 / 0.3 / 0.4 / 0.6 / 0.8 / 1.0	**Brass / Steel / Specialty**

Extruder 2	
Nozzle size	Nozzle type
0.2 / 0.25 / 0.3 / 0.4 / 0.6 / 0.8 / 1.0	**Brass / Steel / Specialty**

Extruder 3	
Nozzle size	Nozzle type
0.2 / 0.25 / 0.3 / 0.4 / 0.6 / 0.8 / 1.0	**Brass / Steel / Specialty**

Extruder 4	
Nozzle size	Nozzle type
0.2 / 0.25 / 0.3 / 0.4 / 0.6 / 0.8 / 1.0	**Brass / Steel / Specialty**

Extruder 5	
Nozzle size	Nozzle type
0.2 / 0.25 / 0.3 / 0.4 / 0.6 / 0.8 / 1.0	**Brass / Steel / Specialty**

Extruder 6	
Nozzle size	Nozzle type
0.2 / 0.25 / 0.3 / 0.4 / 0.6 / 0.8 / 1.0	**Brass / Steel / Specialty**

Extruder 7	
Nozzle size	Nozzle type
0.2 / 0.25 / 0.3 / 0.4 / 0.6 / 0.8 / 1.0	**Brass / Steel / Specialty**

Extruder 8	
Nozzle size	Nozzle type
0.2 / 0.25 / 0.3 / 0.4 / 0.6 / 0.8 / 1.0	**Brass / Steel / Specialty**

Extruder 9	
Nozzle size	Nozzle type
0.2 / 0.25 / 0.3 / 0.4 / 0.6 / 0.8 / 1.0	**Brass / Steel / Specialty**

SwitchFly 1	
Nozzle size	Nozzle type
0.2 / 0.25 / 0.3 / 0.4 / 0.6 / 0.8 / 1.0	**Brass / Steel / Specialty**

SwitchFly 2	
Nozzle size	Nozzle type
0.2 / 0.25 / 0.3 / 0.4 / 0.6 / 0.8 / 1.0	**Brass / Steel / Specialty**

SwitchFly 3	
Nozzle size	Nozzle type
0.2 / 0.25 / 0.3 / 0.4 / 0.6 / 0.8 / 1.0	**Brass / Steel / Specialty**

SwitchFly 4	
Nozzle size	Nozzle type
0.2 / 0.25 / 0.3 / 0.4 / 0.6 / 0.8 / 1.0	**Brass / Steel / Specialty**

SwitchFly 5	
Nozzle size	Nozzle type
0.2 / 0.25 / 0.3 / 0.4 / 0.6 / 0.8 / 1.0	**Brass / Steel / Specialty**

SwitchFly 6	
Nozzle size	Nozzle type
0.2 / 0.25 / 0.3 / 0.4 / 0.6 / 0.8 / 1.0	**Brass / Steel / Specialty**

SwitchFly 7	
Nozzle size	Nozzle type
0.2 / 0.25 / 0.3 / 0.4 / 0.6 / 0.8 / 1.0	**Brass / Steel / Specialty**

Rhino Tools Log

Spindles, lasers, knives and your own tools

Hot Joe

Voltage	ESC rating	Motor size	kV rating	Max RPM
12 / 24 / 36 / 48 V	**20 / 30 / 50 / 80 A**			

Hot Joe (spare)

Voltage	ESC rating	Motor size	kV rating	Max RPM
12 / 24 / 36 / 48 V	**20 / 30 / 50 / 80 A**			

Light Saber

Voltage	Wattage	Diode type	Wavelength	Focal distance
12 / 24 / 36 / 48 V				

Light Saber (spare)

Voltage	Wattage	Diode type	Wavelength	Focal distance
12 / 24 / 36 / 48 V				

Drag Knife

Blade holder	Blade angle	Blade offset	Cut depth	Notes

Name:				

Name:				

Name:				

Notes

PART

03

Software and Operation

Installing the software, changing tools, running jobs and the Tool Manager portal.

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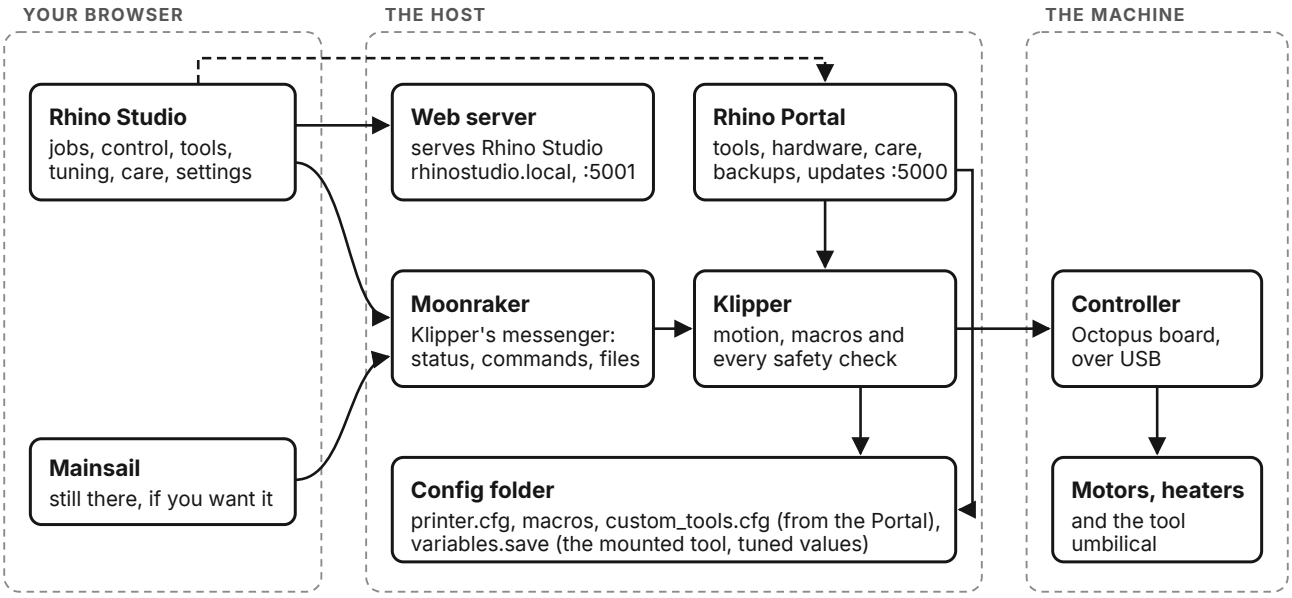
Running a job	**60**
Pause, resume and cancel	**64**
The Tool Manager portal	**65**
Pins, outputs and system hardware	**83**
Admin lists	**86**
Console commands	**89**
Safety systems	**91**

How the software fits together

The Rhino's software runs on a small computer beside the machine, the **host**: a Dell OptiPlex 7050, or a Raspberry Pi on older builds. You use it from a browser on any computer, tablet or phone on the same network.



Rhino Studio talks to Klipper through Moonraker, and to the Portal for everything Klipper doesn't keep

- **Klipper** moves the machine and runs the macros: tool swaps, job start and end, and every safety check. It decides what is mounted, and records it in `variables.save`; no screen can change that except through Klipper's own swap wizard.
- **Moonraker** passes messages between Klipper and the screens: the machine's status, commands, and the job files.
- **Rhino Studio** is the main screen. It sends Klipper only commands it has checked, and asks before anything that moves the machine or powers a tool. It slices models in your browser, or on the host with PrusaSlicer if you choose that.
- **The Rhino Portal** keeps what Klipper doesn't: the tools you add, system hardware, maintenance, backups, slicing profiles and updates. Studio's Assets, Care and Settings screens use it. It writes `custom_tools.cfg` from your tools, and asks Klipper to restart only when no job is running. It still has its own pages at port 5000.
- **Mainsail** is still installed and works as before; you don't need it.

Install and first start

The Rhino software installs from a menu on the Pi, in the style of KIAUH: you upload the zip in Mainsail, type three short commands, then pick numbered options. Every option says what it is about to do and asks before changing anything, and every install backs up your config folder first.

Before you start: the printer idle, and a **print head (BlockOne or SwitchFly) mounted**. The Z check after the first start homes Z, and only a print head can do that. You need the Pi's address (the one you open Mainsail with) and its login and password; on most Klipper installs the login is `pi`.

1. **Upload the zip.** In Mainsail open **Machine**, press **Upload** and choose `rhino-config-<version>.zip` (for example `rhino-config-1.4.8.zip`). Check the file list shows exactly that name; if the browser renamed it (`... (1).zip`), fix the name and upload again.
2. **Open a terminal on the Pi.** On Windows open **PowerShell**, on a Mac open **Terminal**, and log in with your login and the Pi's address. Answer `yes` the first time, then type the password (it stays invisible).

   ```
   ssh pi@192.168.1.50
   ```

3. **Open the Rhino menu.** Paste one line at a time:

   ```
   cd ~/printer_data/config
   ```

   ```
   unzip -o rhino-config-1.4.8.zip -d ~/rhino-1.4.8
   ```

   ```
   bash ~/rhino-1.4.8/rhino.sh
   ```

   If `unzip` is missing, run `sudo apt install -y unzip` first. The menu looks like this:

   ```
   /============================================================\
   |             ~~~~ [ Rhino Tool Manager ] ~~~~               |
   |------------------------------------------------------------|
   | Installed version : not installed                          |
   | This folder has   : 1.4.8   (/home/pi/rhino-1.4.8)         |
   | Portal            : not installed                          |
   |------------------------------------------------------------|
   | 1) Preview the install       (changes nothing)             |
   | 2) Install / update from this folder                       |
   | 3) Install a new zip  (unpack, preview, install)           |
   | 4) Undo an install            (pick a backup)              |
   | 5) Portal status and recent log                            |
   | 6) Restart the portal                                      |
   |                                                            |
   | Q) Quit                                                    |
   \============================================================/
   Perform action:
   ```

4. **Preview: option 1.** It checks your login and config folder, whether a job is running, whether something else uses port 5000 and whether an earlier portal is installed, then lists every step without changing anything. It ends with *Dry run finished*, or with **STOPPED:** and what to do.

5. **Install: option 2**, then `y`. It asks for the password once and works through its stages, each ending in **OK** (a few only run when needed):
   - backs up the config folder to `~/rhino-backup-<date>`;
   - copies the new files, never touching `variables.cfg`, `variables.save`, your positions (`myrhino/positions.cfg`) or the tool files the portal generates;
   - keeps the SAVE_CONFIG section at the bottom of `printer.cfg`, and never doubles it;
   - fixes `/home/pi` paths if your login is different;
   - installs Flask from the Pi's own packages, plus Waitress when it is available;
   - replaces any earlier portal service (its settings are saved in the backup), starts the portal at every boot, and checks it answers.

   It finishes with the portal's address, `http://<printer address>:5000`. Afterwards the menu is always at `bash ~/rhino-manager/rhino.sh`.
6. **First start, in Mainsail.** With the print head still mounted:
   - Run `FIRMWARE_RESTART` and wait for *Ready*. A red box names the file and line Klipper did not accept.
   - Answer the pop-up asking which tool is mounted (or run `RESTORE_TOOL`).
   - Run `CHECK_TOOLHEADS`. When the console line starting `rhino:` says OK, press **Restart now**.
7. **Check Z before the first job.** After any update, check Z (the Rhino uses `gear_ratio: 2:1` on all three Z steppers). Home with the print head, move Z 10 mm and measure it, then run `Z_TILT_ADJUST` and re-check your Z offsets.

> **TIP**
> **A good place to stop** is after the preview (nothing has changed) or after the first start. Avoid stopping between the install and the first start.

Updating: upload the new zip, open the menu, choose **3** and pick it. It unpacks the zip, shows the preview and asks whether to install; answer `y`, then do the first start again. Your tools, photos and maintenance history are kept. **Undoing:** option **4** lists your backups (the oldest is your setup from before the first install), previews the restore and asks first; anything added since is moved aside to `~/rhino-undone-<date>`, not deleted.

> **⚠ WARNING**
> **Never home Z with the laser, spindle or drag knife mounted.** Homing Z brings the bed up to the tool. Their macros refuse to, and jobs end with the bed lowered to the safe park height, so X and Y can be homed after a restart without the tool touching anything.

> **NOTE**
> The portal has no login. Anyone who can reach port 5000 can add or remove tools, but it cannot run G-code and refuses to restart Klipper during a job. Keep it on your home network, or start it with `--host 127.0.0.1` and reach it through an SSH tunnel.

Swapping tools

Always change tools through the swap wizard. Klipper's safety checks (laser lockout, spindle lockout, job/tool match) all trust the slot the wizard records.

1. Run `START_TOOL_SWAP` (or press its macro button). A pop-up lists the other tools; pick the one you are installing, then press **Start the swap** in the confirm window. Nothing moves before that.
2. The machine switches off all tool power, the heaters and the part fan, then waits until the hotend **and the bed** are below 40 °C. After a hot print this can take several minutes.
3. It parks the head at the swap position (`swap_x` / `swap_y` in `myrhino/positions.cfg`), disables the print head's extruder motor and shows **Swap the toolhead**.
4. Unplug the old tool's umbilical at the top rear of the frame. Undo the knurled nut, remove the M3×50 bolt and lift the tool off the carriage.
5. Slide the new tool down into the carriage's V-groove until it seats, then refit the bolt and nut snug.
6. Plug the new tool's umbilical fully home and route the cable clear of the gantry.
7. Confirm in the pop-up. Every toolhead has a thermistor on contacts 3–4, and the wizard checks it before anything is switched on: if the plug is not fully home it shows **Toolhead not detected** – reseat the umbilical and press **Check again**. Only then does a print head's extruder motor get power. While the umbilical is unplugged the nozzle temperature reads about −50 °C; that is expected, and Klipper keeps running.
8. The new slot is saved to `variables.save`, so it survives a restart.

> **TIP**
> **From Rhino Studio:** press **Swap** in the top bar and pick the tool you are fitting. Studio says the machine will cool down and move, and asks first; the wizard's questions then appear as Studio windows, the same as the pop-ups above. After a restart, answer with the confirm button that appears in the top bar.

> **NOTE**
> Tool swaps are refused while a job is running or paused. All tool power is switched off before the swap starts. If X and Y are not homed, the swap homes them first with the tool that is on the machine.

> **TIP**
> **LightSaber and the Y endstop.** Set the Y endstop position for the LightSaber before anything homes Y with it mounted, or the laser reaches the frame first.

After any Klipper restart, a pop-up appears about 8 seconds later: *Klipper restarted – which toolhead is mounted?* Press **Yes** if the tool shown is the one on the machine, or **No – run the swap wizard** if it is not. Until you answer, the tool-mounted safety gates stay locked. You can bring the question back any time with `RESTORE_TOOL`.

Running a job

Print jobs start from the sliced file. Every other tool starts from a guided setup that walks you through safety checks and setting the work zero, then starts the file for you.

Pick the material first (non-print tools): `SET_TOOLHEAD TOOLHEAD=LightSaber MATERIAL=PLYWOOD`. This applies the material's feed, power and acceleration, and the setup wizards read it. It never changes the Z zero.

3D printing (BlockOne, SwitchFly, print-head clones)

1. Slice with the Rhino OrcaSlicer profile for that head (the BlockOne profiles ship in `slicer/orca/`; import them with **File › Import › Import Configs**). Its start G-code calls `START_JOB` with the tool, material, nozzle and extruder, then `CALCULATE_PA` for pressure advance.
2. Upload and print from Mainsail as usual.
3. `START_JOB` refuses to run if the file's tool does not match the mounted tool, then homes and shows a confirmation pop-up. With SwitchFly each filament path has its own material: the file names one (or both, as `MATERIAL_T0` / `MATERIAL_T1`), and any path left out is asked first – *Path 2 (T1) material?*, with **Same as last time** as the first button – while the file waits. The confirmation then lists both paths. It applies the nozzle height saved for that head by the paper test (`NOZZLE_HEIGHT_CALIBRATE`, run once per head and kept for it - never by the job), and keeps it when the file is released. With PLA it asks whether to heat the bed or print on a cold bed; a bed that starts below 65 °C warms in stages before `Z_TILT_ADJUST`.
4. `END_PRINT` switches off the job's tool and every tool output, clears any work offsets, lowers the bed to the safe park height, moves the head to the park spot (X551 Y381 unless you changed `myrhino/positions.cfg`) and runs the staged `BED_COOLDOWN`. After a job longer than 2 hours it also backs up the config (`BACKUP_CFG`), which can hold Klipper for up to 2 minutes. Controls and system hardware ticked *switch off when made safe* (an exhaust fan, say) keep running after a normal finish; switch them off when you are done. Cancel, tool swap and the emergency stop still switch them off (1.4.6).

Filament change mid-print: `FILAMENT_CHANGE` pauses, walks you through unload, load and purge, then resumes. `FILAMENT_CHANGE`, `PURGE` and `PREHEAT` only run with a print head mounted (they heat the nozzle), and filament change needs the printer homed. The nozzle heater itself - `M104`, `M109` or Mainsail's temperature box - only switches on with a print head mounted and its thermistor reading normally, and it is switched off within a second if the thermistor ever reads as unplugged while it is on.

Setting Z zero (laser, spindle, drag knife, tools you added)

Every non-print job sets its Z zero by eye in the same **Set Z zero** window (the `SET_Z_ZERO` macro, also a Mainsail button). The job setups open it for you.

- **Bed up** / **Bed down** 10, 1 and 0.1 mm move the bed; *up* brings the material toward the tool. The window shows the bed height and stops at the travel limits (never lower than Z 290, `zero_max_z` in `myrhino/positions.cfg`).

- Spindle, knife and added tools: bring the material up until the tool tip just touches its top surface. LightSaber: bring it to the focus point - the smallest, sharpest dot - using **Focus dot ON (1%)** with laser glasses on.
- **SET – Z zero here** makes that bed height G-code Z0 (it is saved, and nothing moves). The job then runs with Z0 there; the laser runs at Z0 with no extra focus offset.

After a restart Klipper does not know where the bed is. If the bed was left parked (every job and swap ends with it lowered), the window first asks **Bed position – Is the bed still down where it was parked?** **Yes** takes the saved park height as Z and homes X/Y; choose **No** if the bed was moved by hand with the power off, then home Z with a print head. Choosing a material never changes the Z zero you set. Laser, spindle and knife files must not contain `START_JOB`: it homes Z, which is refused with those tools mounted.

Homing. Z is only homed with a print head mounted and confirmed - the probe is on the print heads. Any other way of homing Z (Mainsail's Home All button, `G28`) is refused with a laser, spindle, knife or added tool mounted, or before the *which toolhead is mounted?* question is answered after a restart. `G28 X Y` is allowed with any tool, but Cancel no longer uses it with a laser mounted (see *Pause, resume and cancel*). When X/Y are homed while Z isn't, Klipper first lowers the bed 10 mm; if the bed's park height was saved, it is updated to match.

Laser (LightSaber)

Run `LASER_JOB_SETUP FILE=<name>.gcode`, then follow the pop-ups:

1. **Laser setup** – confirm ventilation, glasses, no PVC/vinyl, material flat, extinguisher nearby.
2. **Set Z zero** – move the bed to the focus point and press **SET**; a low-power test mark follows.
3. **Check the test mark** – **Yes** to continue, **No – set focus again** to go back.
4. **Ready to cut** – set the pass count, then **Start the cut**. The count is a note for you: the file runs once, so build multiple passes into the file itself.

> **⚠ WARNING**
> Wear laser glasses from the focus dot onwards: the focus dot and the test mark fire the laser too. **Start the cut** powers the rail for the file; `END_PRINT`, cancel, a swap and the emergency stop switch it off.

CNC (HotJoe)

Run `CNC_JOB_SETUP FILE=<name>.gcode`:

1. **CNC setup** – safety confirmations.
2. **Set X/Y zero** – jog to the corner, then **Set X/Y zero here**.
3. **Set Z zero** – bring the material up to the bit tip and press **SET**.
4. The bed drops clear, the spindle relay switches on and the spindle spins up gently, then to your starting throttle.
5. **Ready to cut** – nudge the throttle ±5 %, then **Start the cut**.

Drag knife (DragKnife)

Run `DRAGKNIFE_JOB_SETUP FILE=<name>.gcode`: verify the material and blade dial, set Z zero on the surface in the **Set Z zero** window, cut a test mark, then **Cut looks good – continue** (or **Cut is wrong – help me adjust** for blade-dial help) and **Start the cut**.

Tools you added (slot 6+)

Run `TOOL_JOB_SETUP FILE=<name>.gcode`:

1. **<tool> setup** – the checklist you wrote in the portal; **All checked – set work zero**.
2. **Set Work Zero** – **Set X/Y zero here**, then **Set Z zero** (the Set Z zero window; the material's Z offset is added on top); **Continue**.
3. **Ready to start** – **Test pulse** fires briefly at low power; then **Start – tool power ON** (power on for the whole file) or **Start – file controls power** (the file's own `M3` / `M5` switch it). Passive tools just show **Start**.

`M3 S<value>` / `M5` in any file control whichever tool is mounted: the laser, the spindle or your powered tool. `S` is scaled against that tool's *S value for full power* and never exceeds its power cap.

Slicer and CAM settings

OrcaSlicer (printing). Make one printer profile per print head (BlockOne, SwitchFly, and any print head you add). In each profile, **Printer settings › Machine G-code › Machine start G-code** calls `START_JOB`. Type the tool's name once; OrcaSlicer fills in the rest from the filament and nozzle chosen when you slice:

```
START_JOB TOOLHEAD="SwitchFly" MATERIAL="{filament_type[initial_extruder]}" NOZZLE_SIZE=
{nozzle_diameter[initial_extruder]} EXTRUDER={initial_extruder}
```

START_JOB parameter	OrcaSlicer placeholder	Becomes
`TOOLHEAD`	typed in each printer profile	`SwitchFly` – must match the mounted tool
`MATERIAL`	`{filament_type[initial_extruder]}`	`PLA`, `PETG` … – the filament profile's type
`NOZZLE_SIZE`	`{nozzle_diameter[initial_extruder]}`	`0.4` – the printer profile's nozzle
`EXTRUDER`	`{initial_extruder}`	`0` or `1` – the first filament used (SwitchFly)

Together they pick the tool's material profile, such as `PLA_0_4`. **Filament type** (in the filament profile's settings) must be one of the tool's materials – PLA, PETG, ABS or ASA for BlockOne and SwitchFly; a type such as *PLA-CF* stops the job at the start with *Material PLA-CF_0_4 not defined for toolhead BlockOne*. End the file (**Machine end G-code**) with `END_PRINT`.

Pressure advance. On the line after `START_JOB`, the profile calls `CALCULATE_PA` with the head's bowden length in centimetres: `BOWDEN_LENGTH=5.9` for BlockOne (59 mm), `BOWDEN_LENGTH=3.5` for SwitchFly (35 mm, both paths). The other values come from OrcaSlicer placeholders, as in the shipped BlockOne profile:

```
CALCULATE_PA BOWDEN_LENGTH=5.9 MATERIAL=[filament_type] LAYER_HEIGHT=[layer_height] NOZZLE_SIZE=
[nozzle_diameter] PRINT_SPEED=[outer_wall_speed] FILAMENT_DIAMETER=[filament_diameter] LINE_WIDTH=
[line_width]
```

LaserWeb 4 (laser). Use these machine settings. They replace the older ones in the repository's *Laserweb Setup* note, which drove pins (`LASERPOWER_pin`, `VACPUMP_pin`) that no longer exist; going through `M3` / `M5` keeps the laser lockout, the pause handling and the power scaling.

LaserWeb field	Value
Gcode Start	`G21` and `G90` – homing, focus and work zero are done by `LASER_JOB_SETUP`
Gcode End	`M5`, then `END_PRINT` on the next line (switches the laser rail off and parks)
Gcode Homing	leave blank
Tool On	leave blank
Tool Off	`M5`
Laser Intensity	`M3 S`
PWM MIN S VALUE	`0`
PWM MAX S VALUE	`1000` (matches `laser_s_max` in `variables.cfg`)

Drag-knife files must not contain `M3`: it is refused while the DragKnife is mounted. Use a separate LaserWeb profile with *Laser Intensity* and *Tool Off* left blank.

CNC files. Spindle speed is set in `CNC_JOB_SETUP` (the throttle buttons), not by `M3 S`. An `M3` in the file leaves a running spindle alone; `M5` at the end stops it.

Pause, resume and cancel

Pause and cancel work from Mainsail's own buttons for every tool. Resume works from Mainsail's button only for print jobs; laser, CNC, drag-knife and added-tool jobs resume with the **Resume job** button in the Rhino pop-up (or `TOOL_RESUME`).

Mainsail's Resume refuses whenever the extruder is cold ("not hot enough"), and with a non-print tool mounted it always is. `TOOL_RESUME` does the same job without that check, and adds the steps a cutting tool needs.

	Print head	Laser	CNC spindle	Drag knife / passive	Powered tool you added
On **Pause**	Mainsail parks as usual	Lift 10 mm, laser off, park	Lift 10 mm while spinning, spindle off, park	Lift 10 mm, park	Lift 10 mm, power off, park
Pop-up	None	**<tool> job paused** with **Resume job** / **Cancel job**	same	same	same
Resume with	Mainsail **Resume**	**Resume job** or `TOOL_RESUME`	same	same	same
On resume	Mainsail as usual	Back over the spot, down slowly, laser back to its level	Back over the spot, spindle ramps 40 % → its speed (4 s), then down slowly	Back over the spot, down slowly	Back over the spot, down slowly, power back to its level
On **Cancel**	Tool off, bed down, home X/Y, park	Laser and rail off, bed down, park **without homing X/Y** (1.4.5)	Spindle off, bed down, home X/Y, park	Bed down, home X/Y, park	Power off, bed down, home X/Y, park

How the resume moves. The head travels to the paused X/Y at its lifted height, drops to 2 mm above the paused height, then makes the last 2 mm at 5 mm/s. Power comes back only once the head is on the paused spot — never at the park corner.

If Mainsail's Resume is pressed by mistake on a non-print job, nothing moves: the Rhino pop-up comes back and the console says to use `TOOL_RESUME`.

Refusals. `TOOL_RESUME` refuses if nothing is paused, or if the recorded tool differs from the one the job paused with. On a print job it simply hands over to Mainsail's Resume.

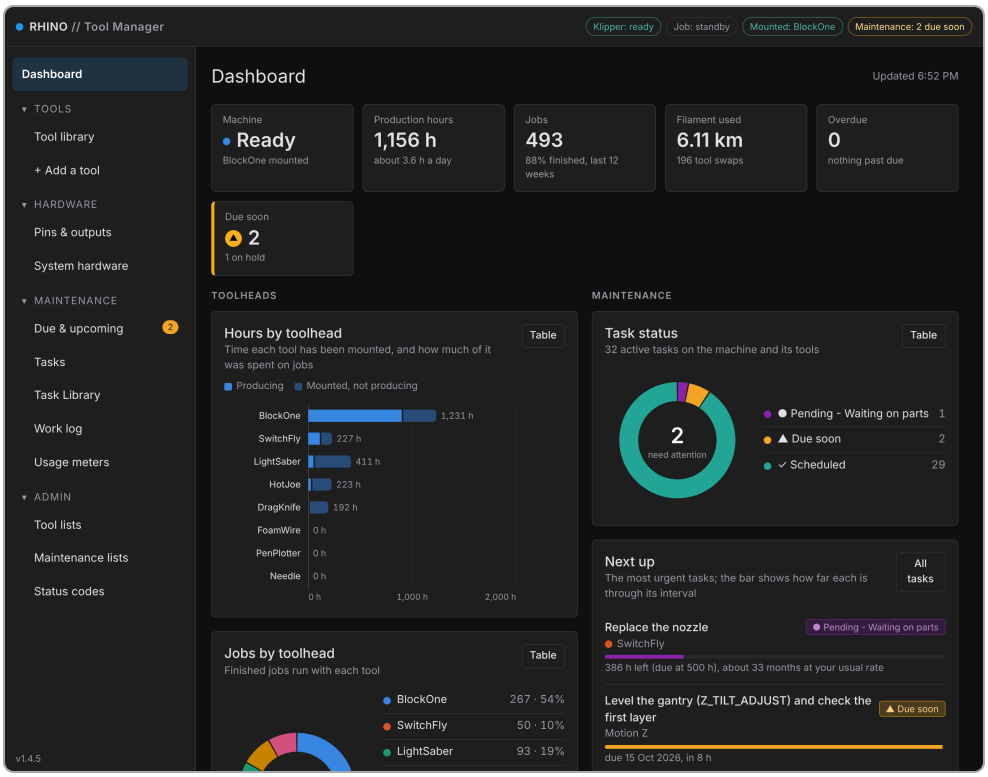
The 10 mm lift is set by `variable_lift` in `_RHINO_PAUSE_HOOK` (`tools/tool_power.cfg`); it is reduced automatically when the bed is near the bottom of its travel.

The Tool Manager portal

Open `http://<printer address>:5000`. The portal adds, edits, renames and deletes your own tools, records which umbilical contacts each one uses, sets up system hardware, tracks maintenance, and restarts Klipper when changes need loading. The sidebar groups it into **Tools**, **Hardware**, **Maintenance** and **Admin**; click a group's name to fold it away. The screenshots come from a test run with three example tools (FoamWire, PenPlotter, Needle), three pieces of system hardware and placeholder photos.

Dashboard

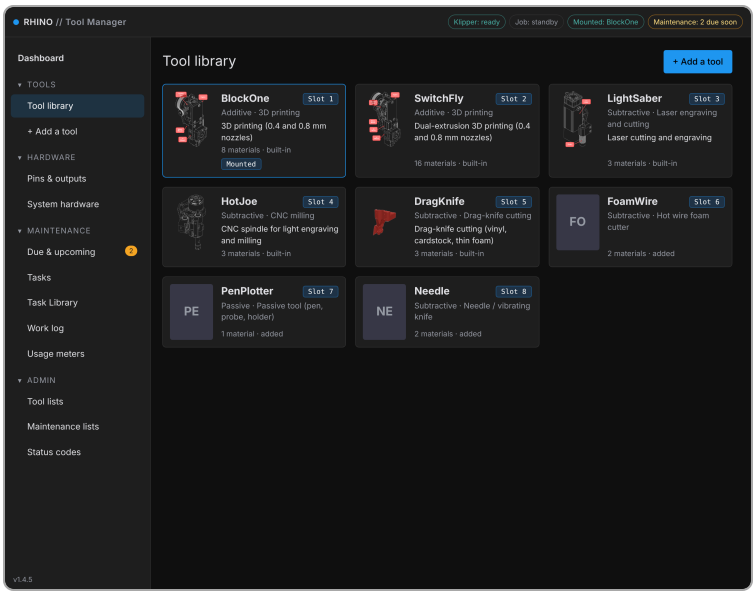
The portal opens on the Dashboard: the machine's state and mounted tool, production hours, jobs, filament used and tool swaps, and anything overdue or due soon. Below are charts of hours and jobs by toolhead (each has a **Table** button for the numbers), the maintenance task status and the most urgent tasks next. The portal follows Mainsail's dark or light theme within 15 seconds.



Dashboard after a year of use (demo data)

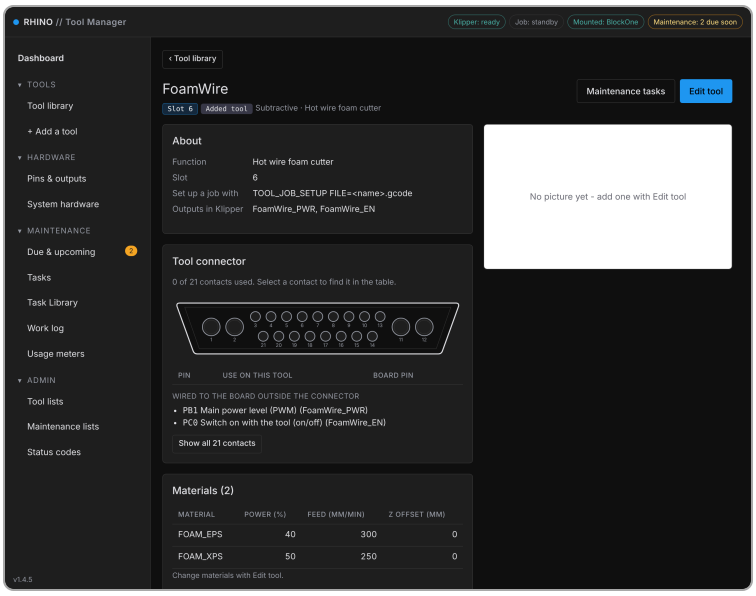
Tool library

Every tool is a card with its picture, name, slot, kind and number of materials; the mounted tool is outlined and marked **Mounted**. The chips at the top right show Klipper's state, the job state, the mounted tool and any maintenance due. Select a card to open the tool's own page.



Tool library with the built-in and added tools

A tool's page shows what it does, how to set up a job with it, its outputs, its connector with the contacts it uses, its materials and its maintenance tasks. Built-in tools are read-only here (they live in `variables.cfg`); your own tools have **Edit tool**.

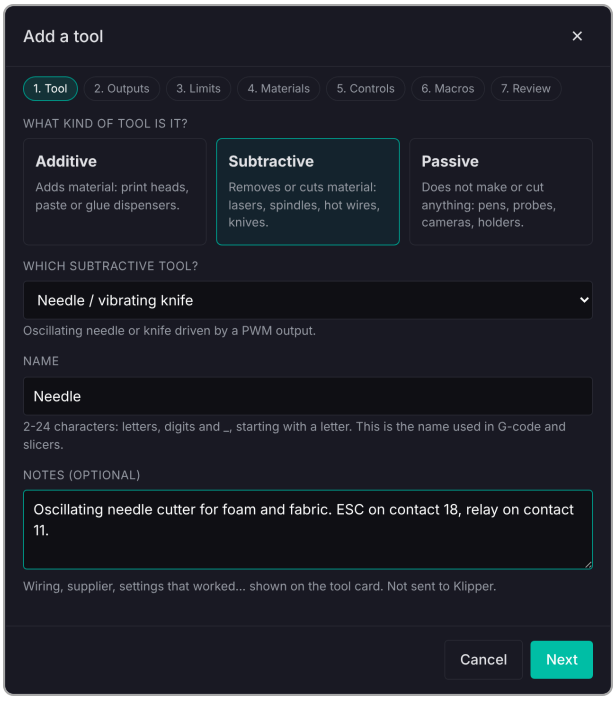


A tool's page (FoamWire, an added tool)

Adding a tool

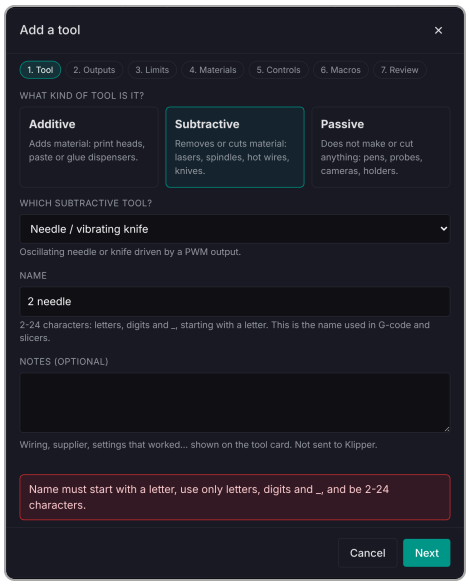
Press **+ Add a tool**. The wizard's steps depend on the kind of tool: powered tools get all seven, passive tools get a Connector step instead of Outputs and Limits, print-head clones get Hardware. Every kind gets Controls and Macros.

Step 1 · Tool – say whether the tool is **Additive** (adds material: print heads, paste or glue dispensers), **Subtractive** (removes or cuts material: lasers, spindles, hot wires, knives) or **Passive** (pens, probes, cameras, holders). Then pick the kind of tool within it, type a name and optional notes. The kind fills in safe starting values; the list of kinds is yours to edit under **Admin › Tool lists**. The name is what G-code and slicers use: 2–24 letters, digits or `_`, starting with a letter.



Step 1: category, kind, name and notes

Mistakes are caught before you move on, with the rule spelled out:

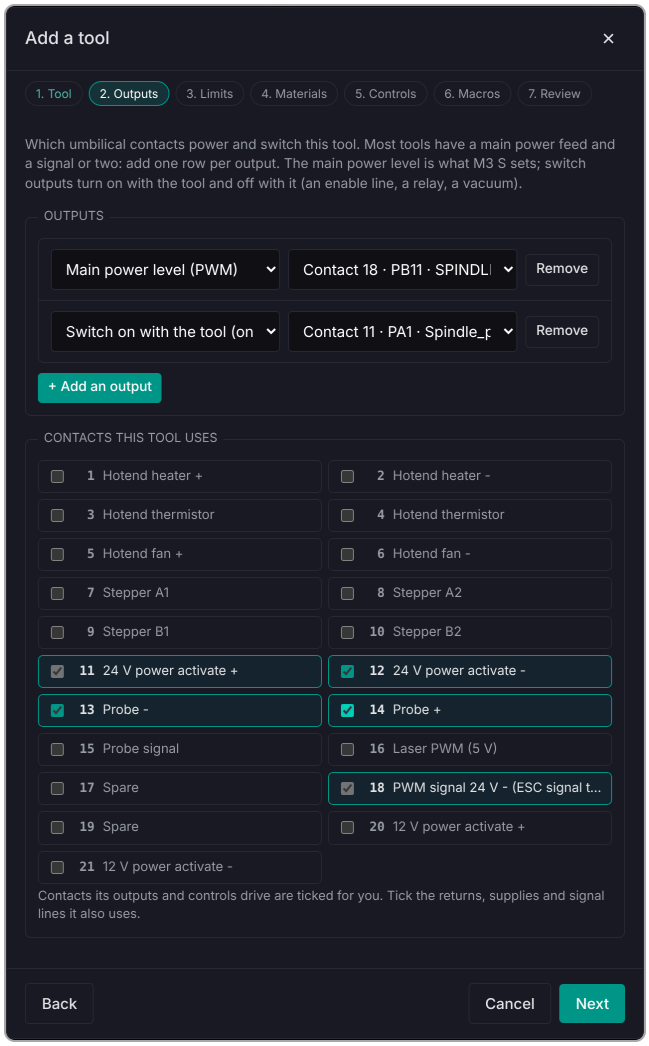


A name that breaks the rules is refused at step 1

Step 2 · Outputs – one row per output the tool needs. Most tools have a main power feed and a signal or two, so press **+ Add an output** for each. Every row has a job and a source:

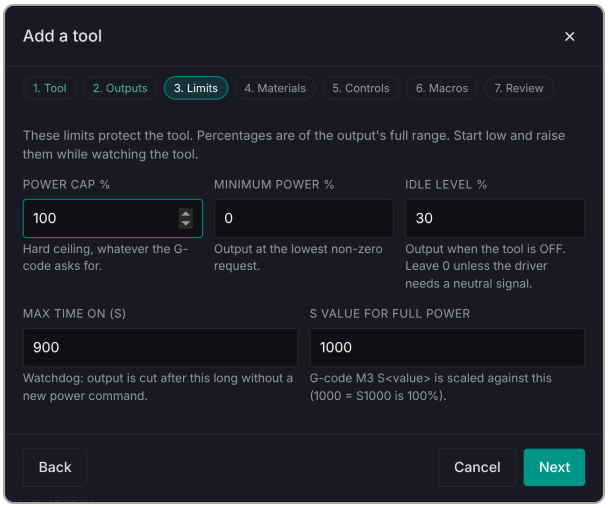
- **Main power level (PWM)** – the output `M3 S` sets, scaled and capped. Exactly one per tool.
- **Switch on with the tool** – an on/off output that turns on when the tool's power does and off with it: a relay, a driver enable line, a vacuum.

The source is a **contact of the umbilical** that a Klipper output already drives (contact 18, the spindle ESC signal, for example), a **new output on a spare pin**, a piece of **system hardware**, or another output in the config. Below the outputs, tick the other contacts the tool uses – returns, supplies, the probe – so its pin map is complete. Contacts its outputs drive are ticked for you.



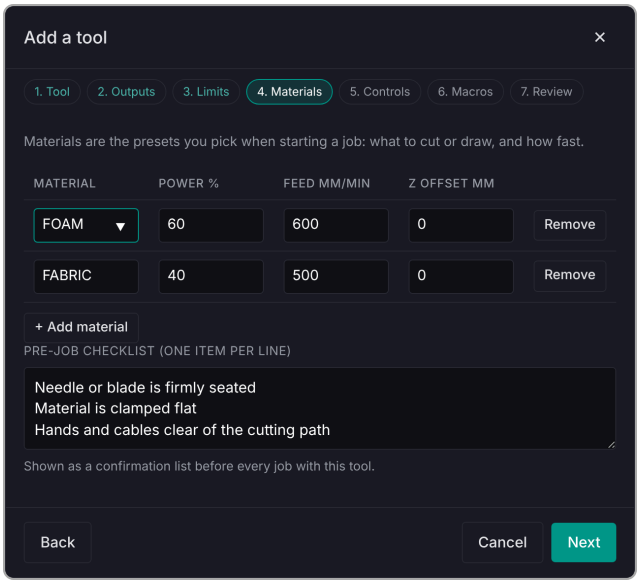
Step 2: a PWM signal on contact 18, a relay on contact 11, and the contacts the tool uses

Step 3 · Limits – the power cap (a hard ceiling whatever the G-code asks), minimum and idle levels, the watchdog's maximum on-time, and the `S` value that means full power. Sharing the ESC channel sets the idle level to its neutral 30%.



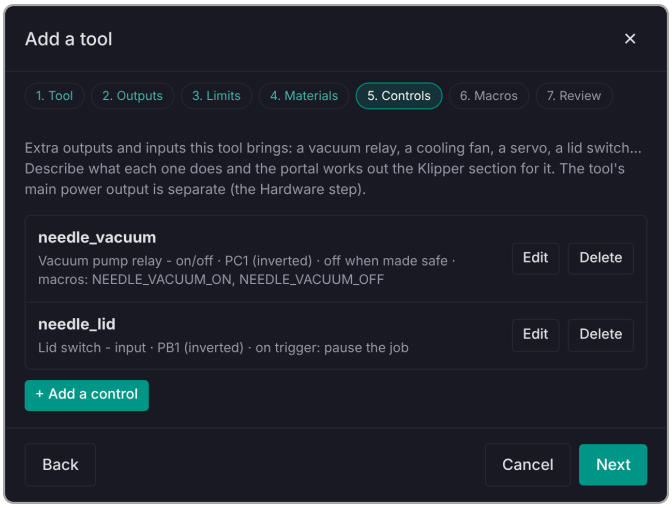
Step 3: power limits and watchdog

Step 4 · Materials – the presets you pick when starting a job (power, feed, Z offset), and the pre-job checklist shown before every job with this tool. The material name box suggests the names listed under **Admin › Tool lists**.



Step 4: materials and checklist

Step 5 · Controls – extra outputs and inputs the tool brings that are not simply switched with it: a fan with its own speed, a servo, a lid switch. Press **+ Add a control** and answer the guide's questions: you describe what the control does, and the portal works out which Klipper section that needs and shows the exact text it will write.



Step 5: the tool's extra controls

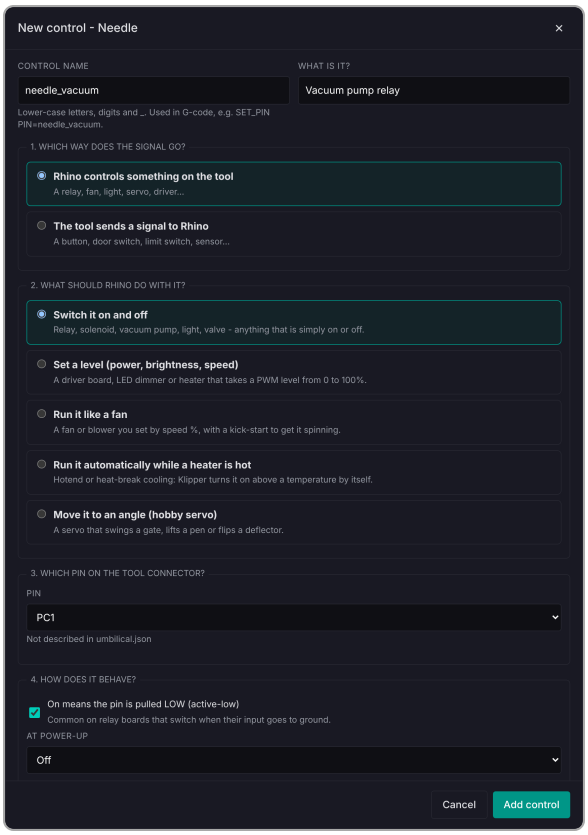
The control guide asks, in order:

1. **Name and purpose.** Lower-case letters, digits and `_`, such as `needle_vacuum`. G-code uses the name (`SET_PIN PIN=needle_vacuum`), so it must not match anything else in your config.
2. **Which way does the signal go?** Rhino controls something on the tool, or the tool sends a signal to Rhino.
3. **What should Rhino do with it?** Pick the closest match:

You pick	Klipper section	Typical use
Switch it on and off	`output_pin`	Relay, solenoid, vacuum pump, light, valve
Set a level	`output_pin`, `pwm: True`	Driver board, LED dimmer, small heater
Run it like a fan	`fan_generic`	Fan or blower set by speed, with a kick-start
Run it while a heater is hot	`heater_fan`	Hotend or heat-break cooling
Move it to an angle	`servo`	Gate, pen lift, deflector
The tool sends a signal	`gcode_button`	Lid switch, limit switch, foot pedal, sensor

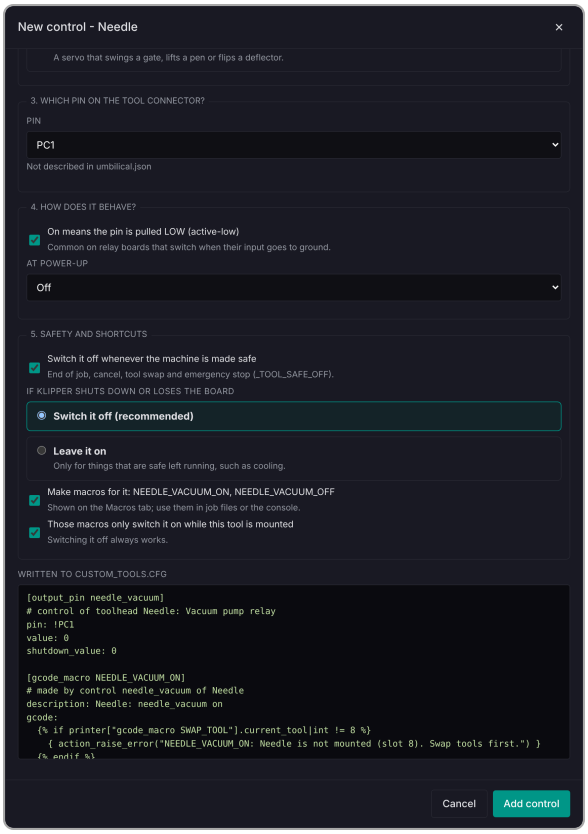
4. **Which pin?** Only free tool-connector pins are listed. If `myrhino/umbilical.json` says what each pin is (`"type": "logic"` or `"type": "mosfet"`), pins that cannot do the job are greyed out.
5. **How does it behave?** Active-low wiring and power-up state for a switch; maximum level and PWM cycle time for a level or a fan; heater and turn-on temperature for a heater fan; pulse widths for a servo.
6. **Safety and shortcuts.** Switch it off whenever the machine is made safe (end of job, cancel, tool swap, `EMERGENCY_STOP`; ticked by default); switch it off or leave it on if Klipper shuts down; and make `NAME_ON` / `NAME_OFF` (or `NAME_SET`, `NAME_ANGLE`) macros, optionally only while this tool is mounted.

The top of the guide: name, direction, what it does and the pin.



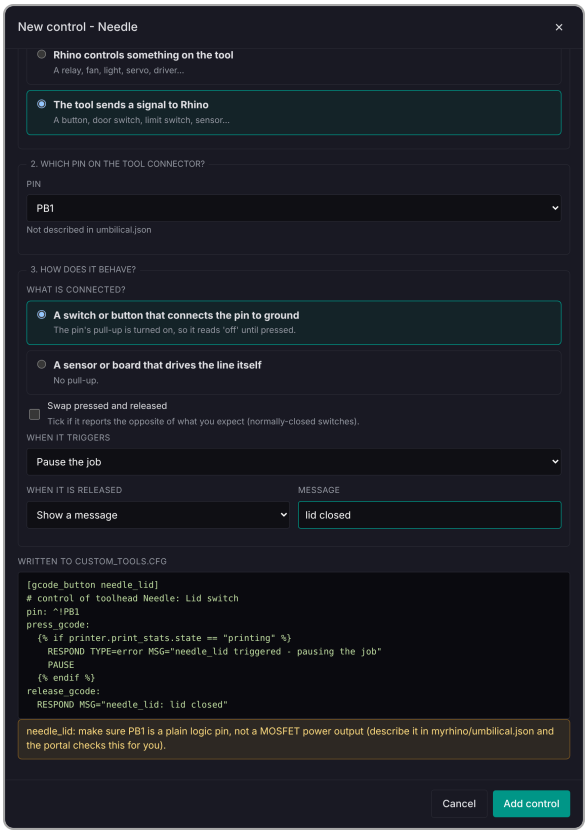
A new control: name, direction and what it does

The bottom of the guide: safety options, the macros it makes, and the exact text written to `custom_tools.cfg`. **Add control** saves only a valid entry; a name already in use, a taken pin or two controls on one pin are refused, with the reason in the preview box.



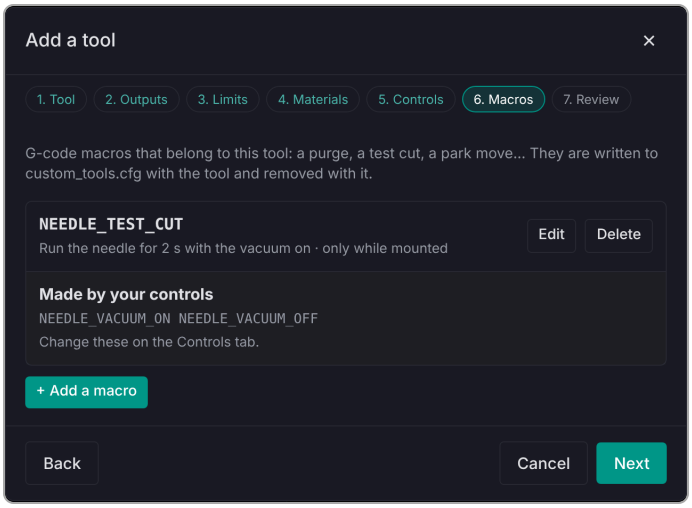
Safety options and the text written to custom_tools.cfg

For a signal from the tool, the guide asks how it is wired. A switch to ground turns on the pin's pull-up; a sensor that drives the line needs none. Then it asks what happens when the signal triggers and when it is released: show a message, pause the job (only while one is running), emergency stop, or run a command or macro.



A lid switch that pauses the job when opened

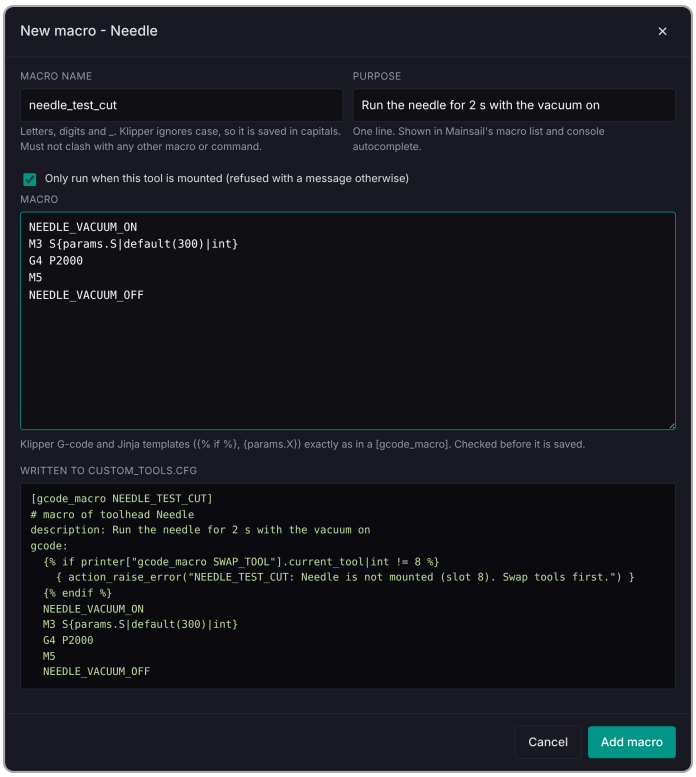
Step 6 · Macros – G-code macros that belong to this tool: a test cut, a purge, a park move. They are written to `custom_tools.cfg` with the tool and removed with it. Macros your controls made are listed underneath.



Step 6: the tool's macros

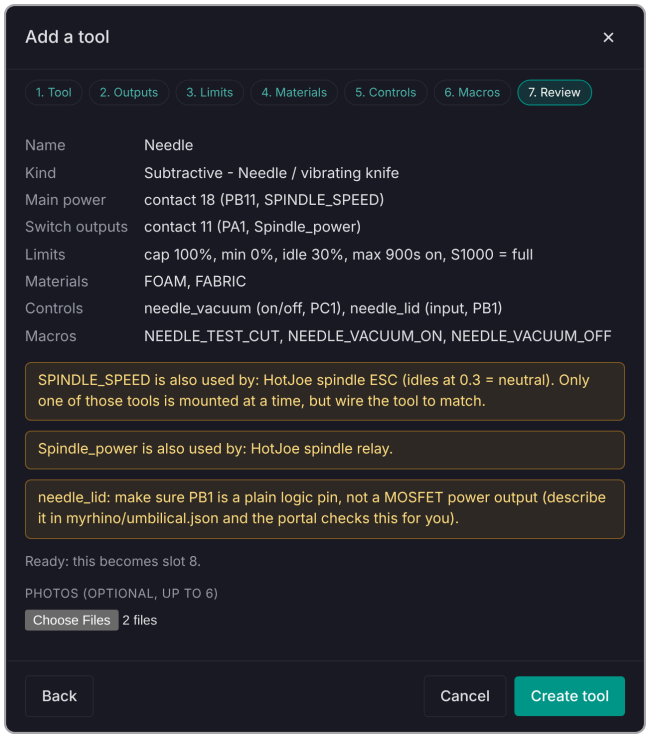
Press **+ Add a macro**. **Macro name**: letters, digits and `_`, saved in capitals because Klipper ignores letter case. **Purpose**: one line, shown in Mainsail's macro list. **Only run when this tool is mounted**: with another tool on the machine the macro refuses with a message. **Macro**: Klipper G-code and Jinja exactly as in a `[gcode_macro]`. The box underneath shows exactly what will be written.

Refused: a name used by any macro or Klipper command (`SET_PIN`, `SWAP_TOOL`, `PAUSE`), G-code words such as `M3`, and template errors such as a missing `{% endif %}`. **Warned:** `SAVE_CONFIG`, `FIRMWARE_RESTART` or `M112` in the body, an unrecognised command (usually a typo), and text after `#` or `;`, which Klipper drops as a comment.



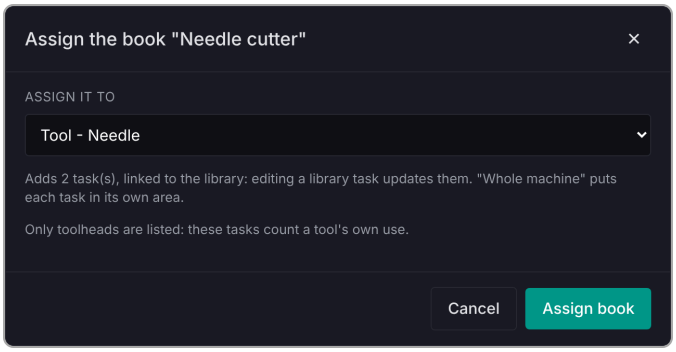
A macro with its live preview

Step 7 · Review – the portal checks everything against your live Klipper config, warns about shared outputs, and tells you the slot the tool will get. Add up to six photos here, then **Create tool**.



Step 7: review, with the outputs, controls and macros listed

Press **Add maintenance tasks** on the confirmation and the portal offers the Task Book that fits the new tool (see *Task Library and Task Books* in Part 4).



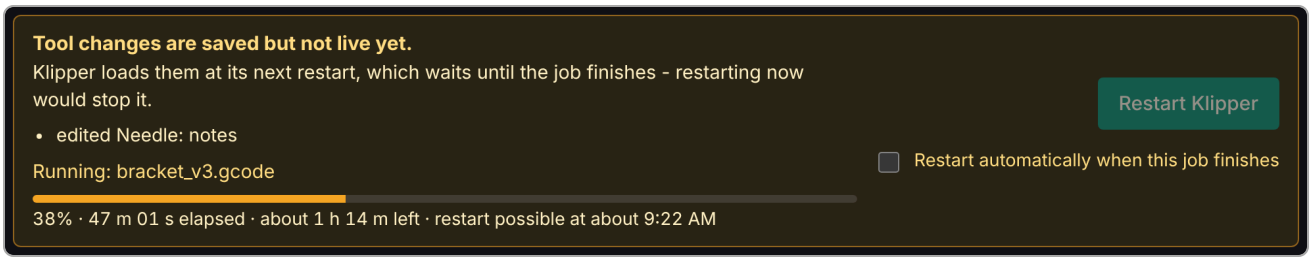
The Task Book offered for a new needle cutter

Restarting Klipper to load changes

A new or changed tool is saved immediately but is not live until Klipper restarts. A yellow banner lists what is waiting, from the portal or the console; press **Restart Klipper** when the machine is idle. Changes Klipper never sees – notes, photos, the pin map – are saved without asking for a restart.

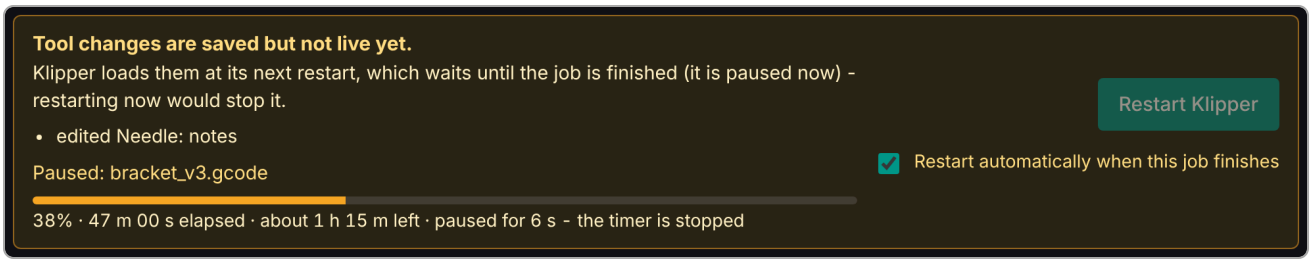
Changes waiting; the machine is idle, so the restart is allowed

While a job is printing or paused, **Restart Klipper** is greyed out and the banner says why, with the file name, a progress bar, time elapsed and time left, and the clock time when a restart becomes possible. Changes made during a job are saved and queued; only the restart waits, so it can never kill a job.



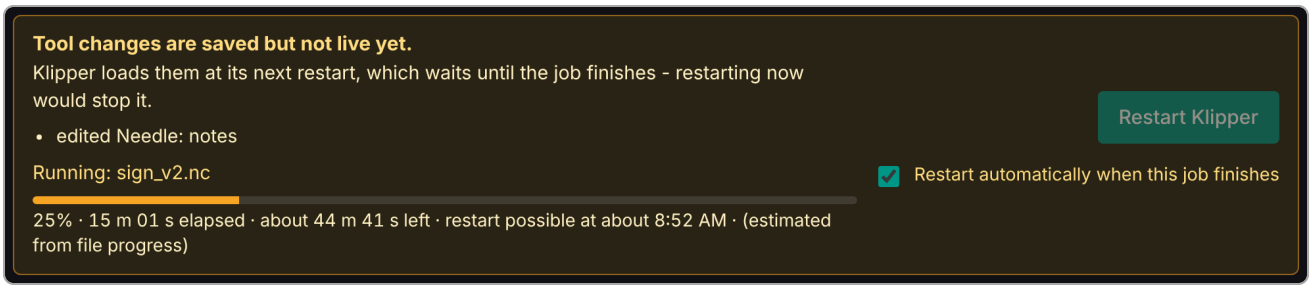
A print running: progress, time left and when a restart is possible

Paused, the countdown stops and the banner shows how long the job has been paused.



The same job paused

Time left comes from the slicer's estimate for 3D prints. Laser, CNC and cutter files usually carry none, so the portal works it out from file progress, the way Mainsail does, and says so.

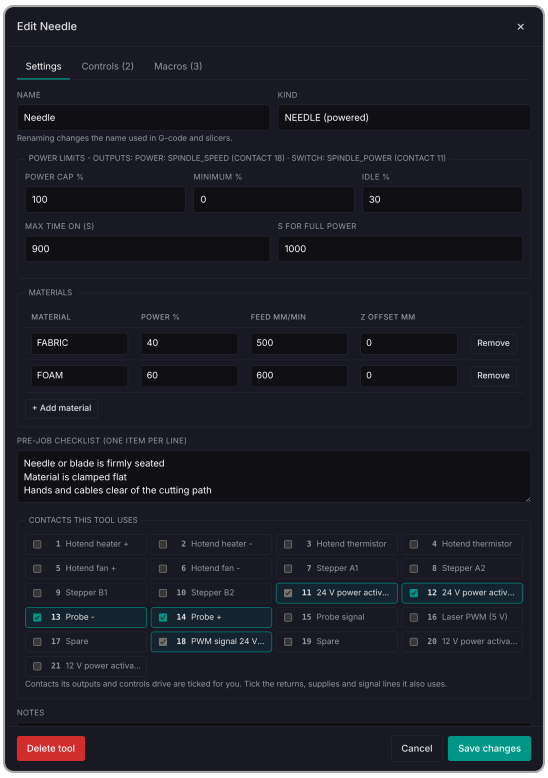


A laser job: time left estimated from file progress

Restart automatically when this job finishes is off by default. Tick it and the portal restarts Klipper 30 seconds after the job completes or is cancelled, which leaves time for `END_PRINT` or `CANCEL_PRINT` to finish. After an error it does not restart, so you can see what happened. A restart started anywhere else (Mainsail, `RESTART_FOR_TOOLS`) also clears the banner. The console commands (`ADD_TOOLHEAD` and the rest) stay refused during a job: they run a shell command that Klipper waits for, which would stall the job's G-code.

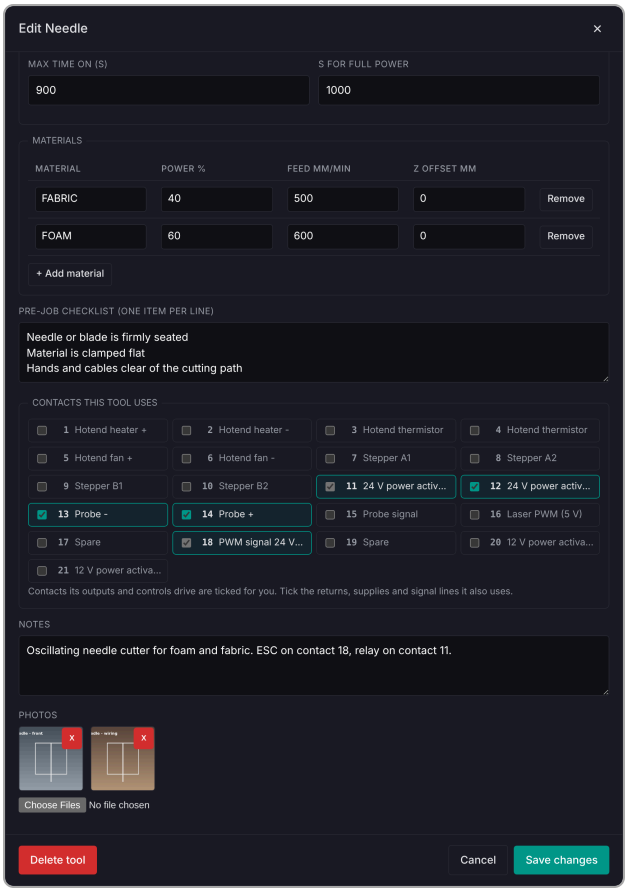
Editing a tool

Open the tool's page and press **Edit tool**. On the **Settings** tab you can rename it and change its power limits and materials; the legend above the limits lists its outputs and the contacts they use.



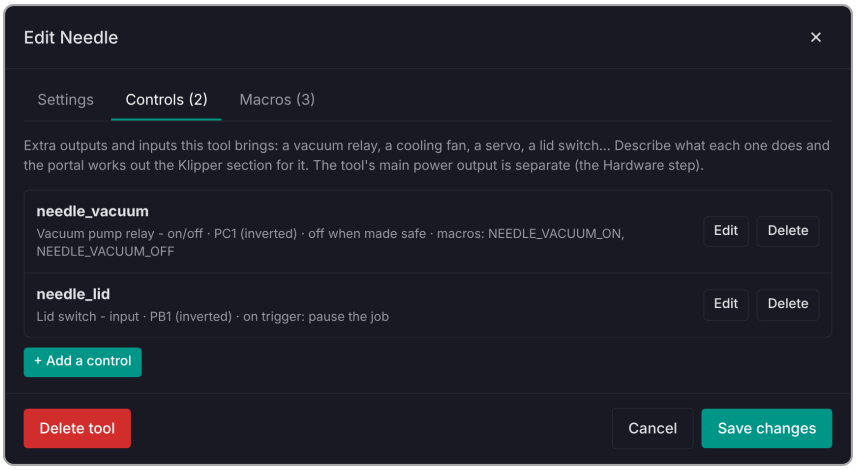
Edit window: name, outputs, limits and materials

Further down are the checklist, the contacts the tool uses (tick more as you wire them), notes and photos. **x** on a photo deletes it; **Choose Files** adds more. Photos are kept in `~/printer_data/rhino_data/images/`, outside the config folder, so config backups stay small.



Edit window: checklist, contacts, notes and photos

The **Controls** and **Macros** tabs work exactly like steps 5 and 6 of the wizard; their changes are saved with **Save changes** and are live after the restart.



Edit window, Controls tab

Deleting a tool

In the edit window press **Delete tool**. The button changes to **Click again to delete <name>**; press it within 4 seconds to confirm. The tool's pins are freed, its photos removed and its maintenance tasks archived. Restart Klipper afterwards.

Two-click delete confirmation

Slicing for Rhino Studio

The portal can run **PrusaSlicer** on the Rhino for Rhino Studio's Prepare, so a phone or tablet can slice a 3D model without a PC. Prusa ships its Linux build only through Flathub, so the host needs Flatpak once: over SSH, `sudo apt install flatpak` (the portal never runs sudo itself). Then install PrusaSlicer from Rhino Studio, **Settings › Software**: it goes in for your login only (`flatpak --user`) and slices a test cube before it is used. The same row offers **Update** when Flathub has a newer build; if a new build can't slice, the one before is put back, and **Go back** does the same by hand. It needs a host with 2 GB of memory or more: on a Pi 3B, slice with Kiri:Moto in Studio instead.

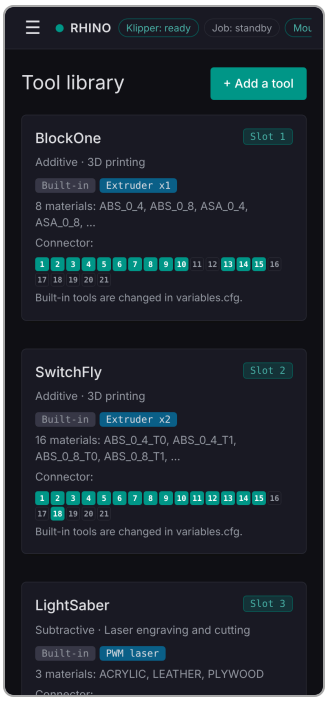
Each slice runs in Flatpak's sandbox, with no network and no files but its own. Slicing always gives way to the printer: it runs at the lowest priority, one slice at a time, with fewer cores and less memory while a job prints. A host under 4 GB (a 2 GB board, or a Pi 3B with an engine put there by hand) waits until the job is finished. Whatever settings Studio sends, the Rhino's start and end G-code, temperatures, bed and nozzle are always its own. PrusaSlicer is © Prusa Research and the Slic3r authors, under the AGPL-3.0 licence, and is downloaded from Flathub on the machine.

Rhino Studio from the Rhino

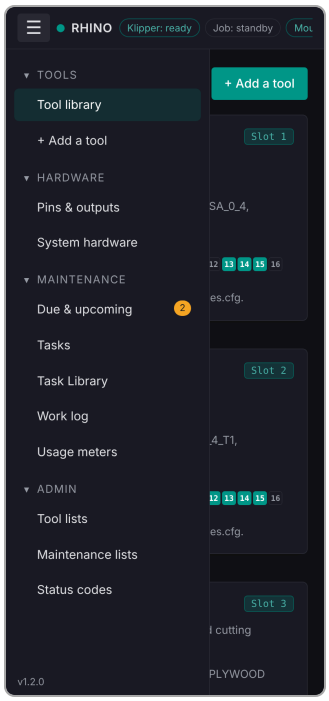
The install also serves **Rhino Studio** from the Pi, at `http://<pi>:5001`: open it from any PC, phone or tablet on your network, with nothing to install on it. The nginx that serves Mainsail serves it on a port of its own, and Mainsail's site is left as it is. Updating the Tool Manager from its zip updates Studio too; undo takes it back out. The menu's header shows the address.

On a phone

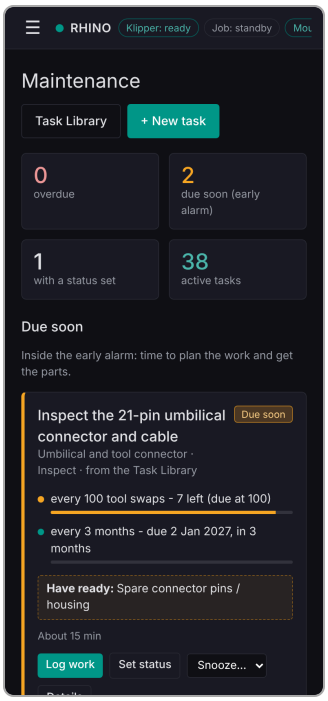
The portal works on a phone: the menu folds behind ☰, and the status chips scroll sideways in the top bar.



Tool library



Menu



Maintenance

Pins, outputs and system hardware

Hardware › Pins & outputs starts with the tool connector: all 21 contacts of the umbilical, the board pin behind each, what drives it in the Klipper config and which tools use it. A contact whose board pin nothing in the config uses is highlighted – it means the pin map and `printer.cfg` disagree (contact 20 here; see *Still to check on the machine*). The contacts come from the Rhino pin map; describe different wiring with `"contacts"` in `myrhino/umbilical.json`.

The 21 contacts, what drives them and which tools use them

Further down: the spare pins a new tool output may claim (Free, In use with the file and section, or Reserved by a tool), the outputs tools can share with their contact and idle level, and the system hardware.

Output channels tools can share

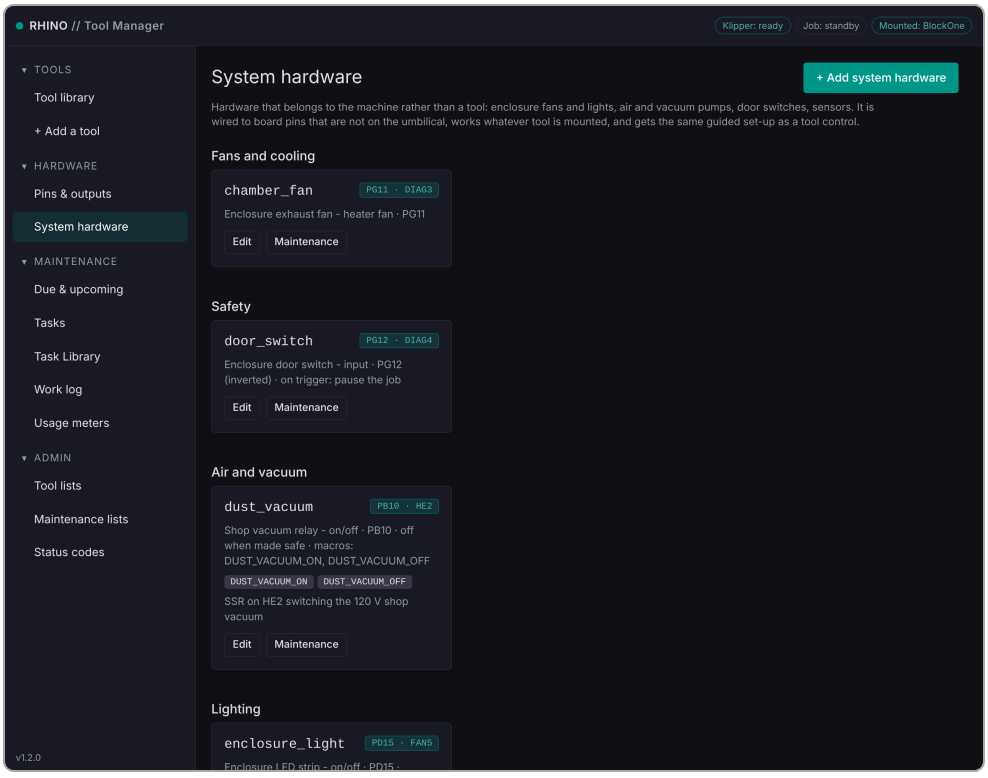
Existing outputs a new powered tool may drive instead of using a new pin.

OUTPUT	TYPE	CONTACT	IDLE LEVEL	ALSO USED BY
FoamWire_PWR	PWM	-	0	-
LASER_INITIALIZE	PWM	-	0	LightSaber laser power rail
SERVO_LASER	PWM	16	0	LightSaber laser intensity / SwitchFly path servo
SPINDLE_SPEED	PWM	18	0.3	HotJoe spindle ESC (idles at 0.3 = neutral)
Spindle_power	On/off	11	0	HotJoe spindle relay
Umbilical_LED	PWM	-	0	umbilical LED strip data line (not a power output!)
dust_vacuum	On/off	system hardware	0	-
enclosure_light	On/off	system hardware	0	-

Output channels tools can share

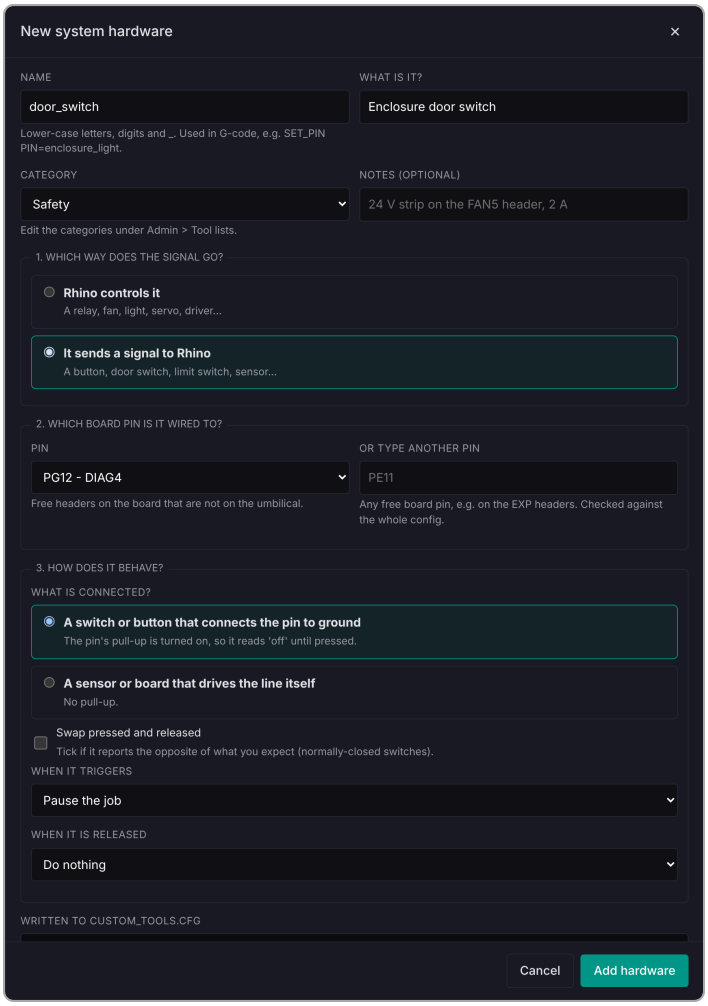
System hardware

Some hardware belongs to the machine rather than a tool: enclosure fans and lights, an air or vacuum pump, a door switch, a sensor. It is wired to board pins that are *not* on the umbilical and works whatever tool is mounted. **Hardware › System hardware** lists it by category; **+ Add system hardware** opens the same guide as a tool control.



System hardware, grouped by category

The differences from a tool control: the pin list offers free **board headers** (FAN5, HE2, DIAG4…) and refuses any pin on the umbilical; you can type another free pin; you choose a **category** (edited under **Admin › Tool lists**); and its macros have no mounted-tool guard. **Switch it off whenever the machine is made safe** is off by default, since lights and fans usually stay on. A powered tool can switch a piece of system hardware on with itself – a shop vacuum with a cutter – by choosing it as a **Switch on with the tool** output in step 2. Each piece can carry maintenance tasks (**Maintenance** on its card).



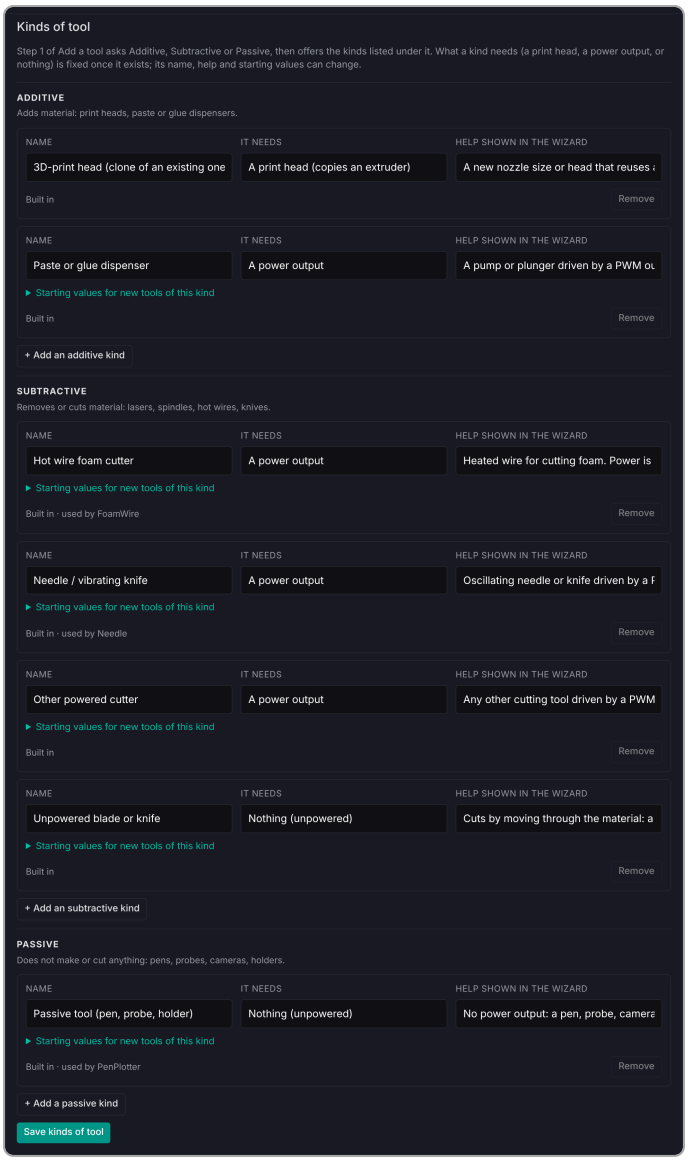
An enclosure door switch that pauses the job

Admin lists

The **Admin** group sets what the portal's drop-down menus offer. Every list is checked as you save: a kind of tool, category or kind of work that something still uses cannot be removed (rename it instead).

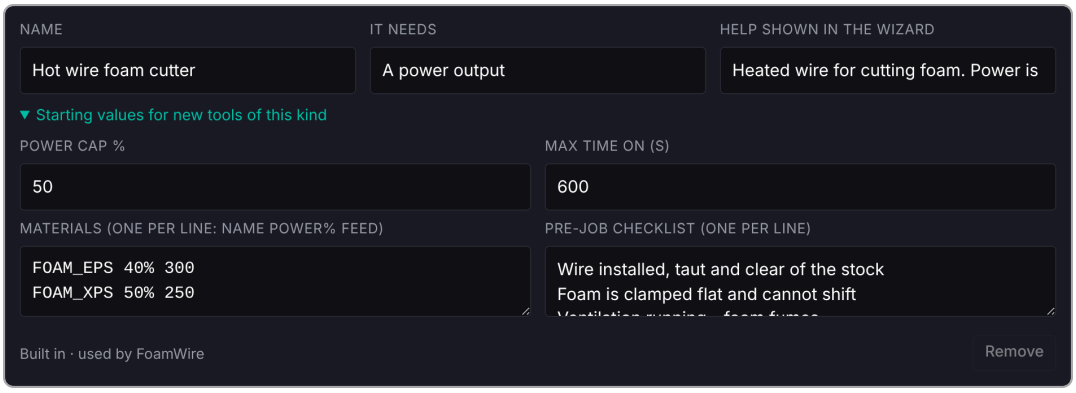
Tool lists

Kinds of tool are grouped under Additive, Subtractive and Passive – what step 1 of Add a tool offers. Each has a name, the help line shown in the wizard and what it needs: a print head (copies an extruder), a power output, or nothing. What it needs is fixed once the kind exists; built-in kinds can be renamed but not removed.



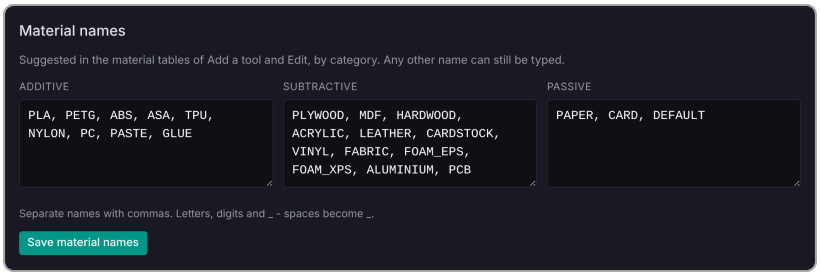
Admin › Tool lists: kinds of tool by category

Open **Starting values for new tools of this kind** to set the power cap, maximum on-time, materials (one per line: name, power %, feed) and pre-job checklist that a new tool of this kind starts with.



Starting values of the hot-wire kind

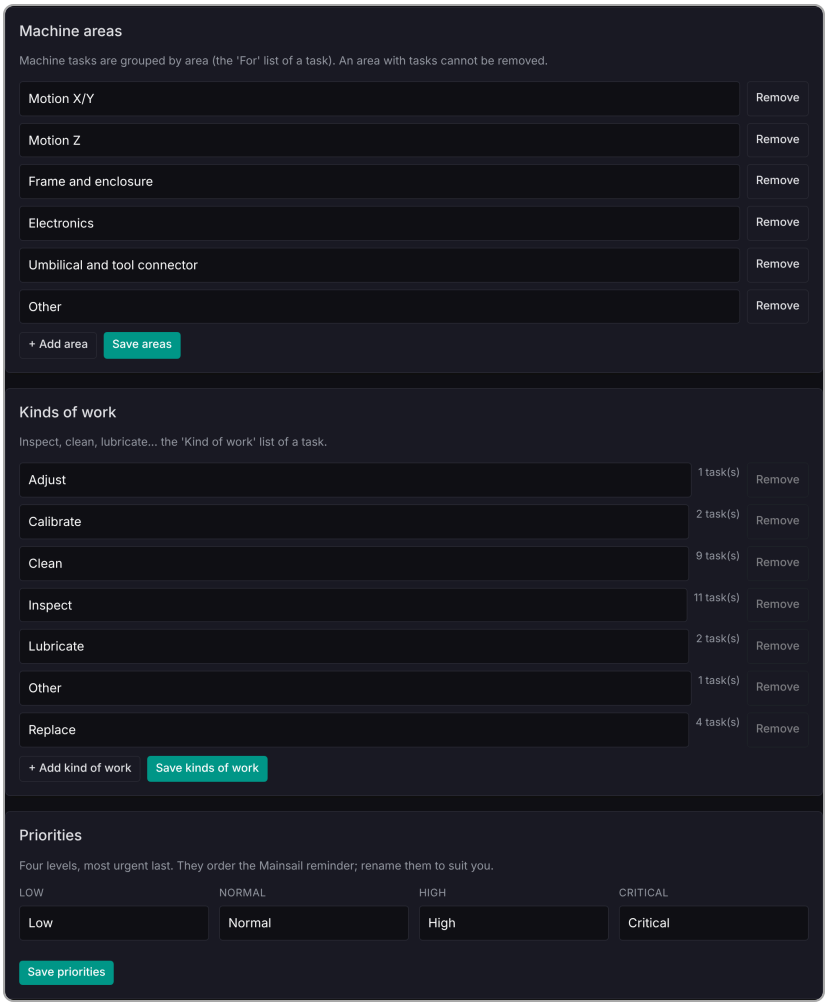
The same page holds the material names suggested in the material tables, by category; the nozzle sizes offered when you add a print head; and the system hardware categories.



Material name suggestions by category

Maintenance lists

The machine areas tasks are grouped by, the kinds of work (Inspect, Clean…) with how many tasks use each, and the names of the four priority levels.



Admin › Maintenance lists

Status codes are described with maintenance tracking in Part 4.

Console commands

Everything the portal does can also be typed in the Mainsail console. Lines starting `rhino:` are the tool manager's replies; nothing changes if a command is refused.

Managing tools

Command	What it does
`ADD_TOOLHEAD NAME=FoamWire KIND=HOT_WIRE PIN=PB1`	Add a powered tool on a new pin
`ADD_TOOLHEAD NAME=Needle KIND=NEEDLE POWER_PIN=SPINDLE_SPEED`	Add a powered tool on a shared output
`ADD_TOOLHEAD NAME=Wide KIND=PRINT_HEAD BASE=BlockOne NOZZLES=0.6,1.0`	Add a print-head clone with new nozzle sizes
`EDIT_TOOLHEAD NAME=FoamWire CAP=0.4 MAX_ON_S=300`	Change limits (also `MIN=`, `IDLE=`, `S_MAX=`, `DEFAULT_POWER=`)
`EDIT_TOOLHEAD NAME=FoamWire MATERIAL=FOAM_EPS POWER=0.45 FEED=280 ZOFF=0`	Add or change a material
`EDIT_TOOLHEAD NAME=FoamWire DELMATERIAL=FOAM_XPS`	Remove a material
`RENAME_TOOLHEAD NAME=FoamWire NEWNAME=HotWire`	Rename
`REMOVE_TOOLHEAD NAME=FoamWire`	Remove (asks you to confirm)
`LIST_TOOLHEADS`	Every tool with slot, type and materials
`TOOLHEAD_PINS`	Which connector pins are free, used or reserved
`CHECK_TOOLHEADS`	Re-check the registry against the config and regenerate `custom_tools.cfg`
`RESTART_FOR_TOOLS`	Restart Klipper to load tool changes (refused during a job)
`TOOLS_MENU`	Reminder of where to add, edit and remove tools

Running the machine

Command	What it does
`START_TOOL_SWAP`	Guided tool change
`RESTORE_TOOL`	Confirm the mounted tool after a restart
`SET_TOOLHEAD TOOLHEAD=<tool> MATERIAL=<material>`	Select tool and material for the next job
`LASER_JOB_SETUP / CNC_JOB_SETUP / DRAGKNIFE_JOB_SETUP / TOOL_JOB_SETUP FILE=<name>.gcode`	Guided job setup for that tool
`RHINO_POSITIONS`	List the positions in `myrhino/positions.cfg` and check them against the travel limits
`G28 / G28 X Y`	Home. Z only with a print head confirmed; `G28 X Y` with any tool

Command	What it does
`SET_Z_ZERO`	Set Z zero by eye for the laser (at focus), spindle, drag knife or an added tool – Bed up/down, then SET
`TOOL_RESUME`	Resume a paused non-print job
`M3 S<value>` / `M5`	Mounted tool on / off
`SET_TOOL_POWER POWER=<0–1>`	Set an added powered tool's output (fraction of its cap)
`FILAMENT_CHANGE`	Guided filament change
`BED_COOLDOWN`	Staged, material-aware bed cooldown
`EMERGENCY_STOP`	All tool power off (see Safety systems)
`BACKUP_CFG MSG=<note>`	Push the config folder to your GitHub backup
`SWITCH_T0` / `SWITCH_T1` `MATERIAL=PLA`	SwitchFly: change filament path. Retracts, moves the servo, heats, loads and purges (the purge lowers the bed to Z5). Without `MATERIAL` it uses the material chosen for that path at the last job start, and asks if there is none.
`NOZZLE_HEIGHT_CALIBRATE`	Paper test for the mounted print head: Down/Up 1, 0.1 or 0.02 mm, then Save. Kept per head; run it once when a head is new or re-seated.
`Z_TILT_ADJUST`	Tram the bed with the three Z motors (home first)
`CALIBRATE_ESC`	Guided spindle ESC throttle calibration (bit removed)
`PM_STATUS`	List maintenance that is overdue or due soon (live)
`PM_REMINDER`	Show the start-up maintenance pop-up again

Safety systems

Each of these is automatic; none needs switching on.

- **Tool must match.** `START_JOB` refuses a file sliced for a different tool. The laser and spindle commands refuse to fire unless LightSaber or HotJoe is the recorded tool, and switch their outputs off when they refuse.
- **Shared pin guard.** `SERVO_LASER` is the laser's intensity signal on LightSaber and SwitchFly's filament servo otherwise; servo commands are blocked while LightSaber is mounted.
- **Network changes undo themselves.** Changing the Pi's name, address, Wi-Fi or power saving (from Rhino Studio, through the Portal) arms a 2-minute undo first; unless you press Keep, it switches back. The only root access is one helper program, `/usr/local/sbin/rhino-net`, that does nothing else (1.4.8).
- **Feed direction follows the tool.** G0/G1 reverse the extruder on SwitchFly path 2 using the saved record of the tool and path. Confirming a swap (or picking another tool after a restart) points that record at the mounted tool, and with SwitchFly turns its selector to path 1 so the servo and the record agree (1.4.7).
- **Power cap.** An added powered tool never exceeds its cap, whatever `M3 S` or `SET_TOOL_POWER` asks for.
- **Watchdog.** Every 5 seconds while an added powered tool is on: with no job running it is cut after the tool's *Max time on*; during a pause it is held off; if the tool is no longer mounted everything is switched off.
- **Everything off on the way out.** Tool swap, `CANCEL_PRINT`, `END_PRINT` and `EMERGENCY_STOP` switch off the tool on the machine: the laser rail and signal, or the spindle relay (ESC to 0.3 neutral). Swap, cancel, the emergency stop and `END_PRINT` also switch off every added tool's outputs (1.4.6: `END_PRINT` for every kind of tool). Controls ticked *switch off when made safe* are switched off by swap, cancel and the emergency stop; after a normal `END_PRINT` they keep running, except with an added powered tool, where they go off as before.
- **Nozzle heater guard.** The nozzle heater only switches on with a print head mounted and its thermistor reading normally, and is cut within a second if the thermistor reads as unplugged.
- **Z homing guard.** Z is only homed with a confirmed print head; with any other tool, or before the restart question is answered, homing Z is refused.
- **Pause makes it safe.** See *Pause, resume and cancel*: lifted clear and powered off before parking.
- **Nothing changes mid-job.** Tool swaps, the tool console commands and Klipper restarts are refused while a job is printing or paused. Changes made in the portal are saved and wait for the next restart.

> ⚠ **WARNING**
> **`EMERGENCY_STOP` is a power-off, not a halt.** It cuts tool power but does not stop motion. To stop the machine dead, use Mainsail's emergency-stop button (`M112`), which shuts Klipper down; then `FIRMWARE_RESTART`.

Start low. Every starting power value (hot wire 40–50 %, caps, maximum on-times) is a placeholder. Test a new tool from a current-limited supply first, then raise its values in the portal while watching it.

Hardware layer. Heavy loads (laser, spindle, bed, vacuum pump) are switched through relays, so a controller fault cannot hold them on through the board itself. If Klipper shuts down, the spindle ESC's signal stops and its relay opens. Close the enclosure doors while the laser or spindle runs.

> **⚠ WARNING**
> Smoke detection is planned in the build notes but not yet fitted. Never leave a laser, hot-wire or spindle job unattended.

PART

04

Rhino Studio

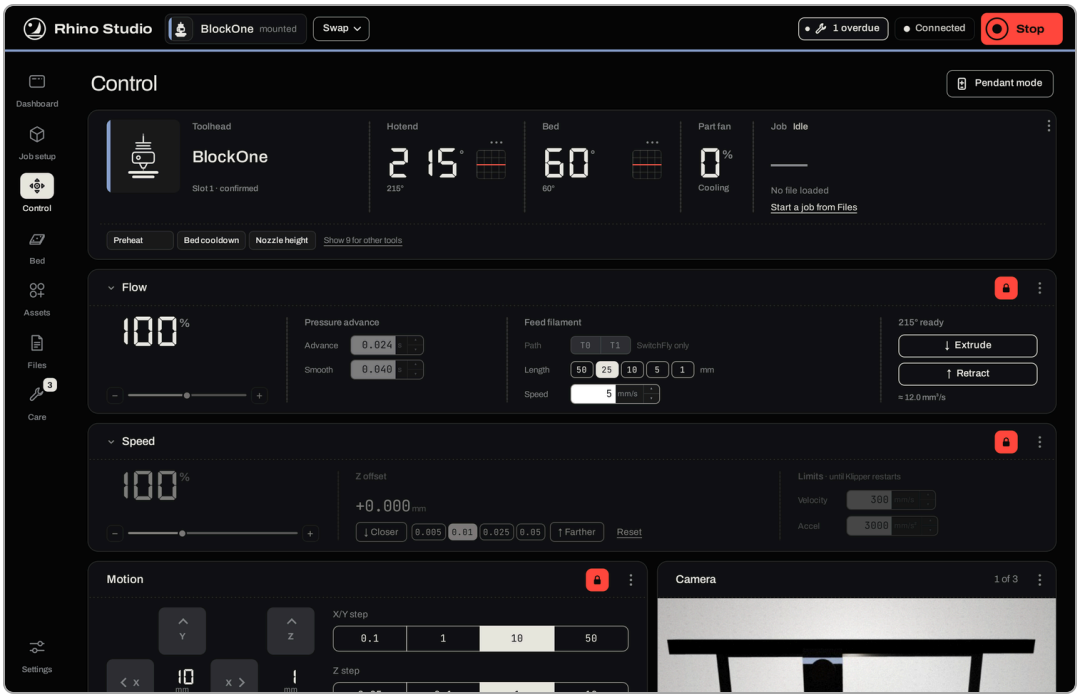
The Rhino's main screen: finding your way, running the machine, tuning a print head and the bed.

Rhino Studio at a glance	**94**
Control	**98**
Tuning a print head	**102**
Bed	**110**

Rhino Studio at a glance

Rhino Studio is the Rhino's main screen. It prepares jobs from your models and drawings, previews them, runs the machine, tunes the print heads, and keeps track of the tools, the hardware and the maintenance, all in one web page. It runs in any up-to-date browser on a computer, tablet or phone. The Rhino serves it, so there is nothing to install.

Open `http://rhinostudio.local`, or the Rhino's address followed by `:5001`, for example `http://192.168.1.50:5001`. Every release of the Tool Manager carries the matching Rhino Studio, so updating one updates both.



Rhino Studio, here on Control: the top bar, the rail on the left, and the panels

Finding your way

- **The rail** on the left holds the screens: **Dashboard**, **Job setup**, **Control**, **Bed**, **Assets**, **Files**, **Care** and **Settings**. A number on **Care** counts maintenance that is due.
- **The top bar** shows the mounted toolhead and whether Klipper has confirmed it. **Swap** lists the other toolheads and starts the swap wizard for the one you pick (see *Swapping tools*). After a Klipper restart, a confirm button appears here until you answer which tool is on the machine. Then come maintenance due, the connection, and **Stop**.
- Many **panels** fold away with the arrow beside their title, and Studio remembers that on this device. The ··· menu holds the rest of a panel's options.
- Screens with several pages (**Assets**, **Bed**, **Care**, **Settings**) have a side menu, which folds away with the arrow at its top.

> **⚠ WARNING**
> **Stop** sends Klipper's emergency stop (`M112`) at once, without asking. Klipper then shuts down and needs a restart; use it when something is going wrong, not to end a job. To end a job normally, use **Cancel print**.

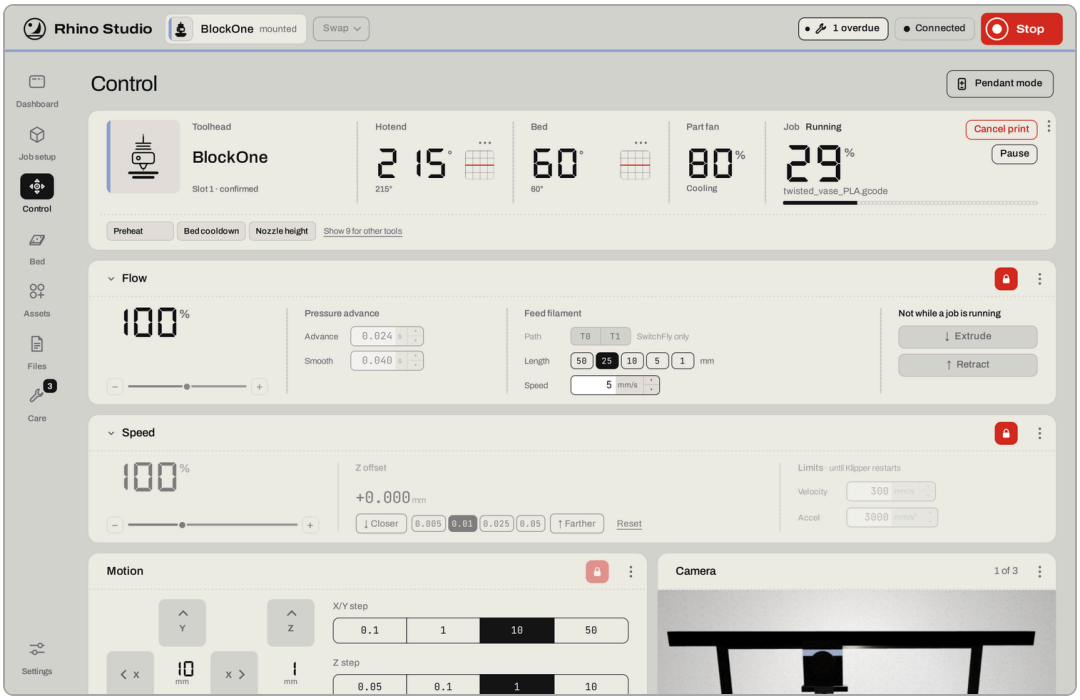
What asks first, and the padlock

Every command Studio sends is checked first. Anything that moves the machine, switches a tool's power, changes the mounted tool, restarts Klipper or starts a job opens a window saying what will happen, with the exact commands, and waits for **Proceed**. Heating and plain macros go straight through. The emergency stop never asks.

Panels that act on the machine live, such as jogging, live tuning, the bed's lead screws and system hardware, sit behind a red padlock. Press it to unlock. The panel turns red while it's live, and locks again by itself: after 60 seconds without a press, when you leave the page, when a job starts, when the tool changes, or when Klipper stops being ready.

Day and night

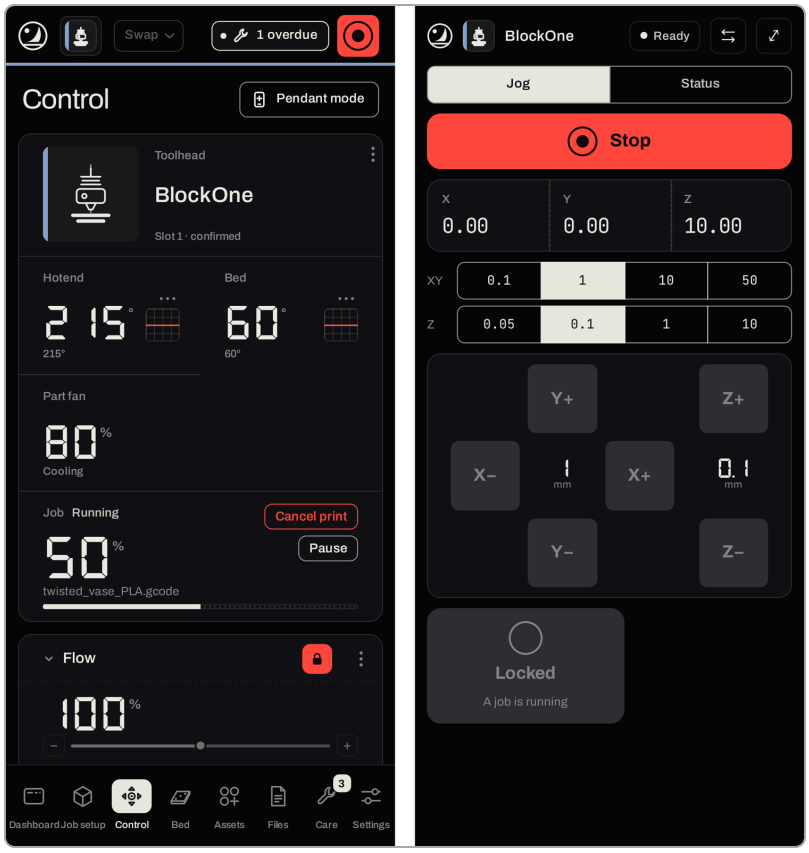
Studio follows your device's light or dark setting. To choose one, or a theme of your own with its own colours, background picture and typefaces, open **Settings › Appearance**. In every look, red keeps one meaning: stop, or a panel that is live.



The same screen in the day look

On a phone

On a narrow screen the rail moves to the bottom and the panels stack in one column. **Pendant mode**, from the top of Control, turns a phone into a hand-held pendant: the position, step sizes, square jog keys and **Stop**. Jogging needs both hands: hold the **Hold** key with one thumb while you press a jog key with the other. Letting go of Hold stops the next press. The ⇄ button swaps the keys round for left-handed use. Jogging stays locked while a job runs.



Control on a phone, and Pendant mode (locked while a job runs)

Dashboard

The Dashboard is for looking, not for running the machine: it has no machine buttons apart from **Stop** in the top bar, so it is safe to leave open on a screen in the workshop. The camera fills the page. The tool is at the top left, the machine's state at the top right, and notices at the bottom left; a red notice needs you, and is answered on Control. A row of figures from the Portal's reports runs along the bottom.

While a job runs, the job's progress, file, time left and finish time show in large type. For a print, the part being printed appears as a turning hologram over the camera: what's done glows, the layer being printed is an amber line, and the rest is a faint outline. The dial under it shows how much is done, the head's speed and the material. Drag the hologram to turn it; **Hide model** hides it on that screen.

The Dashboard while a print runs

NOTE

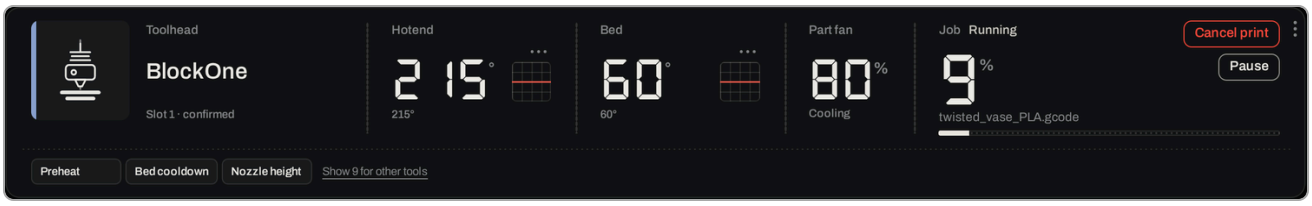
Cameras are there to watch, not to keep you safe: a picture can freeze or lag. Studio pauses a camera when it isn't on screen.

Control

Control runs the machine: what the tool is doing, the job, live tuning, jogging, the camera, macros and the console. The panels shown depend on the mounted tool. **Flow** and **Speed** appear only with a print head.

The status box

Across the top: the toolhead (its slot, and whether Klipper has confirmed it), then the readings for that tool, then the job.

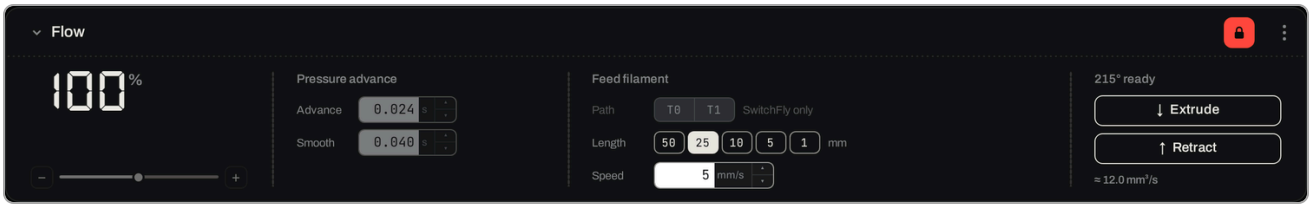


The status box during a print: the head, hotend, bed, part fan and the job

- **Readings.** A print head shows the hotend, the bed and the part fan; a laser shows its power and body temperature; a spindle shows its throttle and body temperature. With no tool mounted, only heaters that are on or still hot show. The small graph beside a temperature shows how it has gone over the last few minutes.
- **Setting a temperature.** Click the number, type the new target and press Enter; 0 switches the heater off, and Esc leaves it as it was. The hotend can be heated only with a confirmed print head mounted; switching off is always allowed.
- **The job.** How far it has got, the file and a progress bar, with **Cancel print** (asks first) and **Pause** or **Resume**. Resume asks first; Pause doesn't. When no job is running it says *Start a job from Files*.
- **Tool macros.** The row under the box holds the mounted tool's own macros. **Show … for other tools** lists the rest.

Flow and Speed: live tuning

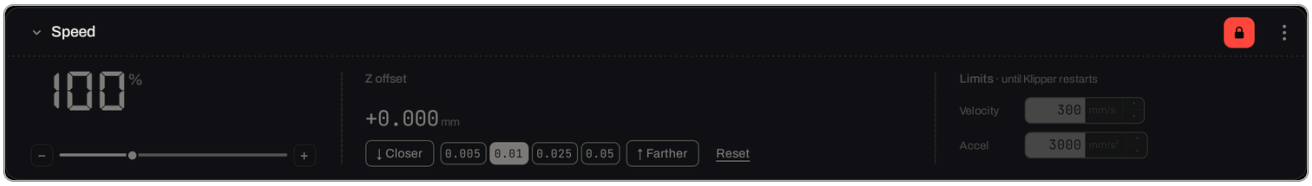
Both panels change the machine as it works, so they sit behind the padlock. The values last until Klipper restarts. A print's own settings from the job file, and what the Rhino applies at job start, come back with the next job.



Flow: flow, pressure advance, and feeding filament

- **Flow**: 50 to 150 %, in 1 % steps or with the slider.
- **Pressure advance**: the advance and its smooth time.

- **Feed filament**: pick a length (1 to 50 mm) and speed, then **Extrude** or **Retract**; the nozzle must be hot. Not while a job is running. **Path** picks T0 or T1 on SwitchFly.

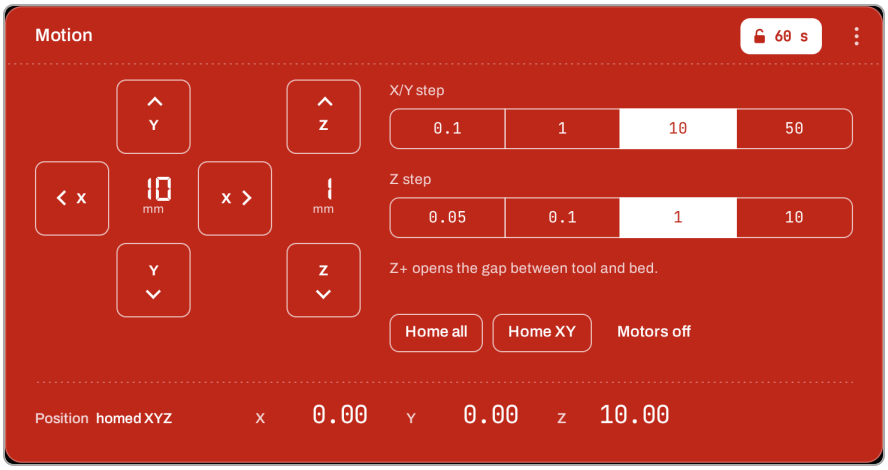


Speed: speed, Z offset and the speed limits

- **Speed**: 10 to 300 % of the job's speeds.
- **Z offset**: **Closer** and **Farther** move the nozzle in the chosen step, at most 0.05 mm a press, also during the first layer of a print. **Reset** takes your changes back out. To keep a first-layer height for good, use *First-layer squish* (see *Tuning a print head*).
- **Limits**: top velocity and acceleration, until Klipper restarts.

Motion

Jogging, homing and switching the motors off. The panel is locked to start with; press the padlock to jog. It stays unlocked for 60 seconds after your last press, with the seconds shown on the padlock.



Motion, unlocked: one press is one step

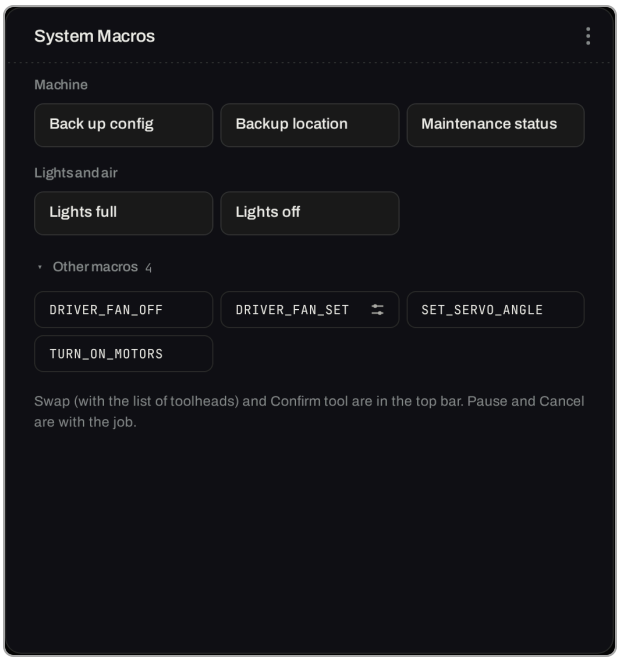
- Pick an X/Y step (0.1, 1, 10 or 50 mm) and a Z step (0.05, 0.1, 1 or 10 mm). One press moves one step. Z+ opens the gap between the tool and the bed.
- An axis can be jogged only once it is homed, and only with a confirmed tool. Moves stop at the machine's travel limits.
- **Home all**, **Home XY** and **Motors off** ask first.
- Jogging locks while a job runs, and when you leave the page, the connection drops, or the tool changes.

Camera

The picture from the Rhino's cameras; the arrows step through them, and the count shows which one you're looking at. Cameras are added under **Settings › Cameras**.

Macros

Each macro is a key. The mounted tool's macros come first; **System Macros** are the same whatever is mounted, grouped by area, with **Other macros** folded at the end. A key with a sliders mark takes values: pressing it asks for them, with the allowed range. Macros you make yourself (**Settings › Your macros**) have their own group.



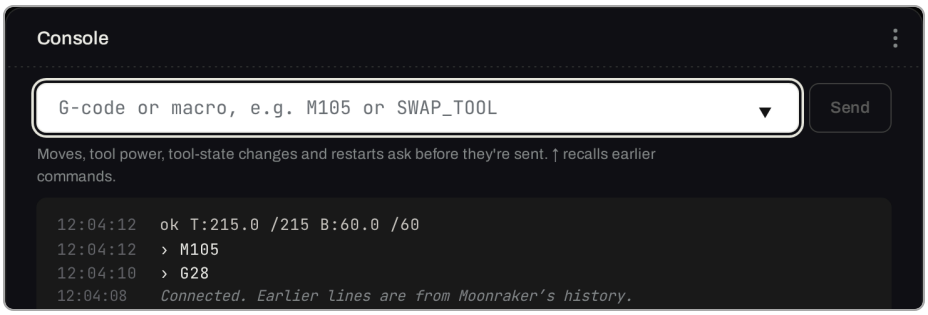
System Macros, grouped by area

Hardware

When the Rhino has system hardware with live controls, such as a case light, an exhaust fan or a chamber heater, the **Hardware** panel shows each one with its reading and a control to set it, behind the padlock. Tool outputs never appear here: a tool's power is switched by its wizard or a confirmed console command.

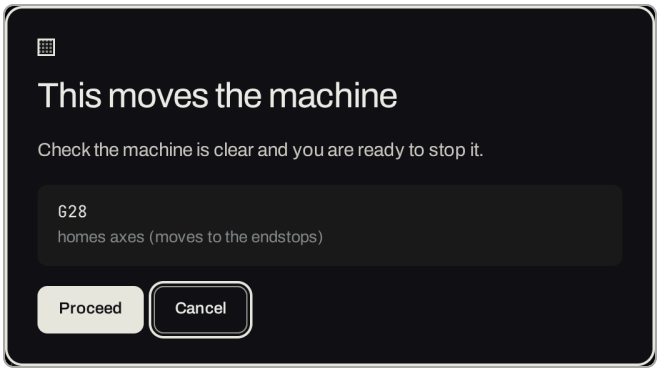
Console

Type a G-code command or macro and press Enter or **Send**; ↑ brings back earlier commands. Klipper's replies appear underneath, newest at the top, together with lines from before you connected.



The console, newest line at the top

A command that moves the machine, switches tool power, changes the mounted tool or restarts Klipper asks first, with what it will do:

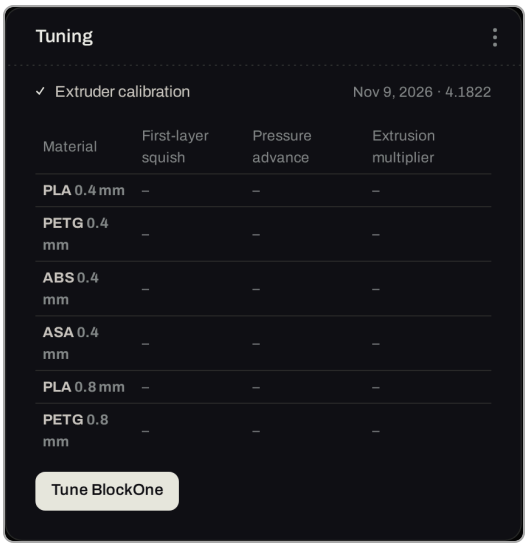


G28 asks first: it homes the axes

Tuning a print head

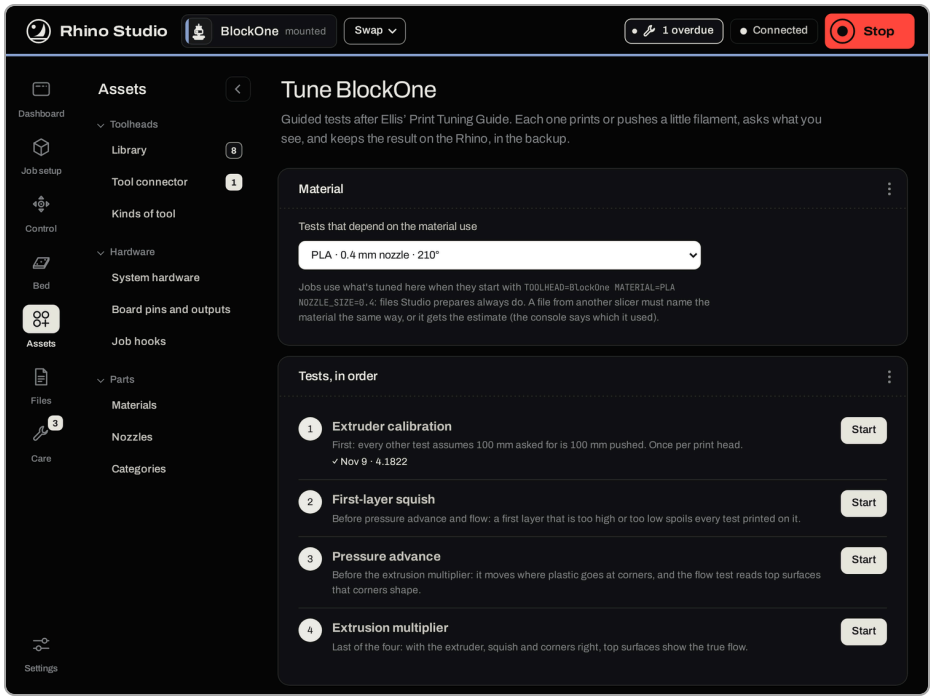
Rhino Studio has four guided tests for a print head, after Ellis' Print Tuning Guide. Each one pushes or prints a little filament, tells you what to look for, and keeps the result on the Rhino, where it is part of the backup. Klipper then applies the results at the start of every job, whichever slicer made the file.

Open **Assets › Toolheads**, select the print head and find its **Tuning** panel. It shows each test with a tick and the date it was last run, for each of the head's materials and nozzles. Press **Tune BlockOne** (the button carries the head's name) to open the tests.



The Tuning panel on a print head's page: the last run of each test, per material and nozzle

The **Tune** page has two panels. **Material** picks the material and nozzle for the tests that depend on them, and shows the line a job file must start with to get these results: `START_JOB TOOLHEAD=BlockOne MATERIAL=PLA NOZZLE_SIZE=0.4`. Files Rhino Studio prepares always have it. **Tests, in order** lists the four tests; press **Start** beside one to run it on this page.



The Tune page: the material to test, and the four tests in order

Test	How often	What it sets	Kept on the Rhino as
Extruder calibration	Once per print head, and after any change to its extruder	Rotation distance	`rhino_rd_<head>`
First-layer squish	Per head and material	Nozzle height over the paper test	`rhino_zfine_<head>_<material>`
Pressure advance	Per head, material and nozzle	Pressure advance	`rhino_pa_<head>_<material>_<nozzle>`
Extrusion multiplier	Per head, material and nozzle	Flow (`M221`)	`rhino_em_<head>_<material>_<nozzle>`

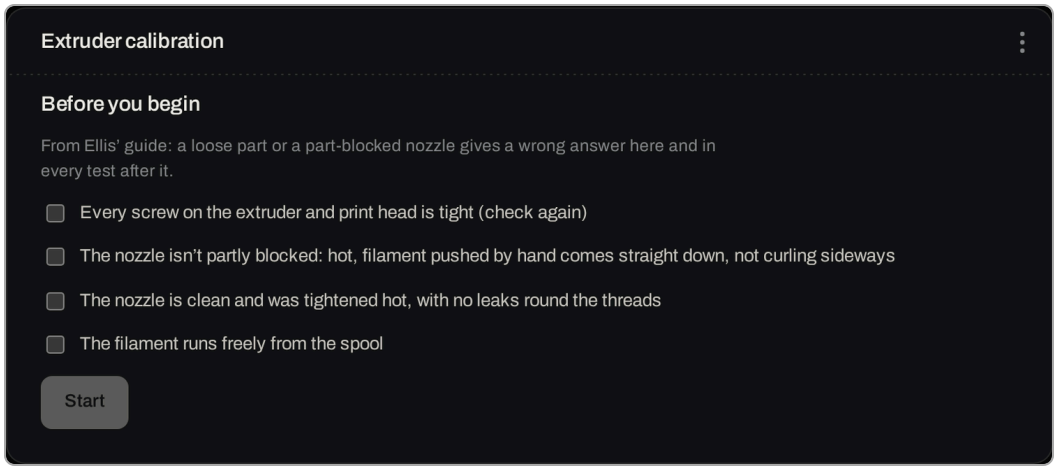
NOTE

Run the tests in the order listed: each one assumes the ones before it are right, and a later test reminds you if an earlier one hasn't been run. Every test needs Klipper ready and the head mounted and confirmed. Run pressure advance and the extrusion multiplier again after changing the nozzle, the hotend or the brand of filament.

Extruder calibration

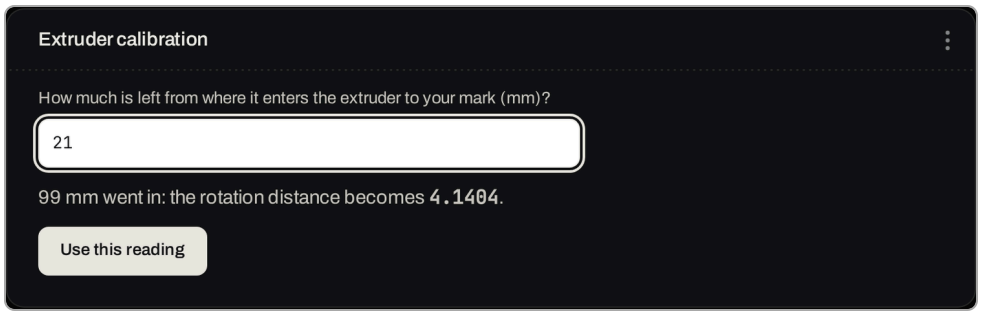
Checks that when Klipper asks for 100 mm of filament, 100 mm goes in. You need a ruler, a fine marker or a strip of tape, and about five minutes.

1. Press **Start** beside **Extruder calibration**. Tick each check under **Before you begin** once it's true, then press **Start**. A loose part or a part-blocked nozzle gives a wrong answer here and in every test after it.



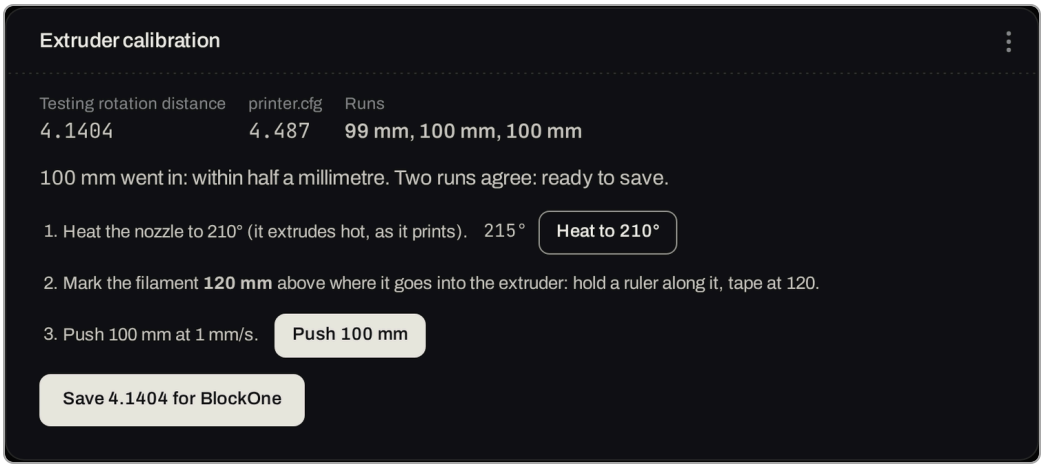
Extruder calibration: the checks before you begin

2. Press **Heat to 210°** (the material's printing temperature). The filament is pushed hot, as it is when printing.
3. Mark the filament **120 mm** above where it goes into the extruder: hold a ruler along it and put the tape or mark at 120.
4. Once the nozzle is hot, press **Push 100 mm**. Studio shows the commands and asks first. The filament goes in at 1 mm/s as two 50 mm moves, Klipper's limit for one move, and takes about 100 seconds.
5. Measure from where the filament enters the extruder to your mark, type it in and press **Use this reading**. Studio works out how much went in and the rotation distance that would have pushed exactly 100 mm.



The reading: 21 mm left means 99 mm went in

6. Mark the filament again and push again. The next push uses the new value. When two runs in a row agree within half a millimetre, Studio says **ready to save**.
7. Press **Save … for BlockOne**. The panel shows the new value beside the old one.

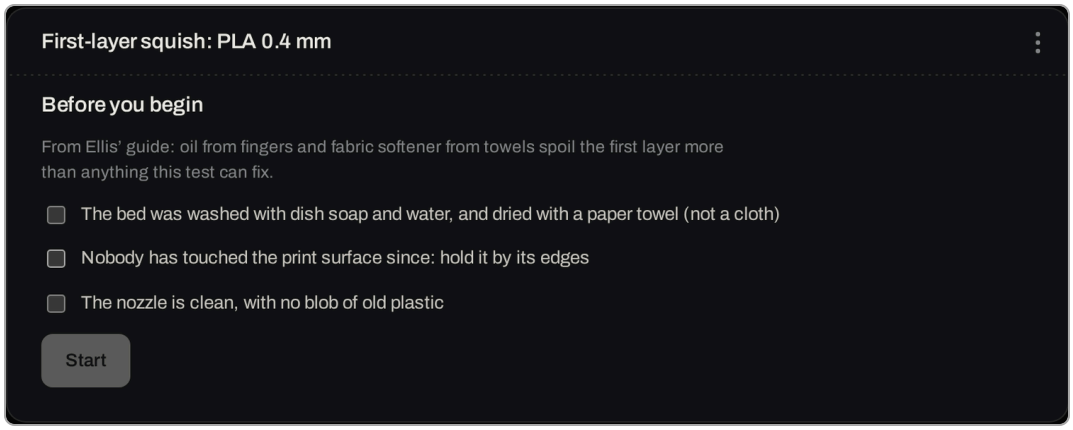


Three runs of 100 mm: ready to save

First-layer squish

Sets how far the nozzle is from the bed on the first layer, for this head and material. It fine-tunes the head's paper test, so run `NOZZLE_HEIGHT_CALIBRATE` from **Control › Macros** first if the head hasn't had one. Studio says so if it's missing.

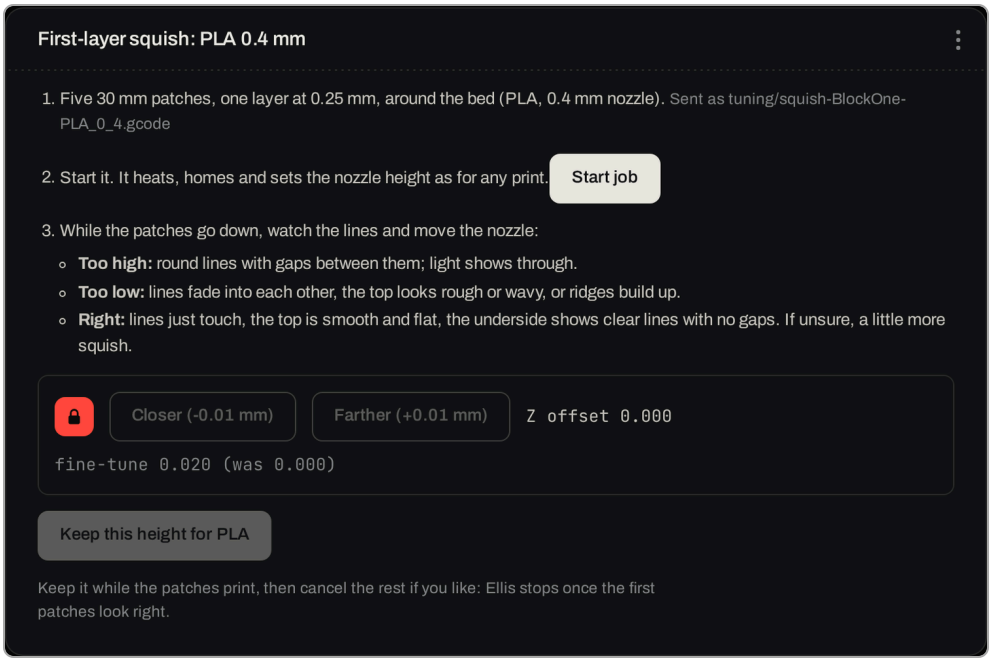
1. Wash the bed with dish soap and water and dry it with a paper towel, not a cloth. Oil from fingers and fabric softener from towels spoil a first layer more than this test can fix. Tick the checks and press **Start**.



First-layer squish: the checks before you begin

2. Press **Make the test**. Studio writes a small job, five 30 mm patches of one 0.25 mm layer spread around the bed, into the `tuning` folder on the Rhino. Press **Start job**: it heats, homes and sets the nozzle height as for any print.
3. While the patches go down, unlock the padlock (the panel turns red) and press **Closer** or **Farther**. Each press moves the nozzle 0.01 mm. Look at each new patch:
   - **Too high:** round lines with gaps between them; light shows through.
   - **Too low:** lines fade into each other, the top looks rough or wavy, or ridges build up.
   - **Right:** lines just touch, the top is smooth and flat, and the underside shows clear lines with no gaps. If you're unsure, a little more squish.

4. When a patch looks right, press **Keep this height for PLA** while the job is still printing. You can then cancel the rest of the print with **Cancel the test print**.



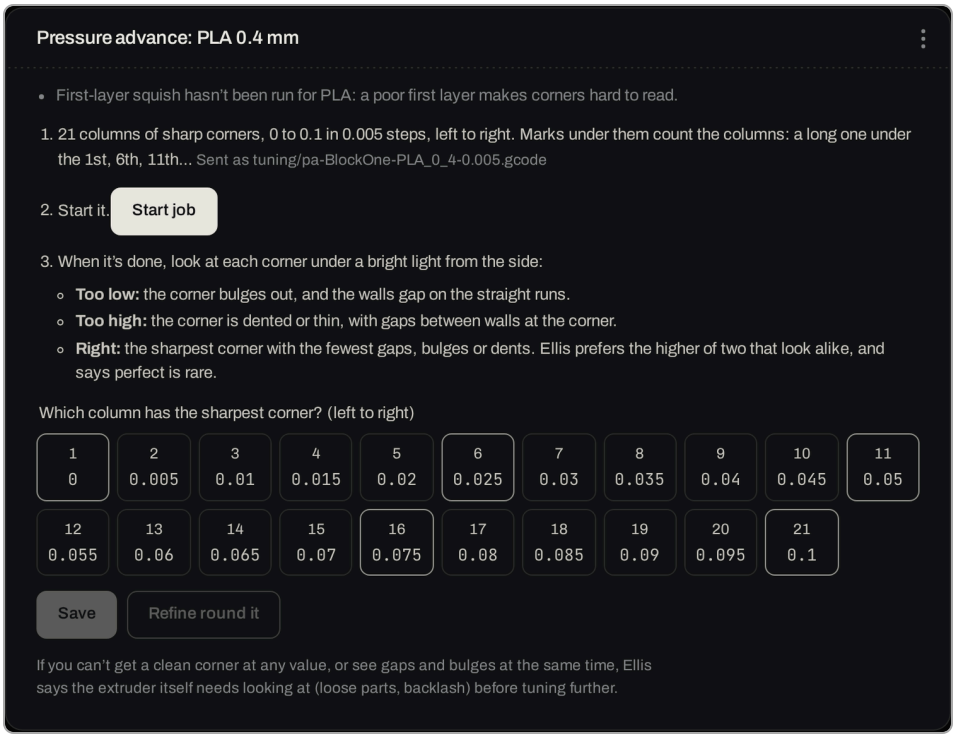
Squish: Closer and Farther while the patches print

What is kept is the difference from the paper test, up to 1 mm either way. At every job start, Klipper adds it to the head's paper-test height for that material.

Pressure advance

Pressure advance eases the extruder off before a corner and back on after it, so corners stay sharp.

1. Press **Make the test**, then **Start job**. The pattern is 21 columns of sharp corners, with pressure advance from 0 to 0.1 in steps of 0.005, left to right. Marks under the columns help you count: a long mark under the 1st, 6th, 11th, 16th and 21st.
2. When it's done, look at each corner under a bright light from the side.
   - **Too low:** the corner bulges out, and the walls gap on the straight runs.
   - **Too high:** the corner is dented or thin, with gaps between the walls at the corner.
   - **Right:** the sharpest corner with the fewest gaps, bulges or dents. If two look alike, take the higher one.
3. Select that column's number. Press **Save**, or **Refine round ...** to print 11 columns 0.002 apart around your pick, and pick again.



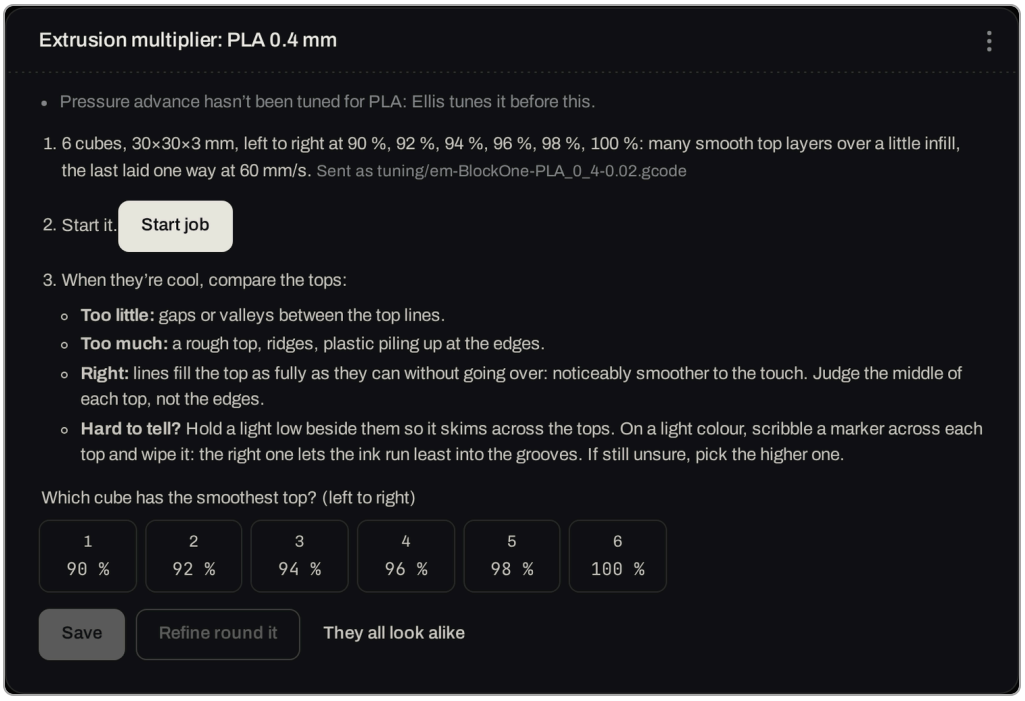
Pressure advance: pick the column with the sharpest corner

> **TIP**
> If no column gives a clean corner, or you see gaps and bulges in the same corner, check the extruder itself for loose parts and backlash before tuning further.

Extrusion multiplier

Sets the flow, so top surfaces fill fully without piling up.

1. Press **Make the test**, then **Start job**. It prints six 30 × 30 × 3 mm cubes. From left to right they print at 90, 92, 94, 96, 98 and 100 % flow.
2. When they're cool, compare the tops. Judge the middle of each top, not the edges.
   - **Too little:** gaps or valleys between the top lines.
   - **Too much:** a rough top, ridges, plastic piling up at the edges.
   - **Right:** the lines fill the top as fully as they can without going over. It's noticeably smoother to the touch.
3. Select that cube's number. Press **Save**, or **Refine round ...** to print finer steps (0.5 %) around it. If all six look alike, press **They all look alike** for a wider spread, 86 to 106 %.



Extrusion multiplier: pick the cube with the smoothest top

> **TIP**
> **Hard to tell?** Hold a light low beside the cubes so it skims across the tops. On a light colour, scribble a marker across each top and wipe it off: the right one lets the least ink run into the grooves. If you're still unsure, pick the higher one.

What the Rhino does with the results

Every job starts with `START_JOB`, which sets up the extruder from scratch for the head, material and nozzle in the job file. Nothing carries over from the job before.

Setting	When a result is kept	When there isn't one
Rotation distance	The head's result	`printer.cfg`'s value
Nozzle height	Paper test plus the material's squish	The paper test alone
Pressure advance	The material and nozzle's result	An estimate from `CALCULATE_PA`
Flow	The material and nozzle's result	100 %

The console shows one line per job saying which values it used and where each came from: *tuned*, or *printer.cfg*. `CALCULATE_PA` runs just after `START_JOB` in every slicer profile. When the material has a tuned pressure advance it keeps that one and says *(tuned)*; otherwise it estimates one from the Bowden length and the material.

> **⚠ WARNING**
> **The material name must match.** A file from another slicer must call the material by the Rhino's name, such as `PLA` and not `PLA+`. If the slicer's name differs from the job's, a red console line says so and the job's material is used. If no head is mapped to the tool, `CALCULATE_PA` says so in red and leaves pressure advance as it is. A kept value outside its sane range (rotation distance 0.1–100, pressure advance 0–1, flow 50–150 %) is ignored and reported, and `printer.cfg` 's value is used. Tune it again if you see that message.

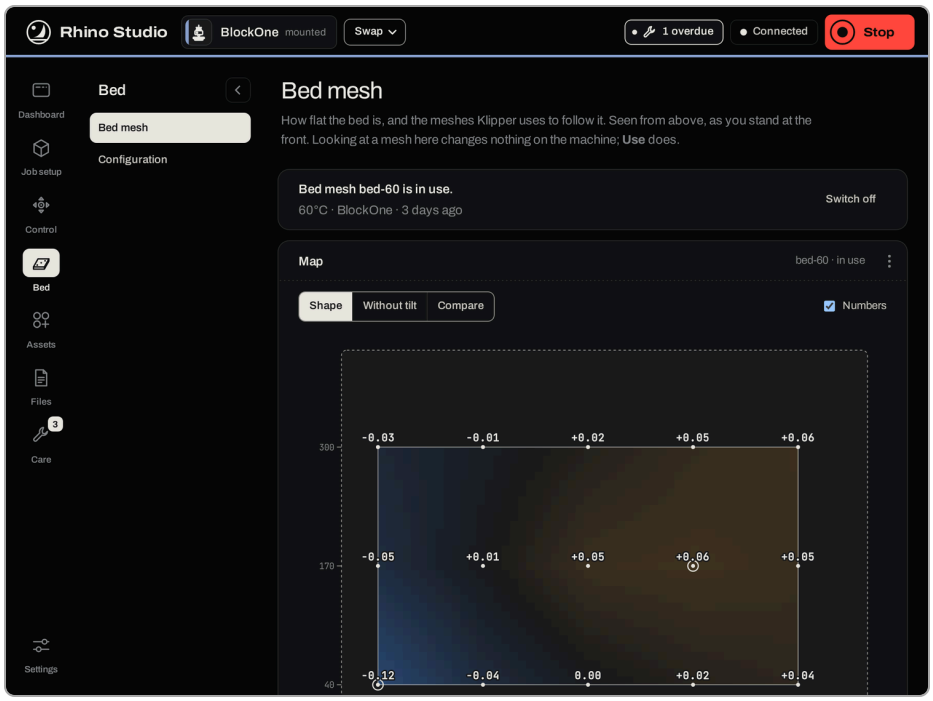
Results saved before Tool Manager 1.9.17 were kept only in Studio. The **Not on the Rhino yet** panel on the Tune page lists them; press **Store on the Rhino** so Klipper uses them too.

Bed

Bed on Rhino Studio's left rail has two pages, chosen from its side menu. **Bed mesh** shows how flat the bed is and the meshes Klipper follows. **Configuration** moves single Z lead screws when the bed is too far off level for Z-tilt to fix.

Bed mesh

The page shows the bed from above, as you stand at the front. The top line says which mesh Klipper is using, with its bed temperature, the head that probed it and when; **Switch off** stops moves from following it. Looking at a mesh here changes nothing on the machine. Only **Use** does.



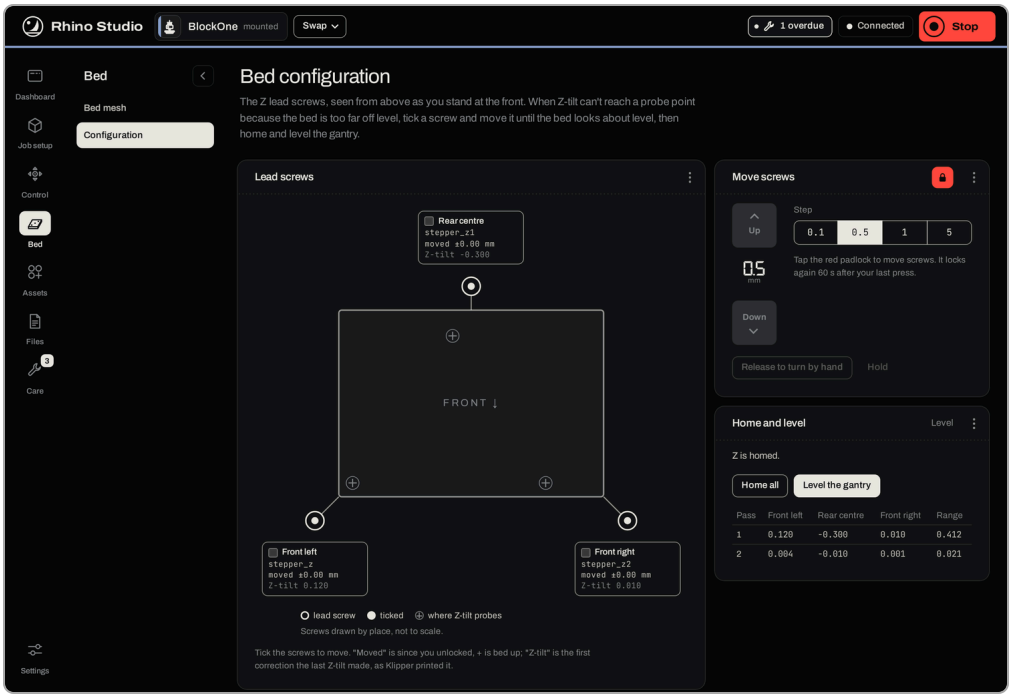
Bed mesh: the mesh in use, seen from above

- **Map**: **Shape** shows the heights as probed; blue is lower, amber higher, and the numbers are in millimetres. **Without tilt** takes out the slope that levelling the gantry would remove, leaving the warp of the plate itself. **Compare** shows this mesh minus another one you choose, such as the same bed probed cold and hot. Untick **Numbers** for a plain map.
- **Reading**: **Range**, between the highest and lowest point, and how much of a first layer that is; **Tilt**, the slope to the right and to the back; and **Warp**, what is left with the tilt taken out.
- **Meshes**: the meshes Klipper holds, with **Use** and **Remove**. New and removed meshes last only until Klipper restarts; **Save to printer.cfg** keeps them, and restarts Klipper.
- **Make a mesh**: the bed temperature, the time at temperature before probing, and whether to level the gantry first (`Z_TILT_ADJUST`).

> **NOTE**
> While a job runs, the page only shows meshes: a print loads its own mesh at the start.

Configuration: moving one lead screw

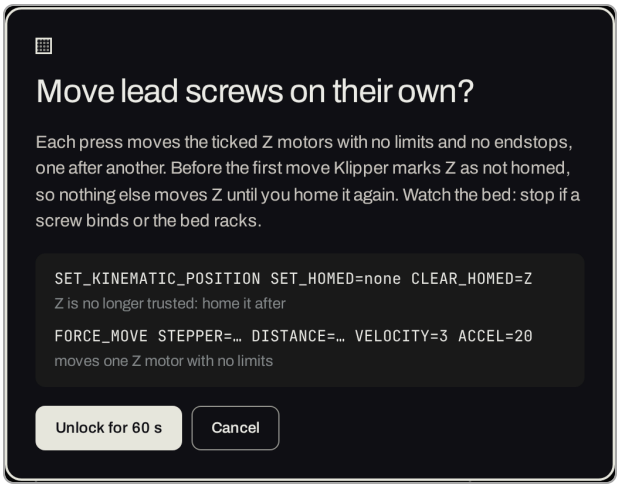
The Rhino's bed rides on three Z lead screws, each with its own motor. `Z_TILT_ADJUST` (**Level the gantry**) probes the bed and turns each screw until the bed is level. If the bed is too far off to start with, Z-tilt stops with *Move out of range*. This page lets you bring the bed close by eye first, by moving the screws you choose. Nothing on this page is saved.



Bed › Configuration: the lead screws, the controls to move them, and Home and level

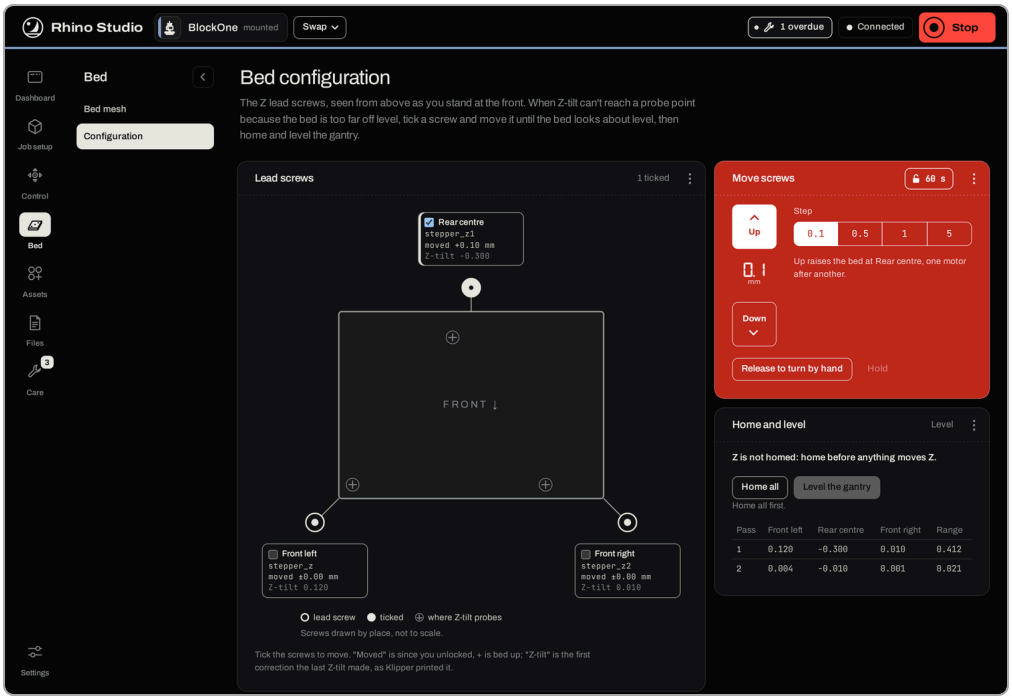
The **Lead screws** panel shows the bed from above with each lead screw just off its corner or edge: front left, rear centre and front right on the Rhino. The screws are drawn by place, not to scale. The crosses are where Z-tilt probes. Each screw's card shows its name, its motor (`stepper_z`, `stepper_z1` …), how far it has moved since you unlocked, and what the last Z-tilt asked of it. Tick a screw's box to choose it.

1. Tick the screw or screws to move.
2. In **Move screws**, press the padlock. Studio asks once and shows what will run.



Unlocking asks once, and says Z will be marked not homed

3. Press **Unlock for 60 s**. The panel turns red.
4. Choose a step: 0.1, 0.5, 1 or 5 mm. **Up** raises the bed at the ticked screws and **Down** lowers it, one motor after another. Each card shows how far its screw has moved (+ is up).
5. Repeat until the bed looks level by eye. The padlock locks again 60 seconds after your last press, and when you leave the page, a job starts, or the tool is swapped or changed.

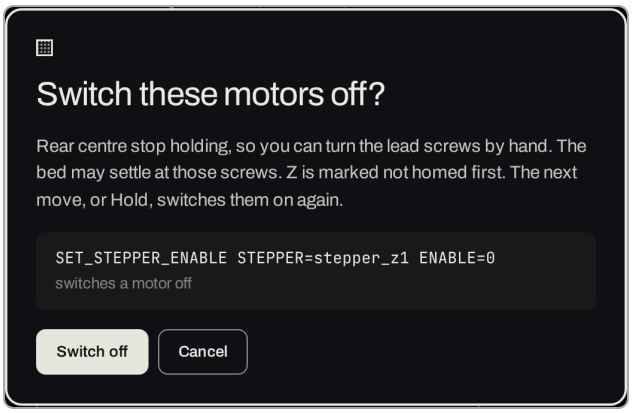


The rear centre screw moved up 0.1 mm, with the panel unlocked

> **⚠ WARNING**
> These moves have no limits and no endstops. Watch the bed while you press, and if a screw binds or the bed racks, press **Stop**. The first move marks Z as not homed and drops the saved park height, so nothing else moves Z until you home again.

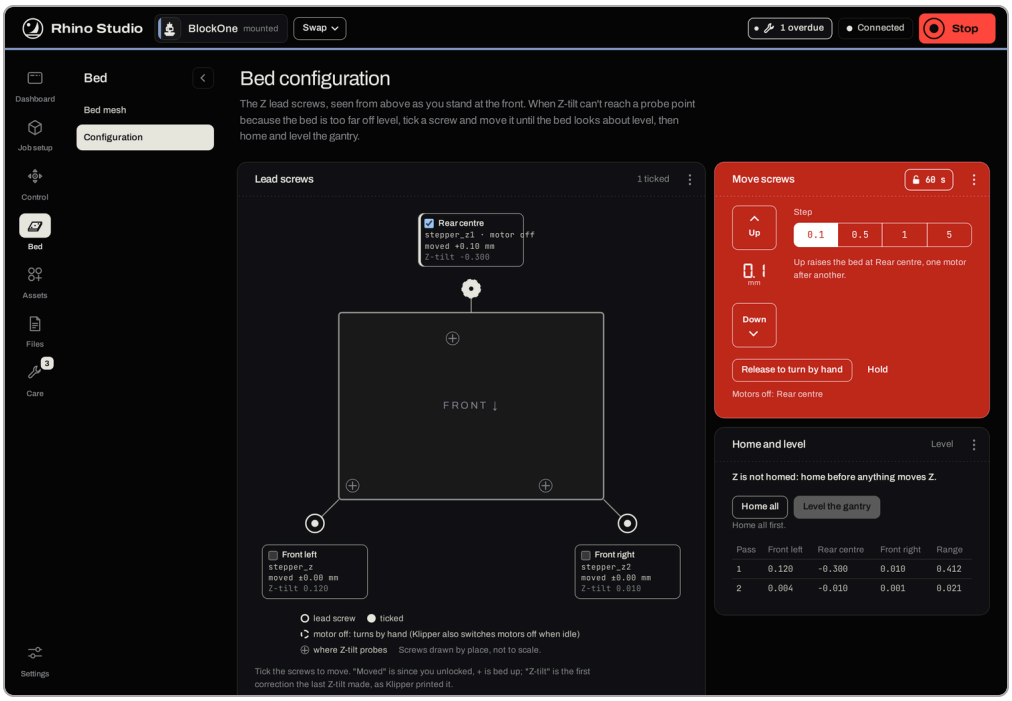
Turning a screw by hand

Release to turn by hand switches the ticked motors off, after asking, so you can turn their lead screws by hand. The bed may settle at those screws. **Hold**, or the next move, switches them back on.



Releasing asks first and shows the command

A screw whose motor is off is drawn with a dashed ring, and its card says **motor off**. Klipper also switches every motor off after the machine has been idle for a while, so **motor off** on all three screws is normal after a pause.

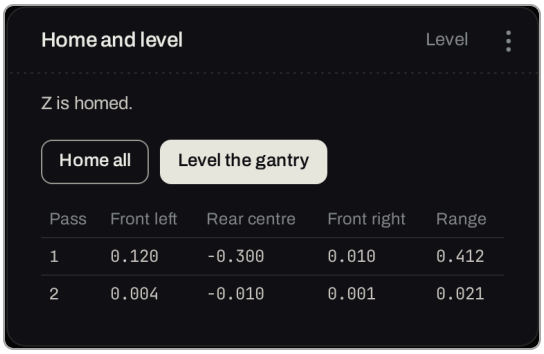


The rear centre motor switched off: a dashed ring and 'motor off'

Home and level

Home all homes every axis. **Level the gantry** runs `Z_TILT_ADJUST`. It needs a print head mounted and confirmed, because it probes with the head, and X, Y and Z homed. Both ask first.

The table shows each pass of the last Z-tilt: the correction each screw got and the range left across the probe points. Z-tilt repeats until the range is within its tolerance. If it stops with an error, the panel shows the error instead. Before any Z-tilt since the Rhino restarted, the panel reads **NO Z_TILT PERFORMED SINCE RESTART**.



Two passes of Z-tilt: 0.412 mm range, then 0.021 mm

When Z-tilt stops with "Move out of range"

1. On **Configuration**, look at the last Z-tilt's numbers on the cards: the screw with the largest correction is the one furthest off.
2. Tick that screw, unlock and move it until the bed looks level by eye, or release it and turn it by hand, then press **Hold**.
3. Press **Home all**, then **Level the gantry**. Repeat from step 1 if it stops again.
4. Once Z-tilt finishes, make a new bed mesh if the old one was taken before the bed was moved.

PART

05

Maintenance

Tracking preventive maintenance in the portal, and the procedures themselves.

Maintenance tracking	**116**
Task Library and Task Books	
Status codes	**130**
Maintenance procedures	**132**

Maintenance tracking

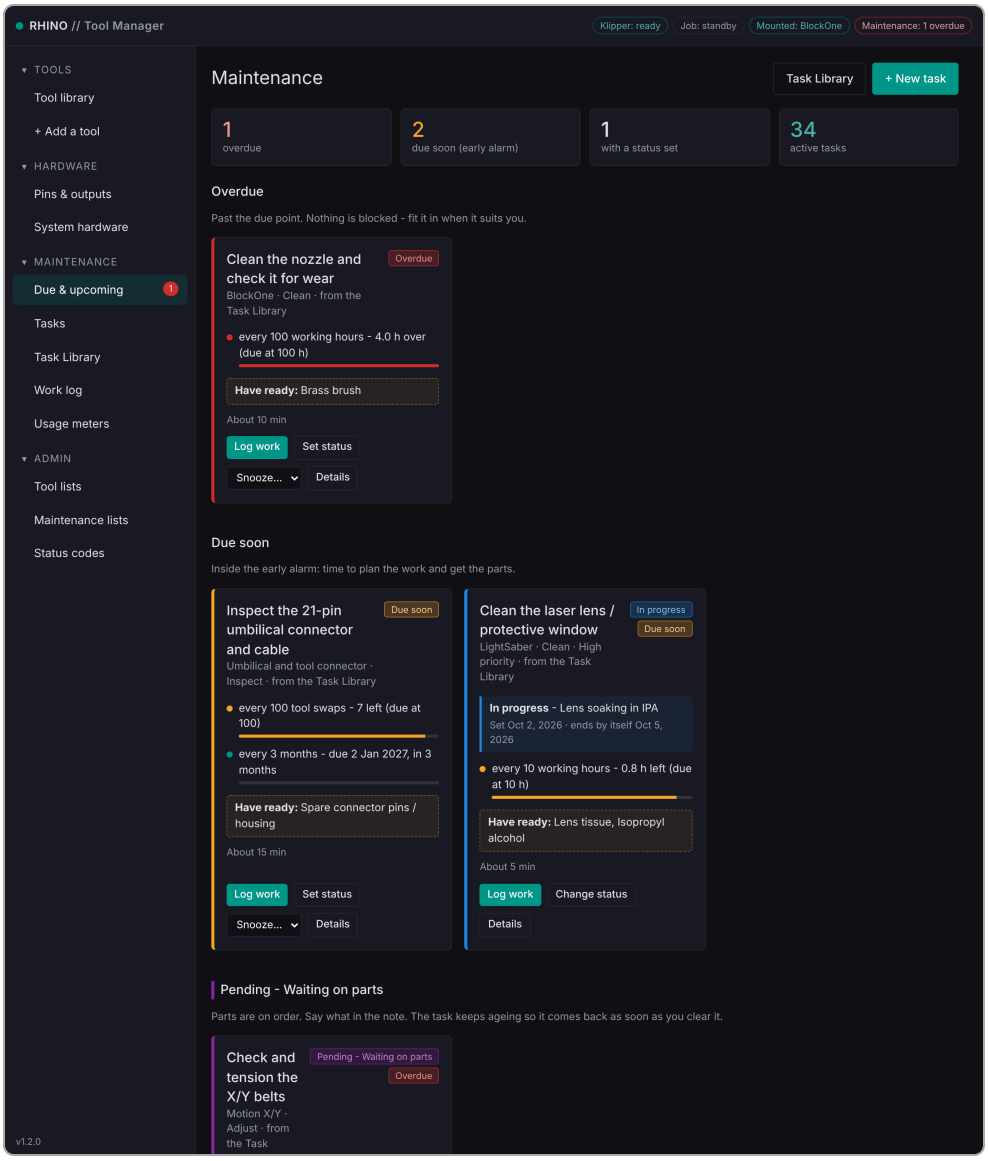
The portal keeps track of preventive maintenance for the machine, every tool and the system hardware. It tells you what is overdue, what is coming up and what parts to have ready. It never blocks a job, a swap or a command. Open it from **Maintenance** in the sidebar: **Due & upcoming**, **Tasks**, **Task Library**, **Work log** and **Usage meters**.

Getting started

Open the **Task Library** and assign a **Task Book** to the whole machine and to each tool: *Rhino motion system and frame* on the machine, *3D print head* on BlockOne and SwitchFly, *Laser module* on LightSaber, and so on. The intervals are starting points; tune them to how your machine wears (see *Task Library and Task Books*). When you add a tool, the confirmation offers the book that fits it.

Due & upcoming

Tasks are grouped by status. The tiles at the top count them. Each card shows every rule with a progress bar, how far off it is, and an estimate in days at your usual rate of use. Once the early alarm goes off, the card lists the parts to **Have ready**.

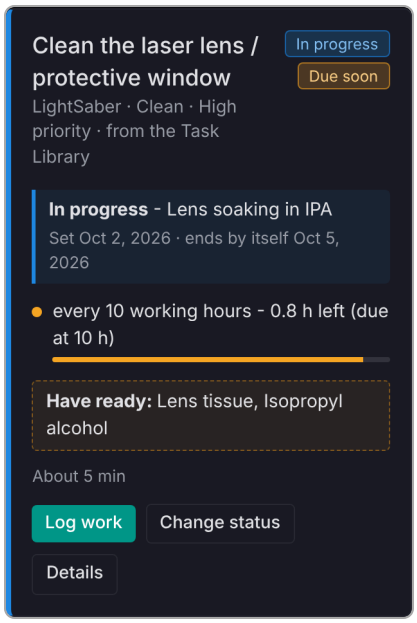


Due & upcoming

Group	Meaning
Overdue	Past the due point
Due soon	Inside the early alarm: time to plan the work and order parts
A status code's own group	Tasks set to a status shown in its own group, such as *Pending – Waiting on parts*
Snoozed	Put off on purpose
Coming up	The next eight tasks by how close they are to due, plus tasks set to a status shown with the upcoming ones

Log work records what happened. **Set status** puts a status code on the task. **Snooze** puts an overdue or due-soon task off for 1 day to 2 weeks. **Details** opens the task. The sidebar badge and a chip in the top bar show the count: red for overdue, amber for due soon.

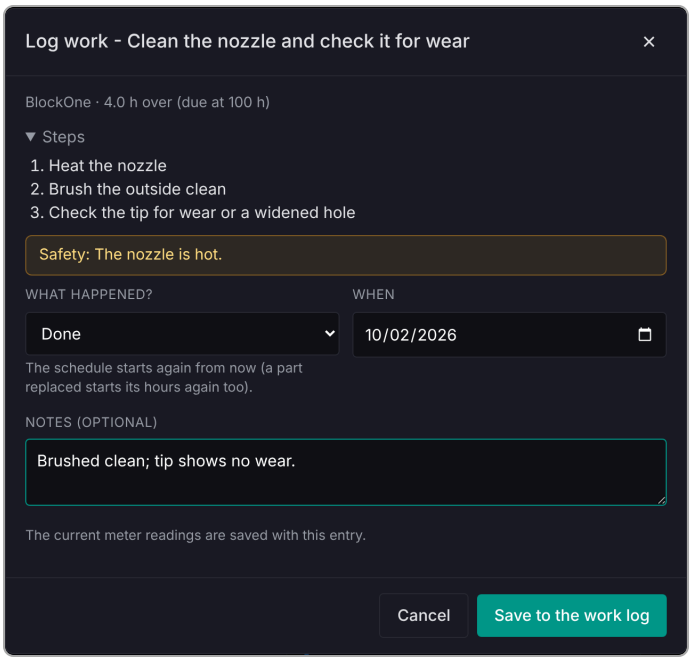
A card with a status code set shows it at the top in its colour, with the note, when it was set and when it ends by itself. Here *In progress* ends after three days; the task is still due soon underneath.



A task set to In progress, with its note

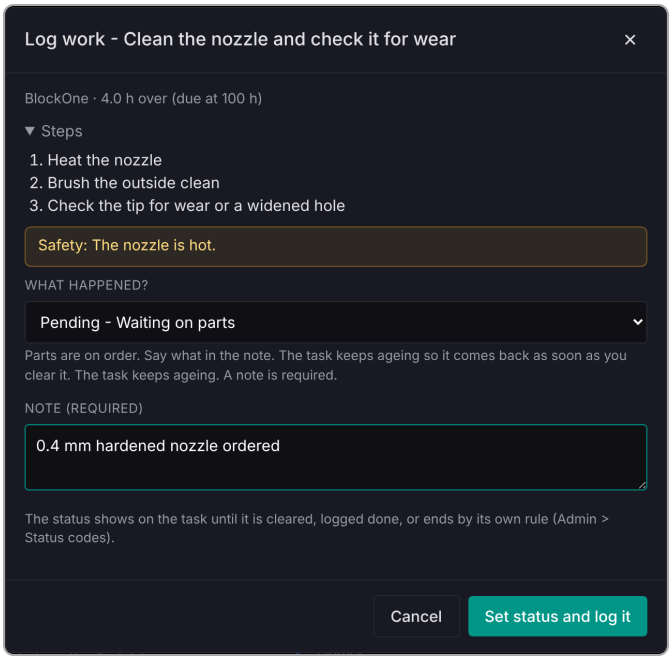
Logging work

Log work shows the task's steps and safety notes. **What happened?** is a list: **Done**; **Skipped – not needed this time** if you checked and nothing needed doing; or one of your status codes. Done and Skipped start the schedule again from here, and so do a part's hours, so a new nozzle starts at zero. You can backdate the entry and add notes; the current meter readings are saved with it.



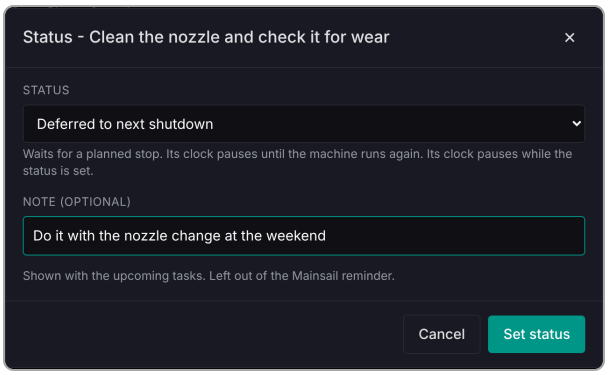
Logging a nozzle clean as done

Choosing a status code instead sets that status on the task and logs it. The help line says what the status does; *Pending – Waiting on parts* needs a note saying what is on order.



Logging a task as waiting on parts

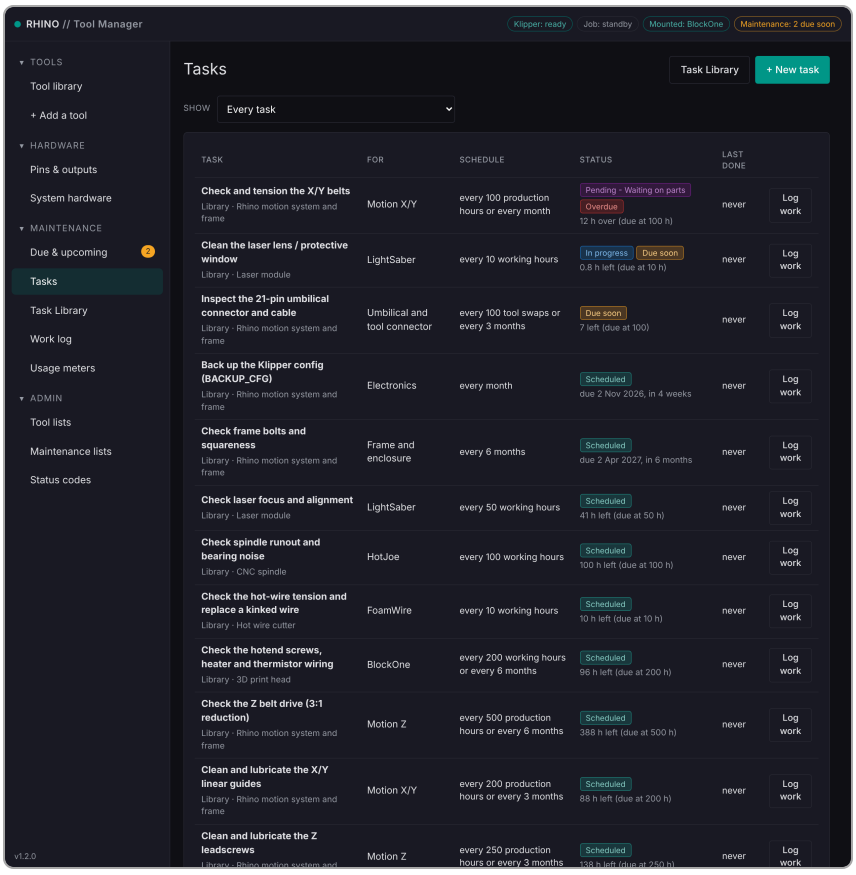
Set status on a card does the same without the steps, and **Clear the status** takes it off again.



Setting a status from the card

Tasks

Every task in one table, sorted overdue first, with its status code and status. Library tasks show *Library* and the Task Book they came from. **Show** filters it: overdue and due soon, with a status set, one machine area, one tool or piece of system hardware, or archived. Choosing an area or a tool also shows its usage meters.

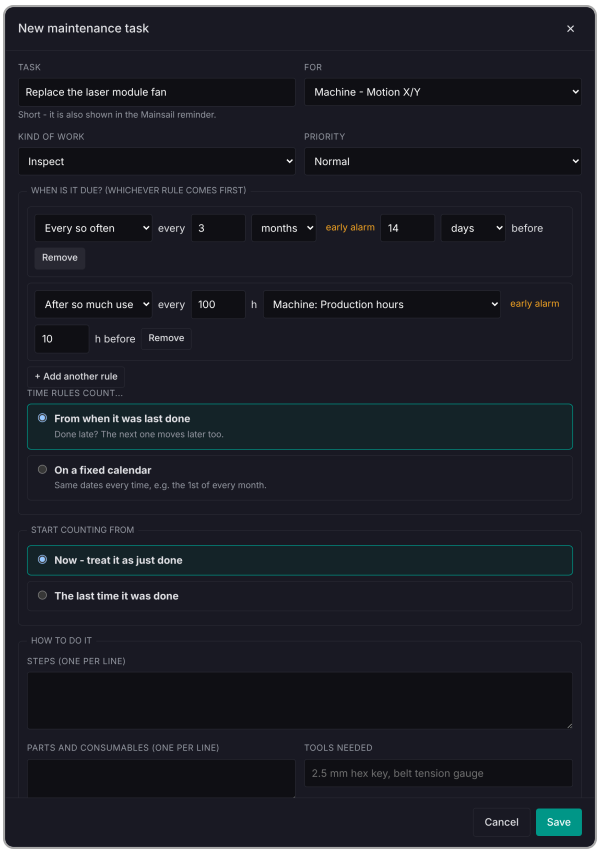


Tasks

Creating a task

+ New task opens the task editor:

- **Task** – a short title. It is also shown in the Mainsail reminder.
- **For** – a machine area, a tool or a piece of system hardware.
- **Kind of work** and **Priority** – used for sorting and the card's label (the lists are under **Admin › Maintenance lists**).
- **When is it due?** – up to four rules. The task is due when the first of them is reached.
- **Start counting from** (new tasks only) – now, or the date it was last done, with the meter readings at that time.
- **How to do it** – steps, parts and consumables, tools needed, time needed, safety notes and links.
- **Reminders** – include it in the start-up reminder, or pause the task.



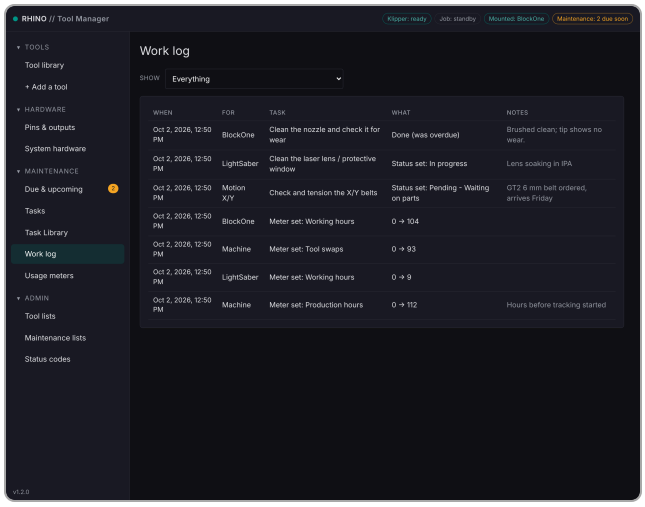
A new task with two rules

Rule	Example	Early alarm
Every so often	every 3 months	14 days before
After so much use	every 100 working hours of this tool	10 h before
After each event	after each emergency stop or Klipper shutdown	due straight away
Once, on a date	1 Dec 2026	7 days before

Time rules count from when the task was last done, so doing it late moves the next one later. Choose **On a fixed calendar** to keep the same dates every time, such as the 1st of every month. A usage rule's early alarm is in the meter's own unit; leave it empty for 10% of the interval. Events are a tool swap, this tool being mounted, a job, and an emergency stop or shutdown. **Add this task to the Task Library**, at the bottom of an existing task, turns it into a library task and links it.

Work log

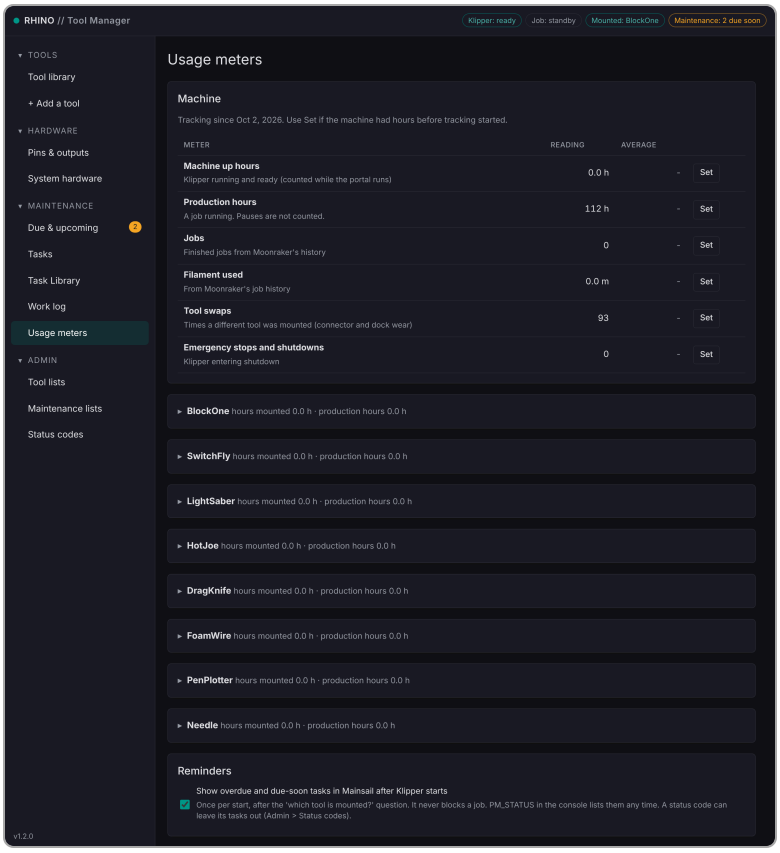
Every entry, newest first: work done or skipped, status codes set and cleared, and meter changes. Filter it by machine area, tool or system hardware.



Work log

Usage meters

Usage rules count these meters. The portal reads them from Moonraker every few seconds and from Moonraker's job history.



Usage meters

Meter	Machine	Each tool	Counts
Machine up hours	yes	–	Klipper running and ready
Production hours	yes	yes	A job running. Pauses are not counted.
Hours mounted	–	yes	The tool on the machine while Klipper is up
Working hours	–	yes	Print heads: nozzle heater on. LightSaber: laser on. HotJoe: spindle running. Added powered tools: output on. Passive tools and DragKnife: running a job.
Jobs	yes	yes	Finished jobs, from the job history
Filament used	yes	print heads	From the job history
Tool swaps / Times mounted	yes	yes	A different tool mounted. A restart is not a swap.
Emergency stops and shutdowns	yes	–	Klipper entering shutdown

The first time the portal runs, it imports the jobs already in Moonraker's history as the machine's starting reading. If the machine had hours before that, press **Set** on a meter and note why; the change goes in the work log. Jobs that ran while the portal was stopped are filled in from the history the next time it starts. System hardware tasks count the machine's meters. The page also has the start-up reminder switch.

Reminders in Mainsail

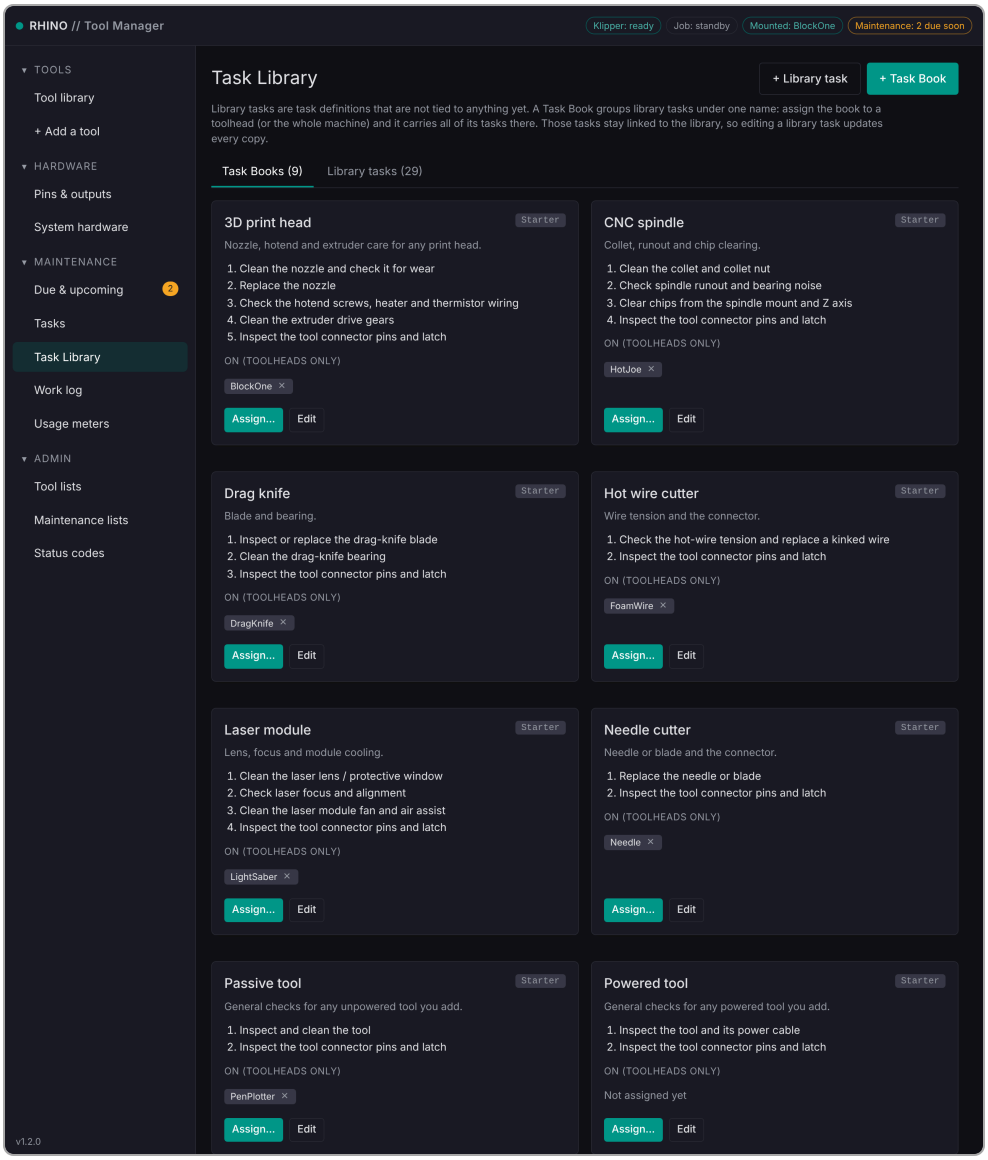
After Klipper starts, once you answer *which toolhead is mounted?*, Mainsail shows a **Maintenance due** pop-up with up to eight overdue and due-soon tasks, overdue first. It appears once per start; close it with **OK**. Snoozed tasks, tasks with the reminder unticked and tasks set to a status code that leaves them out of the reminder are not shown.

The portal keeps `myrhino/maintenance_status.cfg` up to date, and Klipper reads it when it starts. At any other time, type `PM_STATUS` in the console for the live list, or `PM_REMINDER` to show the start-up pop-up again. Renaming a tool moves its tasks and meters with it. Removing a tool or a piece of system hardware archives its tasks (filter **Archived**) and keeps their work-log entries.

Task Library and Task Books

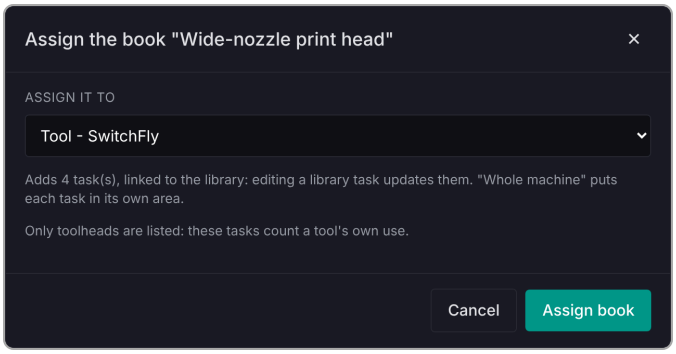
The **Task Library** holds task definitions that are not tied to anything yet – the starter tasks for the Rhino and each kind of tool, and any you add. A **Task Book** is a named group of library tasks. Assign a book to a toolhead, a piece of system hardware or the whole machine, and it carries all of its tasks there.

The **Task Books** tab shows each book, its tasks and where it is assigned. **Assign...** puts it on something more; the **×** on an assignment takes it off, which archives the tasks it added there (their work log is kept). Books whose tasks count a tool's own use, such as working hours, can only go on toolheads.



Task Books, with where each is assigned

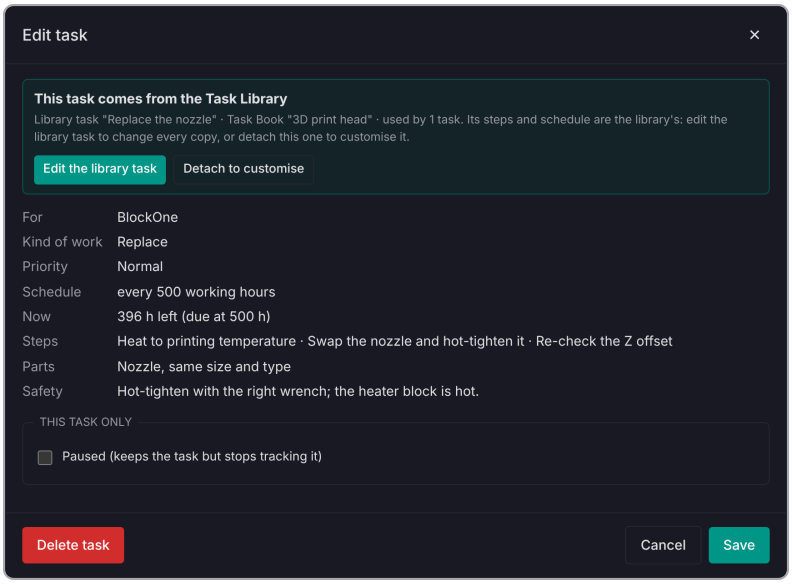
Assign... asks where. **Whole machine** puts each task in its own machine area (belts in Motion X/Y, leadscrews in Motion Z...). Assigning a book twice adds nothing twice.



Assigning a Task Book

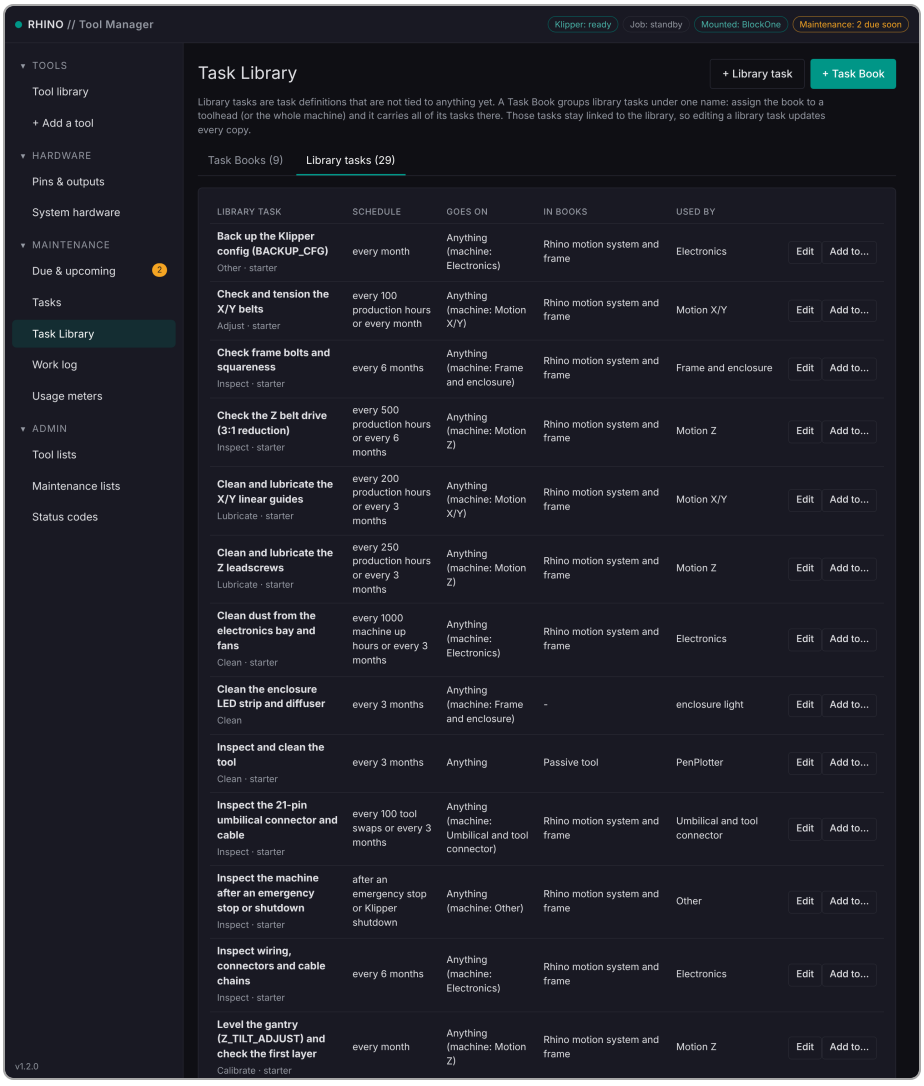
Linked tasks

Tasks that come from the library stay **linked** to it. Their steps and schedule are the library's, so **Details** on a linked task shows them read-only with two choices: **Edit the library task**, which changes every copy at once, or **Detach to customise**, which turns this one task into an ordinary task you can edit on its own. Pausing a linked task affects only that task. Each copy keeps its own schedule: when it was last done, its readings and its status.



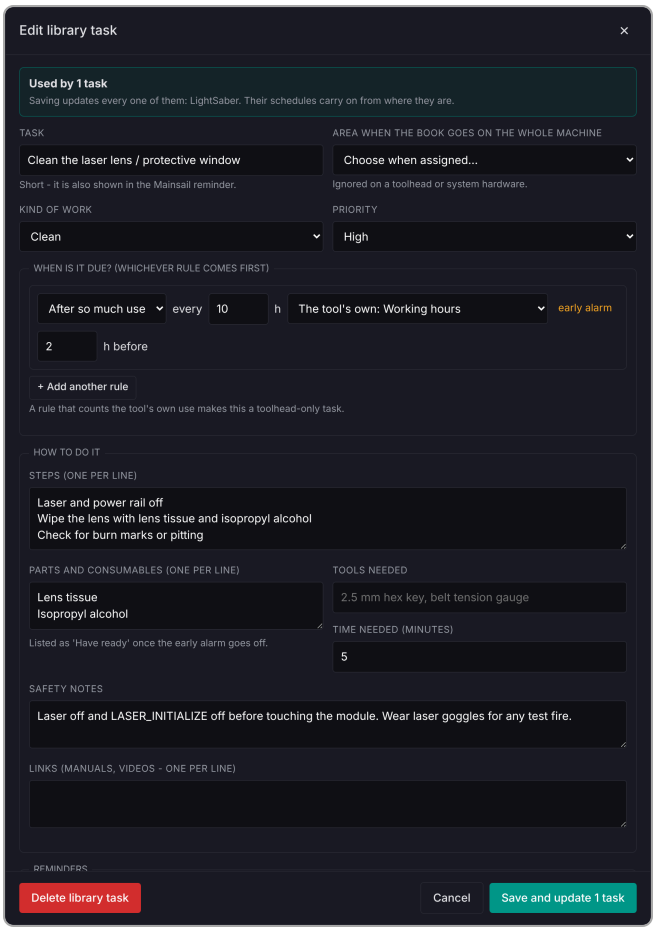
A linked task

The **Library tasks** tab lists every library task: its schedule, what it can go on, the books it is in and the tasks made from it. **Edit** opens the editor; saving says how many linked tasks it updates. **Add to…** puts one library task on something without a book. **+ Library task** adds your own; its **Area when the book goes on the whole machine** decides where it lands with **Whole machine**. Deleting a library task keeps the tasks made from it as ordinary tasks.



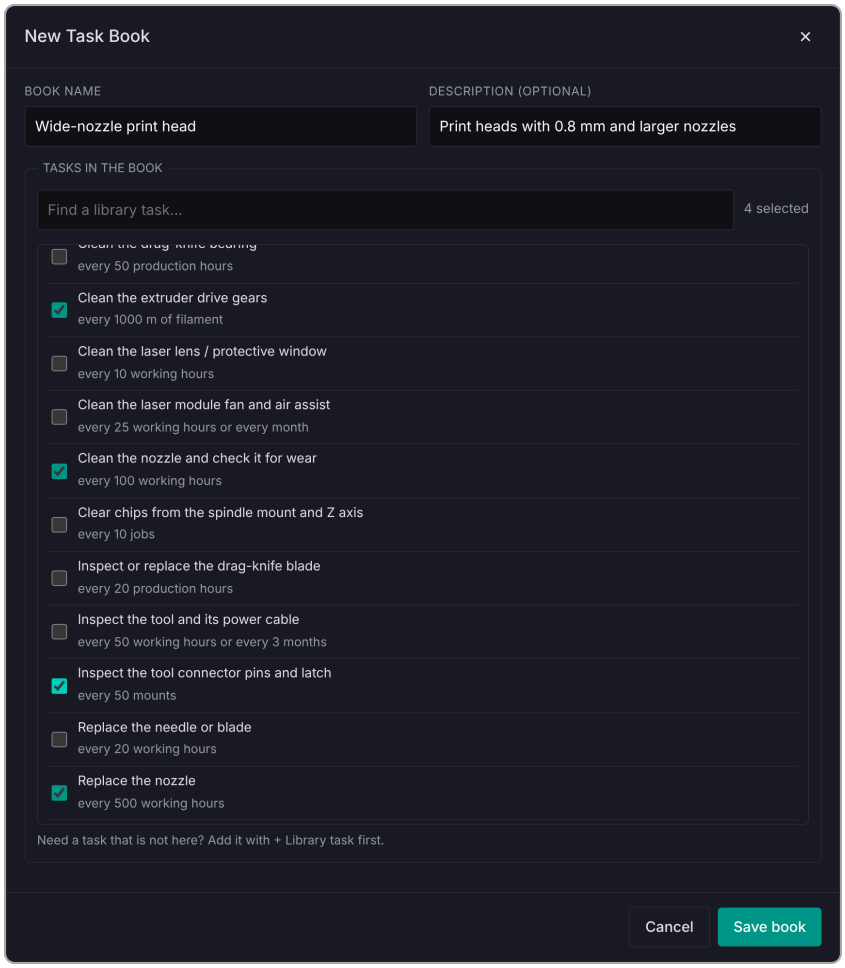
Library tasks

Editing a library task: the box at the top lists the tasks it will update.



Editing the laser-lens library task

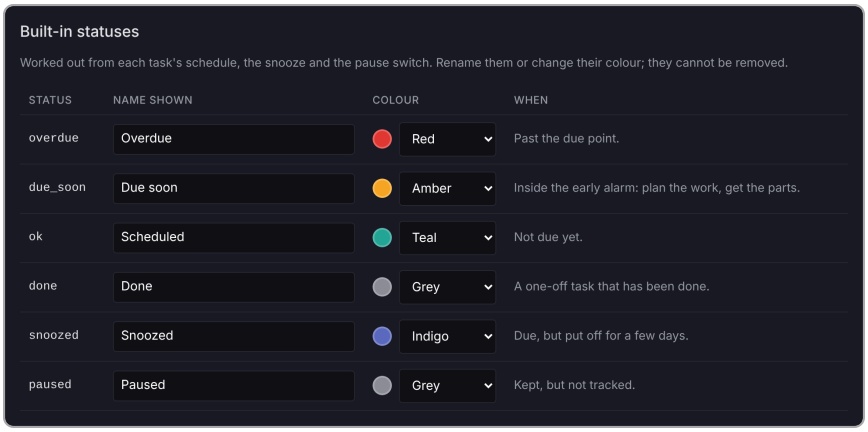
+ Task Book makes a book: a name, an optional description and the library tasks in it, found with the search box. Tasks you add to a book that is already assigned are added there too; tasks you take out are archived there.



A new Task Book for wide-nozzle print heads

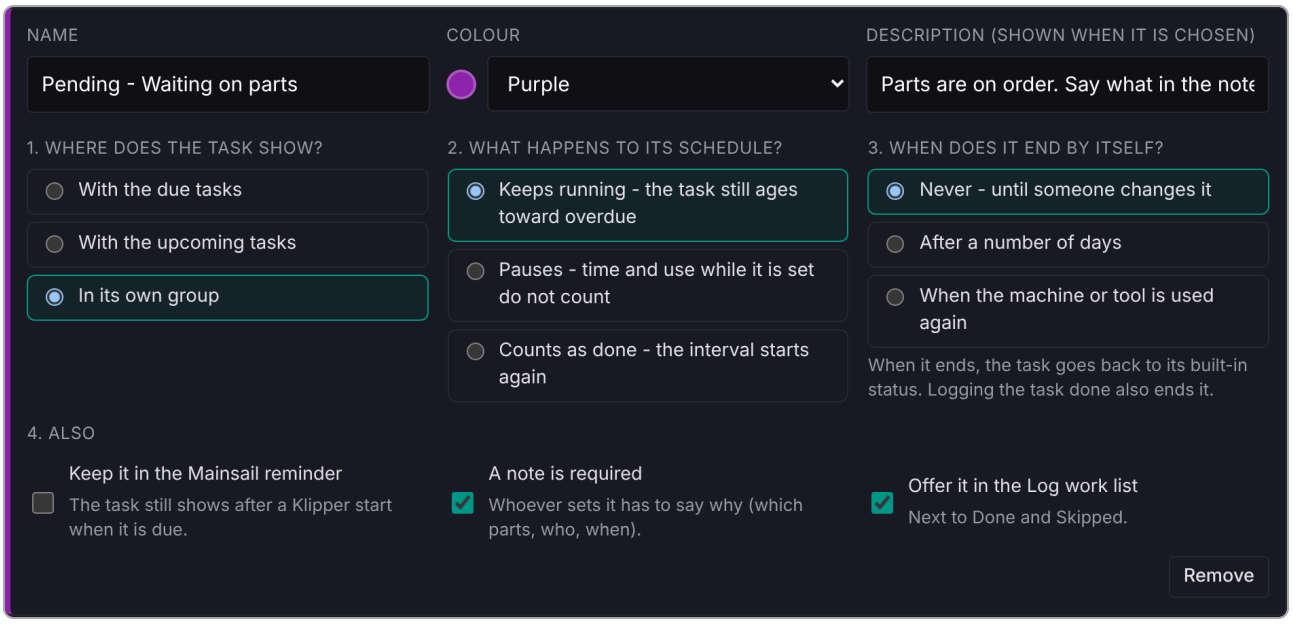
Status codes

Admin › Status codes lists two kinds of status. The **built-in statuses** – Overdue, Due soon, Scheduled, Done, Snoozed and Paused – are worked out from each task's schedule, the snooze and the pause switch. Rename them or change their colour; they cannot be removed.



Built-in statuses: name and colour

Your status codes are set by hand, from a task card or the Log work list: *Pending – Waiting on parts*, *In progress*, *Deferred to next shutdown*, or any you add with **+ Add a status code**. Each answers four questions, and the portal applies the answers as rules.



The rules of Pending – Waiting on parts

Question	Choices	What it does
1. Where does the task show?	With the due tasks · in its own group · with the upcoming tasks	Where Due & upcoming lists it. Only *with the due tasks* keeps it in the overdue and due-soon counts.
2. What happens to its schedule?	Keeps running · pauses · counts as done	*Keeps running*: the task still ages toward overdue. *Pauses*: time and use while the status is set do not count; the due point moves later by that much when it ends. *Counts as done*: the interval starts again when the status is set.
3. When does it end by itself?	Never · after a number of days · when the machine or tool is used again	When it ends, the task goes back to its built-in status and the end is logged. Logging the task done also ends it.
4. Also	Keep it in the Mainsail reminder · a note is required · offer it in the Log work list	Whether the start-up pop-up still shows the task when due; whether whoever sets it must say why; whether it appears next to Done and Skipped.

A status code that tasks are set to cannot be removed until they are cleared. The three codes the portal starts with:

Status code	Shows	Schedule	Ends
Pending – Waiting on parts	own group	keeps running	never (note required)
In progress	with the due tasks	keeps running	after 3 days
Deferred to next shutdown	with the upcoming tasks	pauses	when the machine is used again

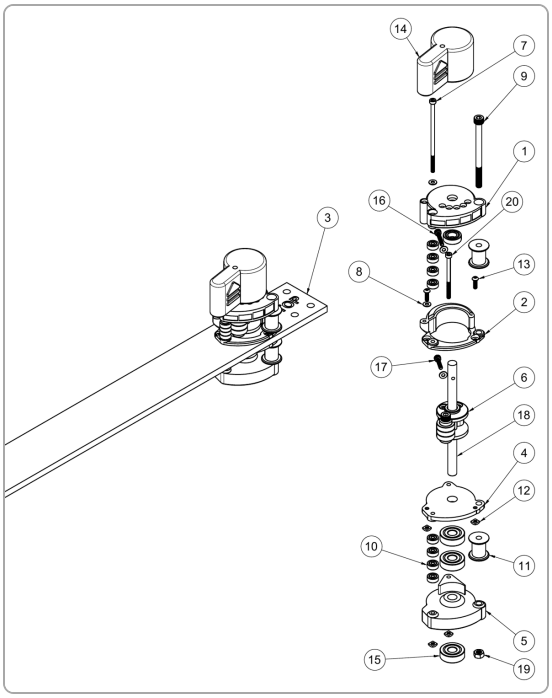
Maintenance procedures

Four items are scheduled in the build notes; no intervals are set yet, so record what you find and when.

Item	What to do
XY belts	Check for stretch and uneven tension; re-tension as below. Replacements: 15 mm GT2, glass-fibre core, 3 m per belt
Bearings and rails	Check the X rail and the bearings in the idlers and tensioners for roughness or play
Z leadscrews	Clean and re-lubricate all three leadscrews
Under the bed	Clear dust, chips and filament from the underside of the machine

Tensioning the XY belts

Each belt has a cam tensioner on the rear frame bar with an adjustment knob that indexes into one of five settings.



Belt tensioner assembly, exploded (item 14 is the adjustment knob)

1. Loosen both belt clamps on the X-carriage.
2. Turn both tensioners to their lowest setting.
3. Pull each belt through its carriage clamp until it is snug with no slack – not tight – while keeping the gantry square to both Y rails. The holes in the linear rails are a quick reference.
4. Tighten the belt clamps.

5. Turn both tensioner knobs to the same setting.

Tramming the bed

With a print head mounted, home with `G28`, then run `Z_TILT_ADJUST`. Klipper probes three points and levels the bed with its three Z motors, retrying up to five times until the points agree within 0.05 mm. Run it after any work on the leadscrews or the bed.

Calibrating the spindle ESC

Do this once when an ESC is new or replaced, or if spindle speeds look wrong. **Remove the bit from the collet first.**

1. Mount HotJoe and run `CALIBRATE_ESC`.
2. Press **Step 2: Send full throttle** (signal 1.0). This switches the spindle relay on with a full-throttle signal, so the ESC powers up and plays its start-up tones.
3. As soon as the tones finish, press **Step 4: Send low endpoint** (signal 0.3).
4. The ESC confirms with a tone. Its throttle range now matches Klipper's 0.3–1.0.

Notes

PART

06

Reference

When something goes wrong, where things live, and what is left to check.

Troubleshooting	**136**
Files, backups and tests	**138**
Build reference	**140**
Still to check on the machine	

Troubleshooting

Symptom	Likely cause	Fix
Mainsail's Resume says *extruder not hot enough*	A non-print tool's job is paused	Use **Resume job** in the Rhino pop-up, or `TOOL_RESUME`
Console: *resume it with TOOL_RESUME* after pressing Resume	Same – the extruder happened to be warm	`TOOL_RESUME`
Job refuses to start: *HARDWARE MISMATCH*	File sliced for a different tool than the one recorded	`START_TOOL_SWAP` to the right tool, or answer the restart pop-up correctly
Laser or spindle command: *CRITICAL SAFETY BLOCK*	Recorded tool is not LightSaber / HotJoe	Check the restart pop-up answer; run `RESTORE_TOOL`
New tool not listed in Mainsail / *Unknown command*	Klipper not restarted since adding it	**Restart Klipper** in the portal, or `RESTART_FOR_TOOLS`
Portal shows **Moonraker offline**	Moonraker stopped, or portal pointed at the wrong address	Restart Moonraker; the portal still edits tools but cannot restart Klipper
Portal: *Restarting now would kill it*	A job is printing or paused	Wait for the job to finish or cancel it
`REMOVE_TOOLHEAD` : *is the mounted tool*	You cannot remove the tool on the machine	Swap to another tool first
Tool commands: *blocked while a job is printing*	Registry changes are locked during jobs	Finish or cancel the job
Add wizard offers no free pins	All three free connector pins (PB1, PC0, PC1) are taken	Share an existing output, or list more umbilical pins in `myrhino/umbilical.json`
Swap wizard: *Toolhead not detected*	Umbilical not seated	Re-seat the 21-pin umbilical and press **Check again**
Added tool switched itself off	Watchdog: on longer than *Max time on* with no job	Raise *Max time on* in the portal, or run it from a job file
ABS/ASA bed cooldown stops early	Klipper's 600 s idle timeout	Add `[idle_timeout]` with `timeout: 1800`
`rhino_toolgen` or backup commands fail	Login is not `pi`	Fix the `/home/pi` paths (see *Install*)
Swap wizard stays on Cooling down	It waits until the hotend and the bed are both below 40 °C	Wait; after a hot print the bed is usually the slow one
Drag-knife file stops with M3 rejected	The CAM profile writes laser power commands	Use a drag-knife LaserWeb profile with Laser Intensity and Tool Off blank
Spindle speed range wrong, or ESC will not arm	ESC endpoints not calibrated to Klipper's 0.3–1.0 signal	Run CALIBRATE_ESC with the bit removed

Symptom	Likely cause	Fix
Control or macro refused: *already exists*	The name is used by another macro, a Klipper command or a config object (Klipper ignores case for macros)	Choose another name; the message says who has it
A pin is greyed out in the controls guide	`umbilical.json` marks it as a MOSFET output, which cannot read a signal or drive a servo	Use a logic pin, or correct the pin's type
Banner says *time left unknown*	The file has no slicer estimate and is under 2% done	Wait a moment; it estimates from progress after that
Meters do not move	The portal service is not running (meters are counted by it)	`sudo systemctl status rhino-portal`; missed jobs are filled in from the history when it starts
No maintenance pop-up after a restart	Nothing was due, the reminder is switched off, or the tool-restore question was not answered yet	Answer the question, or run `PM_STATUS`

Files, backups and tests

To change one thing, open the one file that owns it.

To change	Edit
Built-in tools, their materials and slots	`variables.cfg`
Positions: park, tool swap, safe height, Set Z zero limit, prime line	`myrhino/positions.cfg` - the one place; every macro reads it
Your added tools	The portal or console commands (stored in `myrhino/custom_tools.json`; `custom_tools.cfg` is generated – never edit it)
Starting values, caps, checklists, allowed characters	`rhino/presets.py`
Which pins reach the tool connector	`myrhino/umbilical.json` (copy `umbilical.example.json`)
`M3` / `M5`, power cap, watchdog, safe-off, pause/resume	`tools/tool_power.cfg`
What Pause / Resume / Cancel add to Mainsail's buttons	`_CLIENT_VARIABLE` and `_RHINO_CANCEL` in `client_macros.cfg`
Tool swap and restart prompt	`toolchanger.cfg`, `guided/tool_swap_wizard.cfg`
Job setup wizards	`guided/*_workflow.cfg`
Portal screens	`rhino/portal/static/portal.js`, `templates/index.html`, `portal.css`
A tool's controls and macros	The portal (Controls and Macros tabs). They are written into `custom_tools.cfg`; the rules are in `rhino/extras.py`
Maintenance tasks, work log, meters	The portal. Stored in `myrhino/maintenance/` (tasks.json, history.json, meters.json), so the GitHub backup keeps them
Starter task intervals	`rhino/maint/library.py` (only affects tasks added afterwards)
The start-up maintenance pop-up	`guided/maintenance.cfg`; `myrhino/maintenance_status.cfg` is generated - never edit it

Positions. Where the head parks, where it goes for a tool swap, how far the bed lowers before a swap or park, the lowest the Set Z zero window goes, and where the prime line starts and how long it is are all in `myrhino/positions.cfg`. Change a number, save, then `FIRMWARE_RESTART`; every macro (and Mainsail's own pause park) uses the new number. Type `RHINO_POSITIONS` in the console to list them - it also checks each one against the travel limits in `printer.cfg`, and the same check runs at every start with any problem shown in red. Updates never overwrite this file. The travel limits themselves, where Z homes and the probe offset stay in `printer.cfg` because Klipper needs them; the paper test goes to the probe's spot worked out from those.

Backups. `BACKUP_CFG MSG=<note>` pushes the config folder to your GitHub backup through klipper-backup. The tool registry keeps its last 10 versions in `myrhino/registry_backups/`. Photos live outside the config folder in `~/printer_data/rhino_data/images/`, so they are not in the GitHub backup.

Tests. `sh dev/run_all.sh` runs a static check of the whole config and 619 automated checks: the macros in a simulated Klipper (against the real Mainsail macros); motion tests that follow where the bed and head go and run every macro and button from seven machine states with each toolhead against five safety rules; the portal, tool controls and macros, maintenance and the Admin lists; every portal screen opened in a real browser, with an accessibility scan and keyboard checks; and the installer run on a fake Pi. It needs `pip install jinja2 flask` (and Playwright for the browser test), and runs on every change pushed to GitHub. The simulator knows the travel limits, not the shape of your tools or stock, so every release also has on-machine checks - the plan is in `docs/testing.md`, and the ones for the current version are in *Still to check on the machine*.

Trying the portal without the printer. `python3 dev/demo_server.py` runs the portal against a copy of the config and a pretend Moonraker. Every screen in this manual can be tried there without touching the machine.

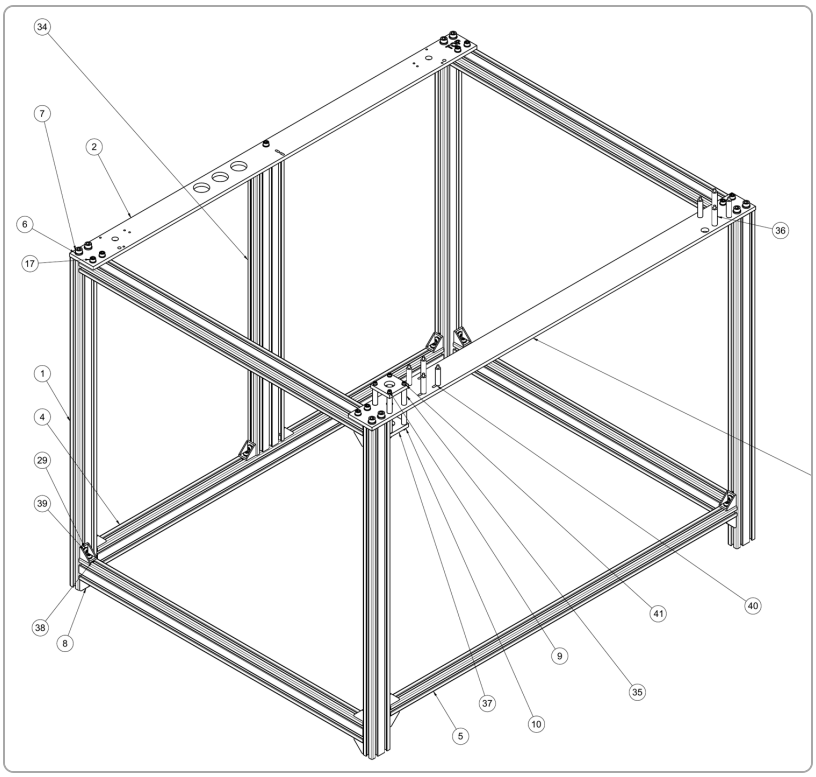
Build reference

The printable parts, CAD (STEP) files and assembly drawings are in the `Rhino-3d-Printer` repository. Print structural parts in ABS or ASA.

Repository folder	What is there
`frame`	Frame assembly drawing and STEP model; `frame/panels` – 3 mm ACP enclosure panels, doors and hinges, front spacer assembly
`motion/xy`	XY joiners, X-carriage and Back Bone, stepper mounts, rear centre idler, belt tensioners, gantry router-template guide
`motion/z`	Leadscrew carriages and Z stepper mounts
`bed`	Vacuum bed heat spreader
`toolheads`	BlockOne (build files to come), SwitchFly, LightSaber laser mount and height gauge, HotJoe spindle parts, DragKnife; Jack Rabbit kept as legacy
`electrical`	Controller, relays, umbilical and wiring diagrams
`filament-box`	“Cheeky Monkey” roll stand and runout switch (unfinished)
`docs`	Software, safety (planned smoke detection) and maintenance notes
`archive`	2023 Klipper configs for the old SKR GTR board, ESC, fan and heated-bed notes – superseded by this config package

Frame

The frame is 2020/2040/2060 aluminium extrusion with 2″ × ¼″ aluminium flat bar across the top; some components mount straight to the flat bar, using printed drilling templates for hole placement.



Frame assembly drawing

Item	Qty	Part
1	8	2040 extrusion, 24 in
2	1	Rear top frame flat bar, 2 in × 0.25 in aluminium
3	1	Front top frame flat bar, aluminium
4	1	2040 extrusion, 744 mm
5	1	2020 extrusion, 744 mm
6	8	M6 washer
7	8	M6 × 20 mm hex socket head screw
8	22	Cast corner bracket
9	4	M3 × 85 mm socket head screw
10	4	M3 hex nut
17	9	M5 × 12 mm screw
29	53	M5 washer
34	1	2060 extrusion, 550 mm

Item	Qty	Part
35	9	M3 × 30 mm spacer
36	8	M3 × 30 mm standoff
37	2	Motion system bracket, aluminium
38	44	M5 hammer nut
39	44	M5 × 10 mm button head hex drive bolt
40	8	M3 × 12 mm socket head screw
41	16	M3 washer

The frame notes recommend socket-head rather than button-head bolts wherever a bolt is torqued hard, because button heads strip more easily.

Still to check on the machine

The automated tests can't move the machine, so each release is checked on it. Work through these in order, with a hand on the emergency stop; stop at any test that doesn't pass and note its number. The full plan, with what counts as a pass, is in `docs/testing.md`.

Install and start

- [ ] **A1** Install 1.4.6 with menu option 3 – the preview shows no STOPPED, and the portal answers on port 5000.
- [ ] **A2** `FIRMWARE_RESTART` with a print head mounted – Klipper ready, no config errors.
- [ ] **A3** The *which toolhead is mounted?* question appears; `CHECK_TOOLHEADS` lists every tool.
- [ ] **A5** The portal opens on port 5000, shows 1.4.6, and follows Mainsail's dark/light switch within 15 s.

Motion and homing

- [ ] **B1** Jog X and Y 100 mm and measure – 100 mm ±0.5. About 50 mm means a step setting is wrong: stop and report it.
- [ ] **B2** Jog Z 10 mm and measure – 10 mm ±0.1 (gear_ratio 2:1). Then `Z_TILT_ADJUST`.
- [ ] **B5** Cancel a print – the bed lowers to Z150 or more, X/Y home, the head parks at X551 Y381 without touching the frame.
- [ ] **B5b** Cancel a laser job (new in 1.4.5) – laser and rail off, bed lowers, no X/Y homing, the head goes straight to X551 Y381. If X or Y starts homing, report it.
- [ ] **B6** Z travel (new in 1.4.3): home with BlockOne, `G1 Z290 F600` – the bed stays clear of its mounts – then `G1 Z300` is refused and nothing moves. If the bed touches a mount at Z290, stop and send the Z where it is 10 mm clear.

Homing guard and positions (new in 1.3.8 and 1.4.0)

- [ ] **A6** LightSaber mounted: Mainsail's Home All is refused with a message and nothing moves. Home X and Home Y work once the Y endstop is set for the LightSaber. BlockOne mounted: Home All homes as before.
- [ ] **C4** Restart with the bed parked, then swap: the console says homing X/Y lowered the bed 10 mm and the park height was updated; `SET_Z_ZERO` then shows the new height.
- [ ] **A7** `RHINO_POSITIONS` lists the positions and says they are inside the travel limits. Change `park_x` in `myrhino/positions.cfg` to 540, restart, cancel a print: the head parks at X540. Put it back.

Tool swaps

- [ ] **C1** Swap BlockOne to LightSaber – waits for 40 °C, lowers the bed, never homes Z.
- [ ] **C2** During a swap, unplug the umbilical – Klipper keeps running (the nozzle reads about −50 °C). Confirm while it is unplugged: *Toolhead not detected*; plug in, **Check again**. **C5** With LightSaber mounted, 200 in the nozzle temperature box is refused.
- [ ] **C3** Start a BlockOne file with the laser mounted – refused, nothing moves.

Printing

- [ ] **D1–D3** Paper test, then a test cube with the Orca profiles: the PLA question closes when answered, 210/65 °C, a 120 mm prime line, a slow first layer, the bed parked and the head at X551 Y381 at the end (1.4.4). The first layer is laid at the paper-test height (fixed in 1.3.8).

SwitchFly materials and the end of a job (new in 1.4.6)

- [ ] **D4** Start a SwitchFly file – it waits on *Path 2 (T1) material?* with path 1 from the file shown; nothing moves after homing.
- [ ] **D5** The confirmation lists both paths and the path it prints from; Proceed heats, primes and prints.
- [ ] **D6** Start it again – **Same as last time** is the first button, also after `FIRMWARE_RESTART`.
- [ ] **D7** Cancel at the path question – the job shows cancelled, nothing heats.
- [ ] **D8** `SWITCH_T1` by hand loads and purges with path 2's material (no PLA default); `SWITCH_T0 MATERIAL=NYLON` is refused and nothing moves.
- [ ] **D9** An item ticked *switch off when made safe* stays on after a normal finish (print and laser), and goes off on cancel.

The record follows the tool (new in 1.4.7)

- [ ] **D10** Swap SwitchFly back on after leaving it on path 2 – the selector turns to path 1 and `G1 E5` feeds in.
- [ ] **D11** Restart, answer *Yes – SwitchFly* – nothing moves, path 2 still feeds in.
- [ ] **D12** Swap to BlockOne after SwitchFly path 2 – `G1 E5` feeds in.

Set Z zero

- [ ] **E1–E2** `SET_Z_ZERO` with LightSaber or HotJoe – the buttons move the bed the amount shown, Bed down stops at Z290, SET moves nothing, and `G1 Z5 F300` then lifts the tool 5 mm above the surface.
- [ ] **E3** Restart with the bed parked, run `SET_Z_ZERO` – the *Bed position* question; **Yes** – the bed doesn't move, X/Y home, then the Set Z zero window opens. Needs `[force_move] enable_force_move: True` (in `toolchanger.cfg`).
- [ ] **E4** Restart after moving the bed below Z150 – no question; home Z with a print head.
- [ ] **E6** After a CNC or drag-knife setup, swap to BlockOne and print – no X/Y shift.

Laser, CNC and drag-knife jobs

- [ ] **F1–F3** Laser at low power on scrap; a CNC air cut 10 mm above the stock first; a drag-knife test line. Run the first file from the LaserWeb settings in *Slicer and CAM settings* at low power.
- [ ] **F4** Pause and resume each tool type – the 10 mm lift clears the work, and the spindle is back to speed before it re-enters the cut.
- [ ] Power values – tune every placeholder power and on-time on the real tools.

Emergency stop and backups

- [ ] **H1–H2** `EMERGENCY_STOP` with the focus dot on – laser off at once; with HotJoe – relay open, ESC quiet.
- [ ] **I1** Klipper-Backup shows your config files in GitHub again after removing the stray `.git` folder.

Settings to know

- [ ] Idle timeout – add `[idle_timeout] timeout: 1800` if you use the full ABS/ASA cooldown.
- [ ] Connector pins – only PB1, PC0 and PC1 are free; list any other spare umbilical pins in `myrhino/umbilical.json`.
- [ ] `LEDFLASH` takes about 1.35 s, which also delays the macros that call it (cancel, emergency stop).

Wiring to confirm

- [ ] **Spindle and servo PWM contact.** The umbilical pin map puts the ESC signal (PB11) on contact 18, but the original HotJoe and SwitchFly sheets listed the PWM signal on contact 19. This manual and the portal use 18, which matches the pin map and `printer.cfg`; check the connector and correct the pages if it is 19.
- [ ] **Laser 12 V power pin.** The pin map lists PD12 (Fan2) for the 12 V power activate on contacts 20–21; `printer.cfg` drives `LASER_INITIALIZE` from PD13 (Fan3 on the Octopus v1.1).
- [ ] **X and Y endstops.** The wiring sheets give X-limit PG6 and Y-limit PG9; `printer.cfg` has `stepper_x` on PG9 and `stepper_y` on PG6.
- [ ] **SwitchFly servo power.** The SwitchFly page lists 24 V power activate on contacts 11–12, while its servo wiring diagram feeds the regulator from 12 V activate.
- [ ] **Fan labels.** On the Octopus v1.1 PA8 is FAN0 and PE5 is FAN1; the pin sheets label them Fan5 and Fan0. The pins are right; only the labels differ.

Network settings (new in 1.4.8)

- [ ] **N1–N4** Read the network; power saving off with Keep, Undo now, and letting the 2-minute timer switch it back.
- [ ] **N5–N7** Rename, fixed address kept from the new address, and a wrong address that switches back by itself.
- [ ] **N8** Changes refused during a print.

PrusaSlicer on the Rhino (from Flathub since 1.5.2)

- [ ] **P0–P2** Install Flatpak once (sudo apt install flatpak), install the engine from Rhino Studio, Settings / Software, and check it runs over SSH without a screen.
- [ ] **P3–P6** Slice a part, check the Rhino's start and temperatures are in the file, a newer slice stops the older one (and its sandbox), and the same slice again comes from the cache.
- [ ] **P7** Slicing during a print: half the cores on a bigger host, refused on a Pi 3B; the print keeps going smoothly.
- [ ] **P8–P9** Updating the engine from Flathub, and going back to the version before.

Rhino Studio from the Rhino (new in 1.5.1)

- [ ] **S1–S3** The install names Studio's address; Studio opens at `http://<pi>:5001` on a PC and a phone, with the Portal's tasks and tools; Mainsail still opens.
- [ ] **S4–S6** A change through the Portal is accepted, the camera shows, and a job reaches Files.
- [ ] **S7** Undo takes Studio's site away again and leaves Mainsail working.